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Ko et al.

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(54) **METHOD FOR FORMING SEMICONDUCTOR STRUCTURE**

(71) Applicant: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.**, Hsinchu (TW)

(72) Inventors: **Chung-Ting Ko**, Kaohsiung (TW); **Wen-Ju Chen**, New Taipei (TW); **Tai-Chun Huang**, Hsin-Chu (TW)

(73) Assignee: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.**, Hsinchu (TW)

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Related U.S. Application Data

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(51) **Int. Cl.**
H10D 84/01 (2025.01)
H01L 21/02 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 84/017** (2025.01); **H01L 21/0259** (2013.01); **H01L 21/30604** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC H10D 84/017; H10D 30/031; H10D 30/6713; H10D 30/6735; H10D 30/6757;
(Continued)

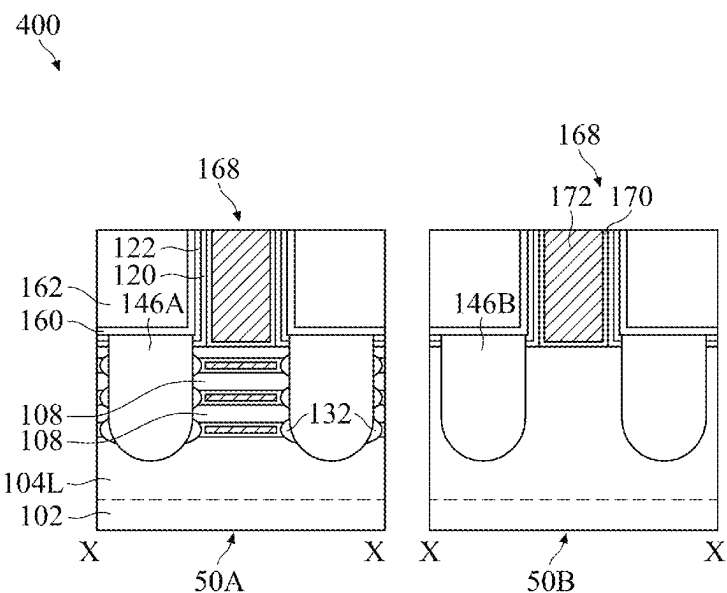
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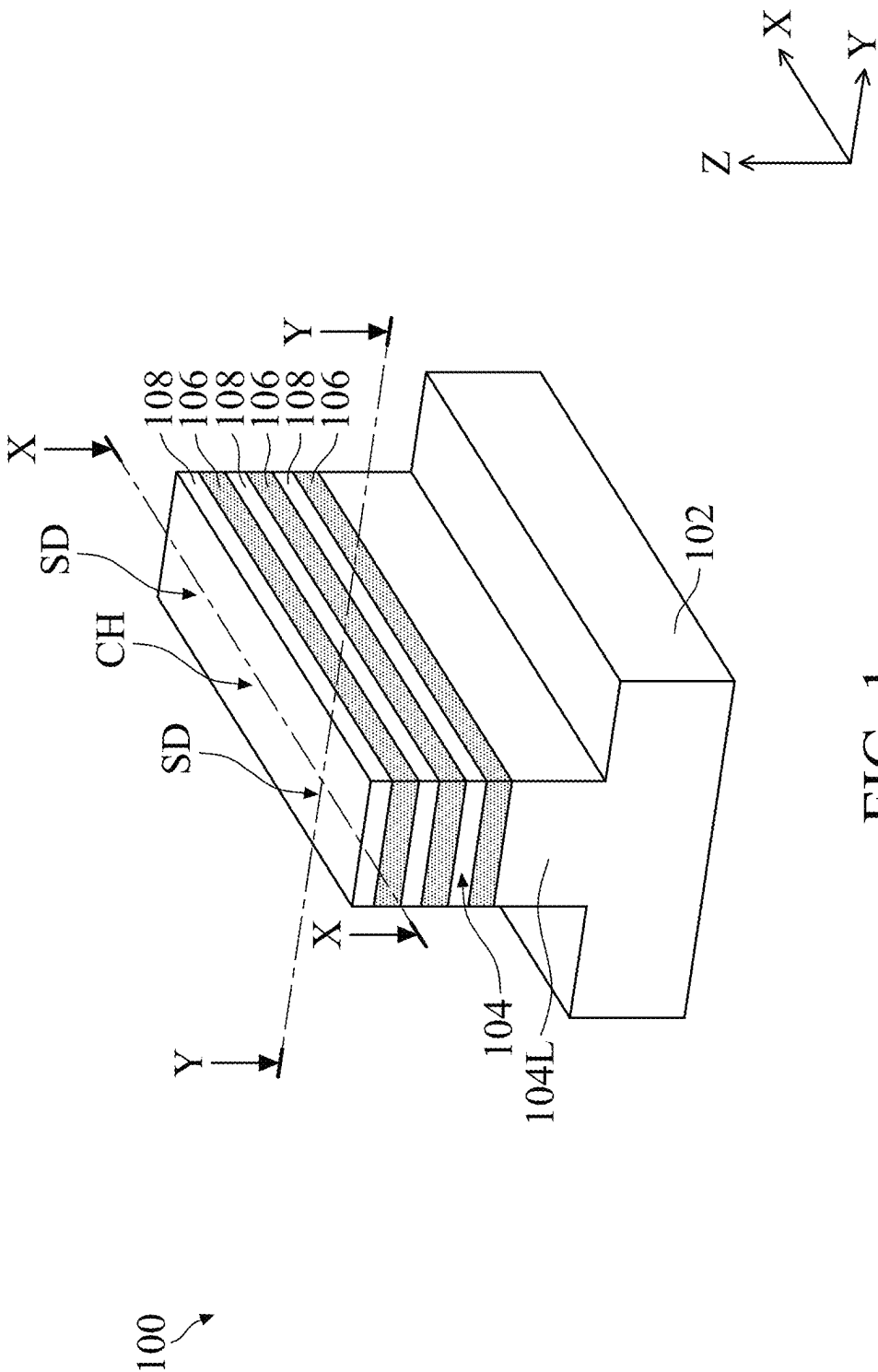
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Primary Examiner — William B Partridge
Assistant Examiner — David Paul Sedorook
(74) *Attorney, Agent, or Firm* — Birch, Stewart, Kolasch & Birch, LLP

(57) **ABSTRACT**
A method for forming a semiconductor structure is provided. The method includes forming a first active region and a second active region, etching the first active region and the second active region to form a first recess and a second recess, respectively, forming the first dielectric layer over the first active region and the second active region, forming a first fill layer over the first dielectric layer to overfill the first recess and the second recess, forming a first dielectric mask over the first fill layer, etching first portions of the first dielectric mask and the first fill layer over the first active region, removing a first portion of the first dielectric layer over the first active region, and forming a first source/drain feature on the first active region.

20 Claims, 35 Drawing Sheets



- (51) **Int. Cl.**
H01L 21/306 (2006.01)
H01L 21/308 (2006.01)
H10D 30/01 (2025.01)
H10D 30/67 (2025.01)
H10D 62/00 (2025.01)
H10D 62/10 (2025.01)
H10D 64/01 (2025.01)
H10D 84/03 (2025.01)
H10D 84/85 (2025.01)
- (52) **U.S. Cl.**
CPC **H01L 21/3081** (2013.01); **H10D 30/031**
(2025.01); **H10D 30/6713** (2025.01); **H10D**
30/6735 (2025.01); **H10D 30/6757** (2025.01);
H10D 62/021 (2025.01); **H10D 62/118**
(2025.01); **H10D 64/017** (2025.01); **H10D**
64/018 (2025.01); **H10D 84/0167** (2025.01);
H10D 84/0172 (2025.01); **H10D 84/0184**
(2025.01); **H10D 84/038** (2025.01); **H10D**
84/85 (2025.01)
- (58) **Field of Classification Search**
CPC .. H10D 62/021; H10D 62/118; H10D 64/017;
H10D 64/018; H10D 84/0167; H10D
84/0172; H10D 84/0184; H10D 84/038;
H10D 84/85; H10D 30/797; H10D
30/014; H10D 62/822; H10D 64/021;
H10D 62/121; H10D 30/43; H10D
84/0193; H10D 84/0128; H10D 84/0158;
- H10D 84/013; B82Y 10/00; H01L
21/0259; H01L 21/3081; H01L 21/31144;
H01L 21/31111
See application file for complete search history.
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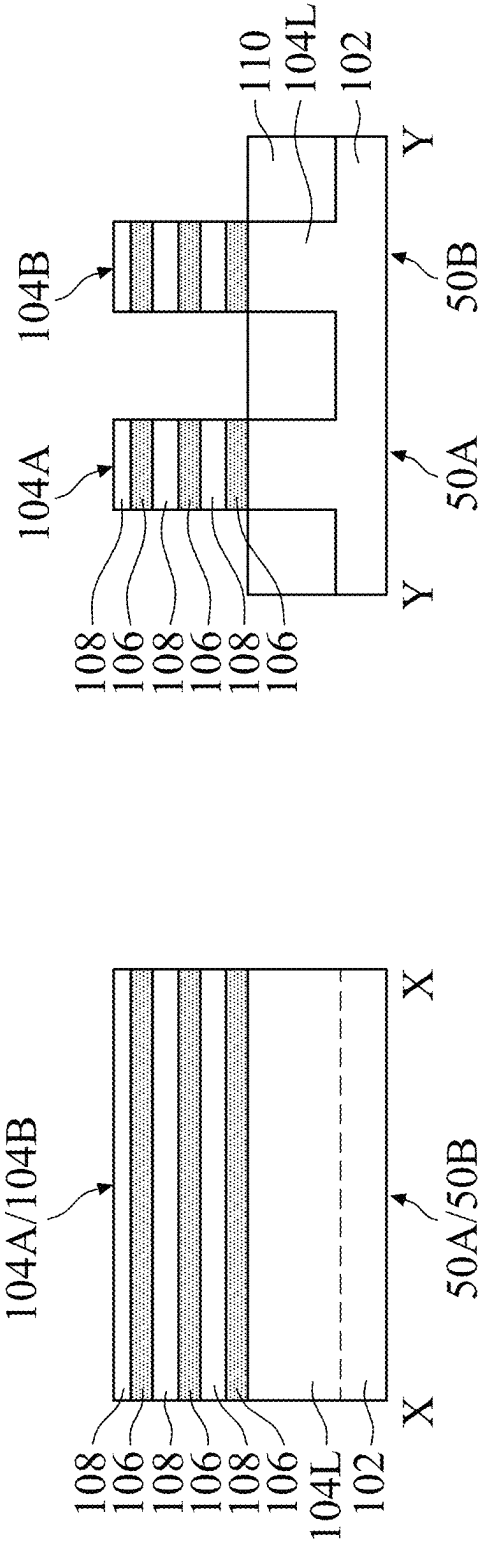


FIG. 2A-1

FIG. 2A-2

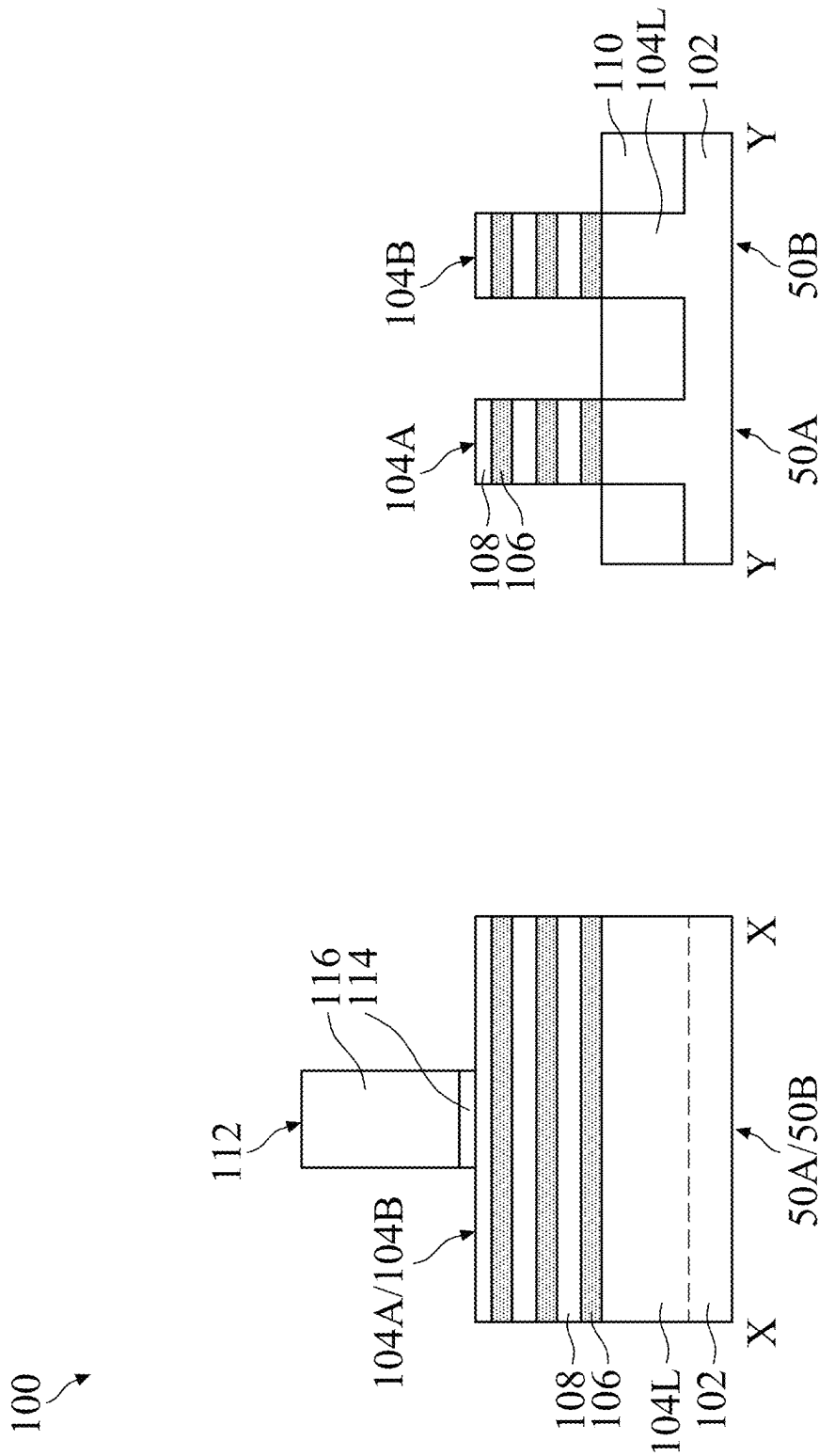


FIG. 2B-1

FIG. 2B-2

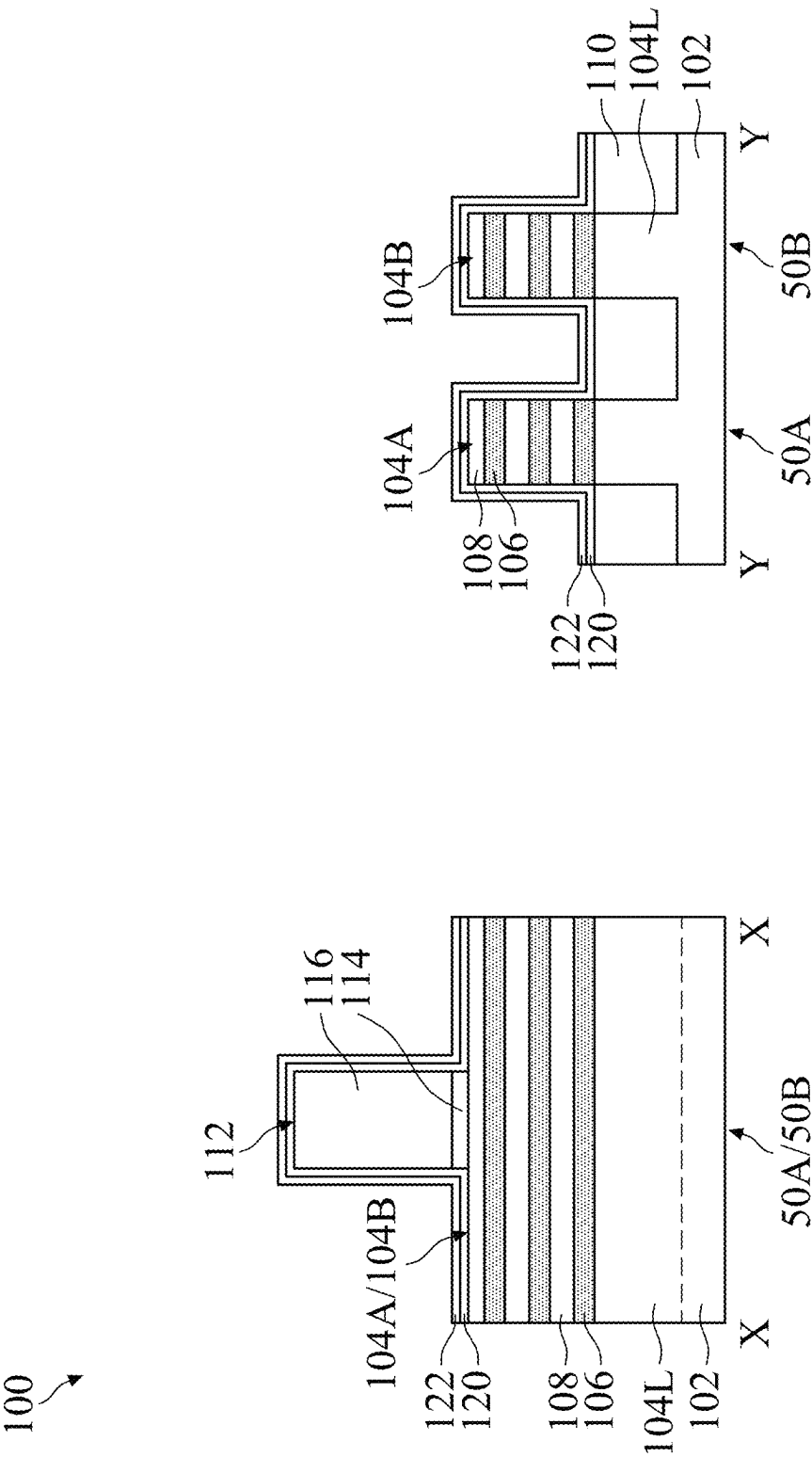


FIG. 2C-1

FIG. 2C-2

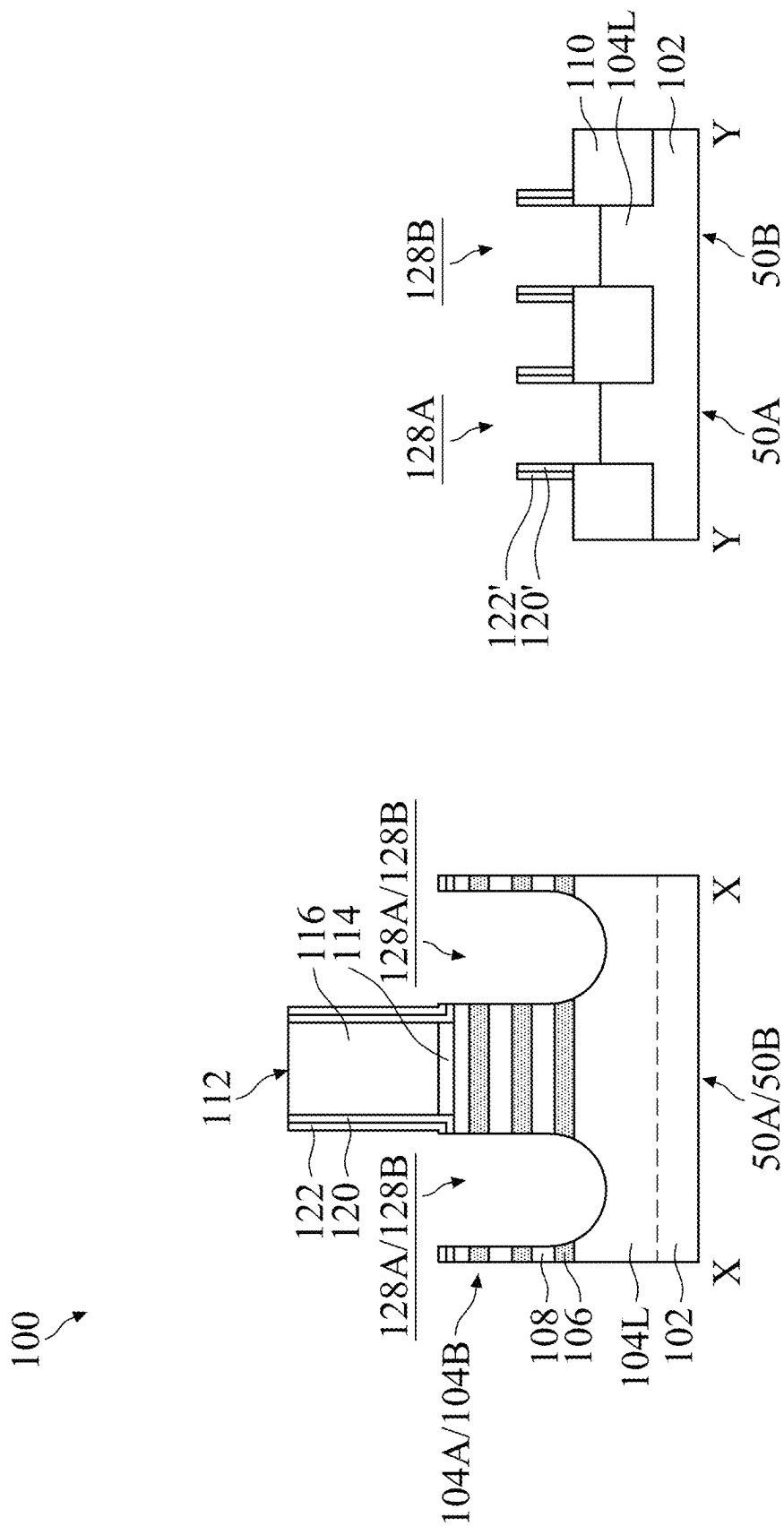


FIG. 2D-1

FIG. 2D-2

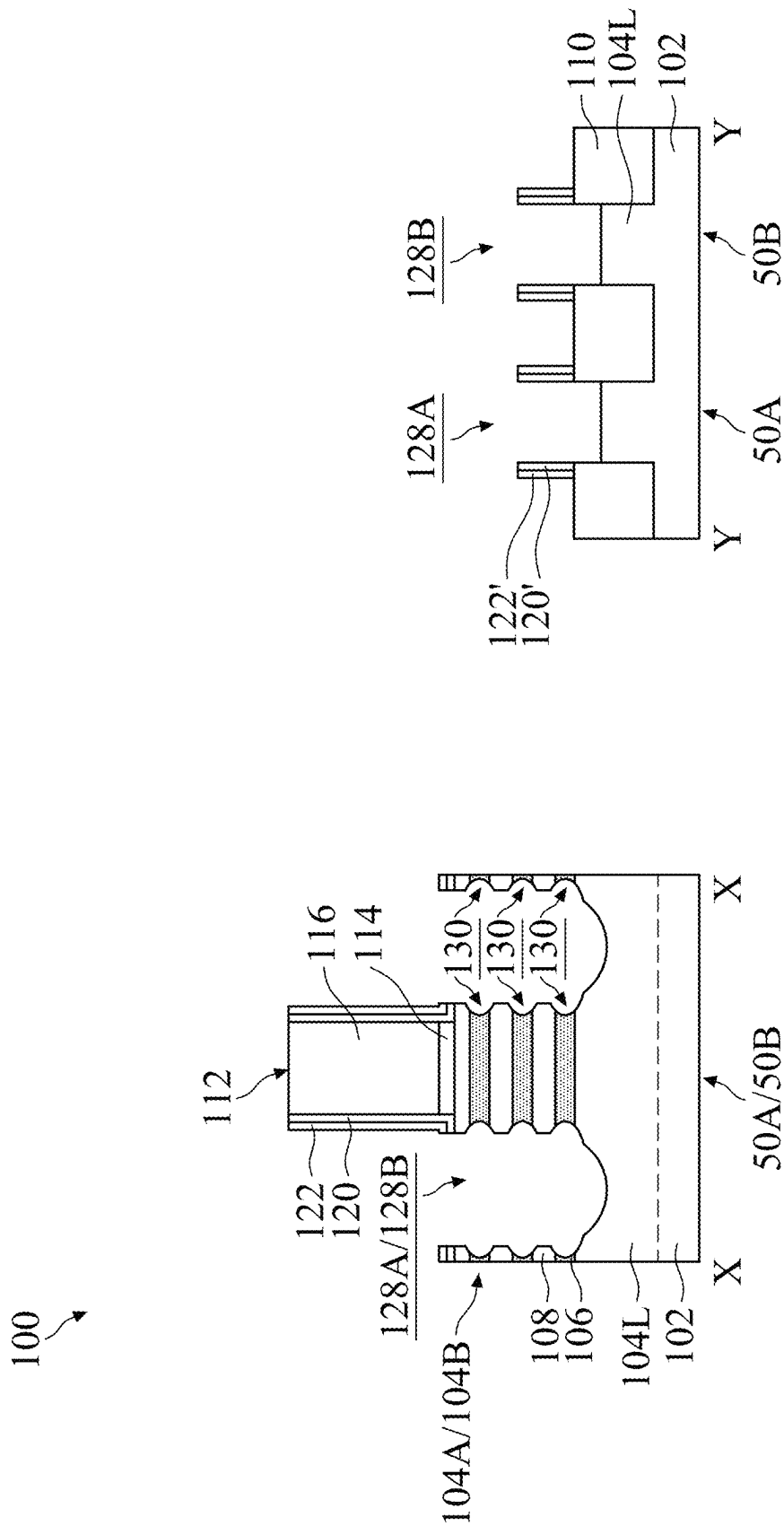


FIG. 2E-1

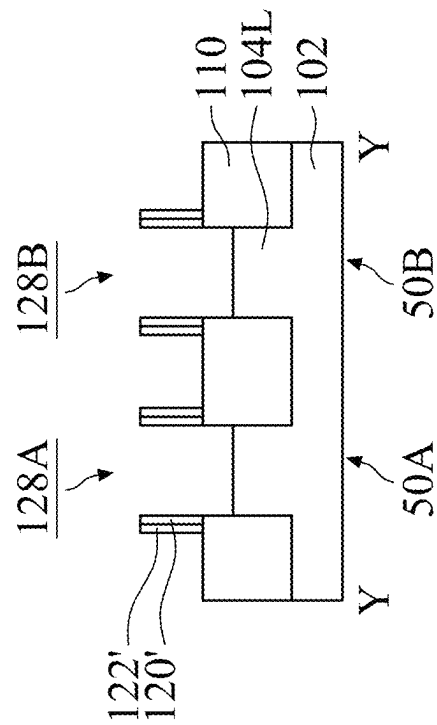


FIG. 2E-2

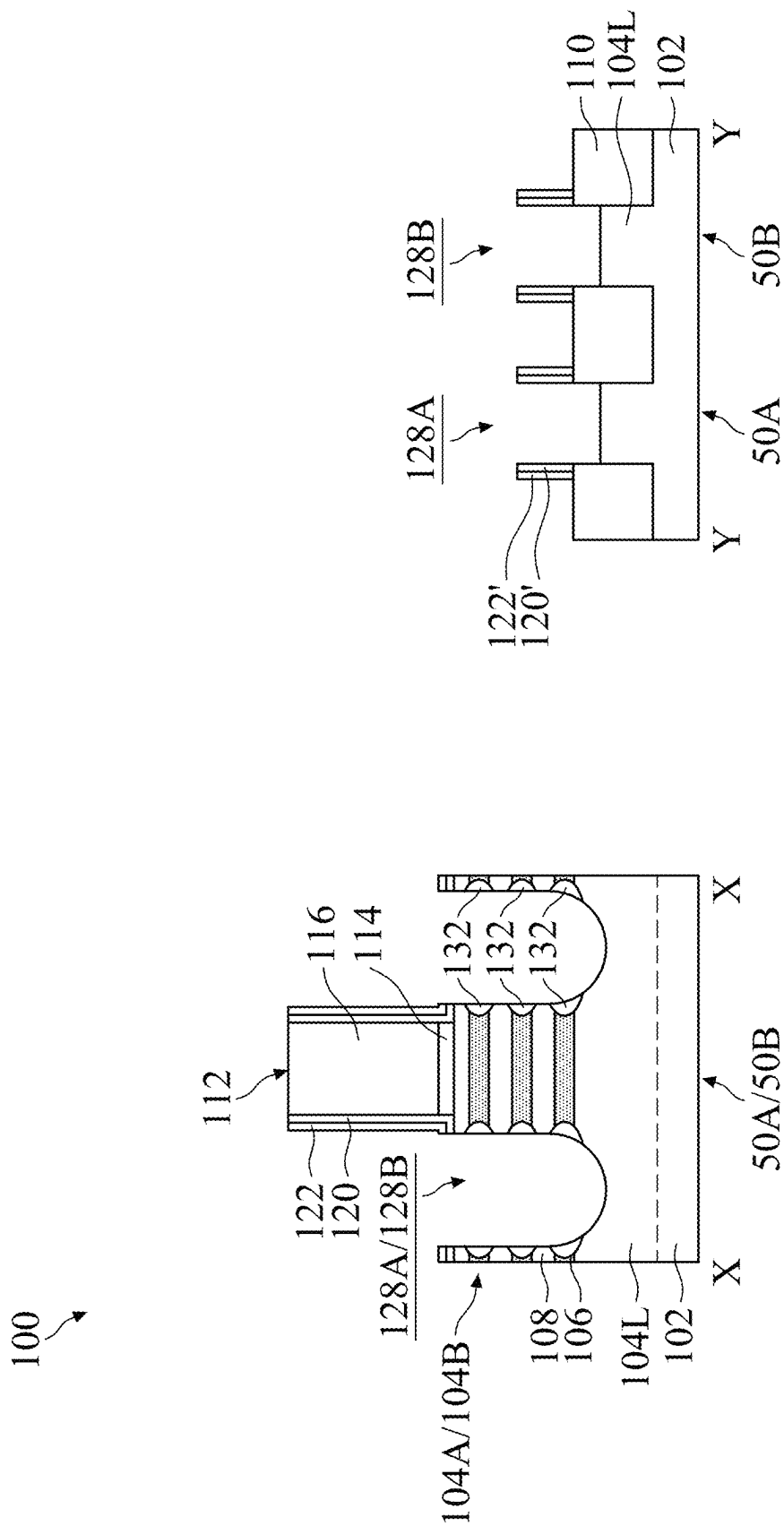


FIG. 2F-1

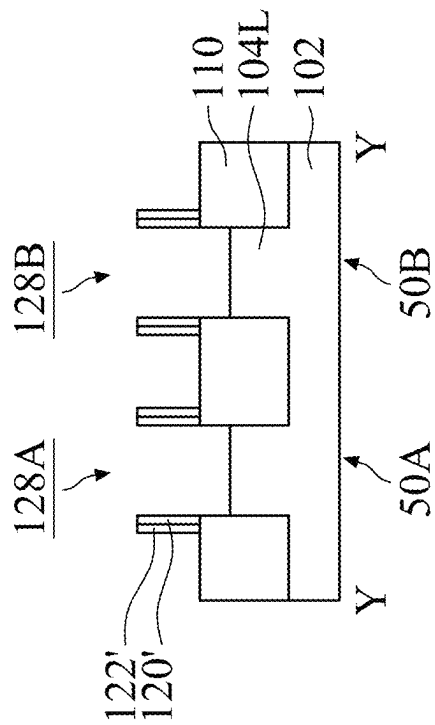


FIG. 2F-2

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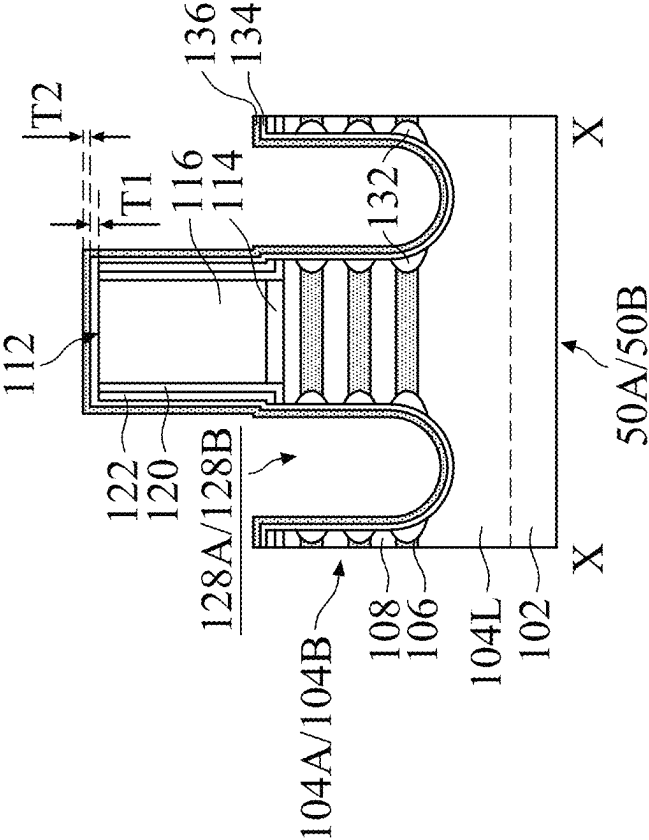


FIG. 2G-1

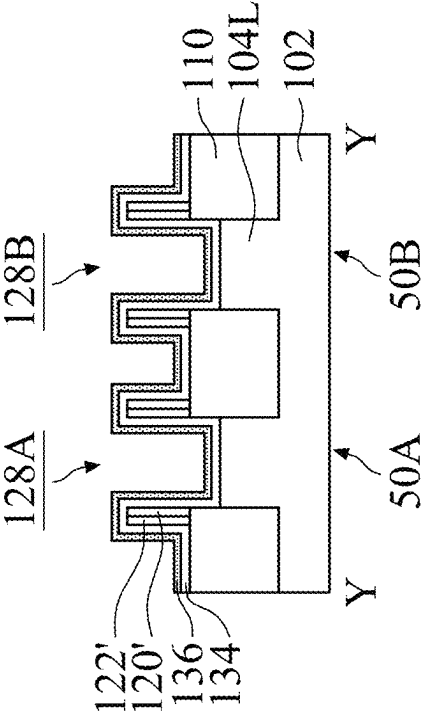


FIG. 2G-2

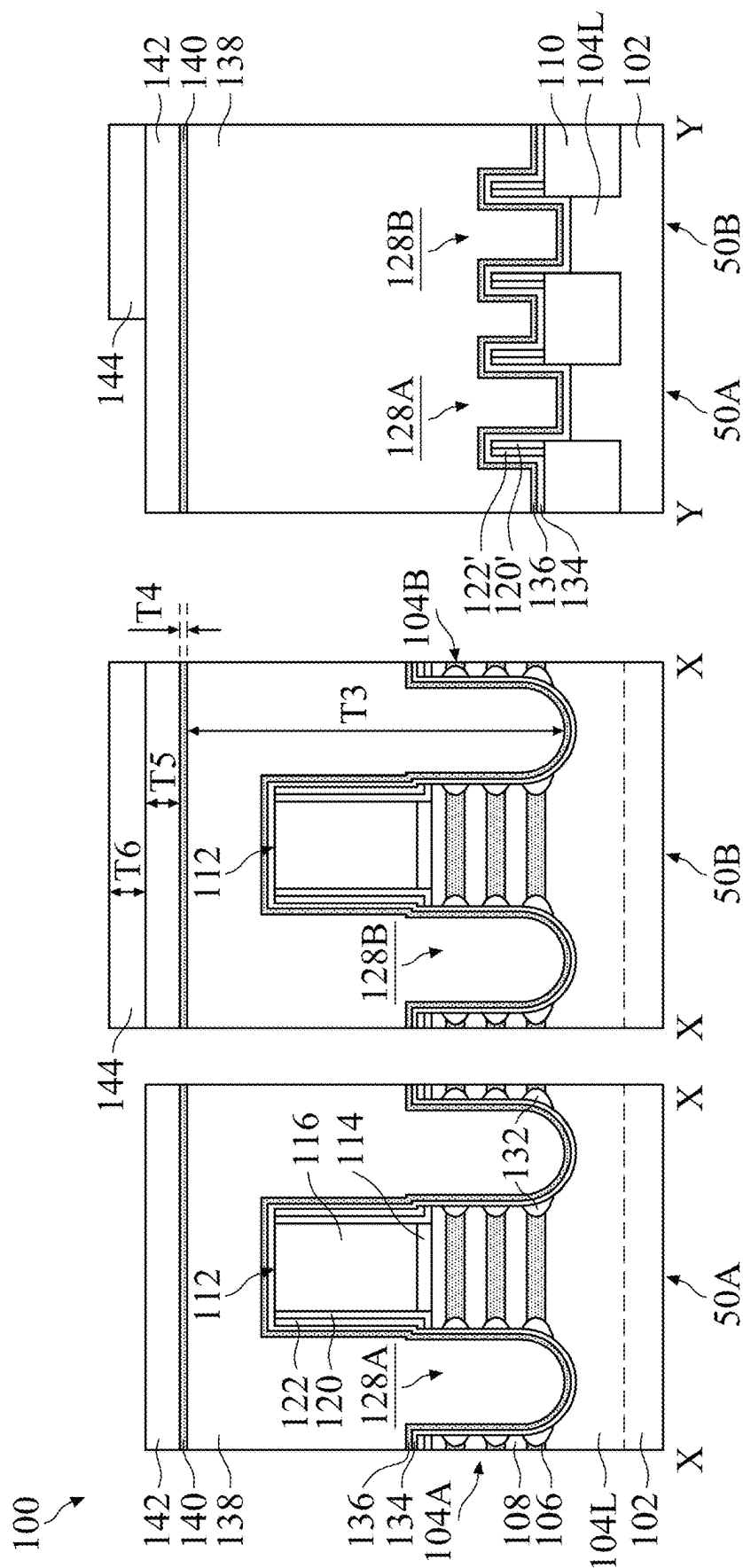


FIG. 2H-1

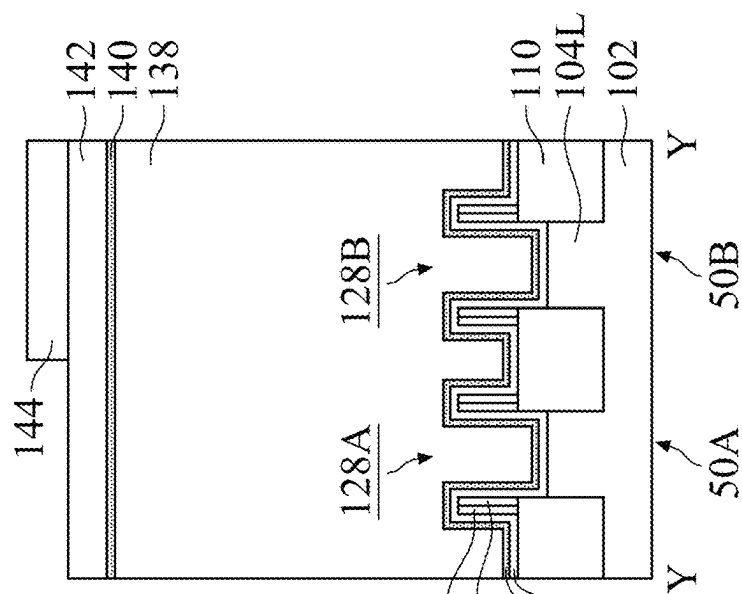


FIG. 2H-2

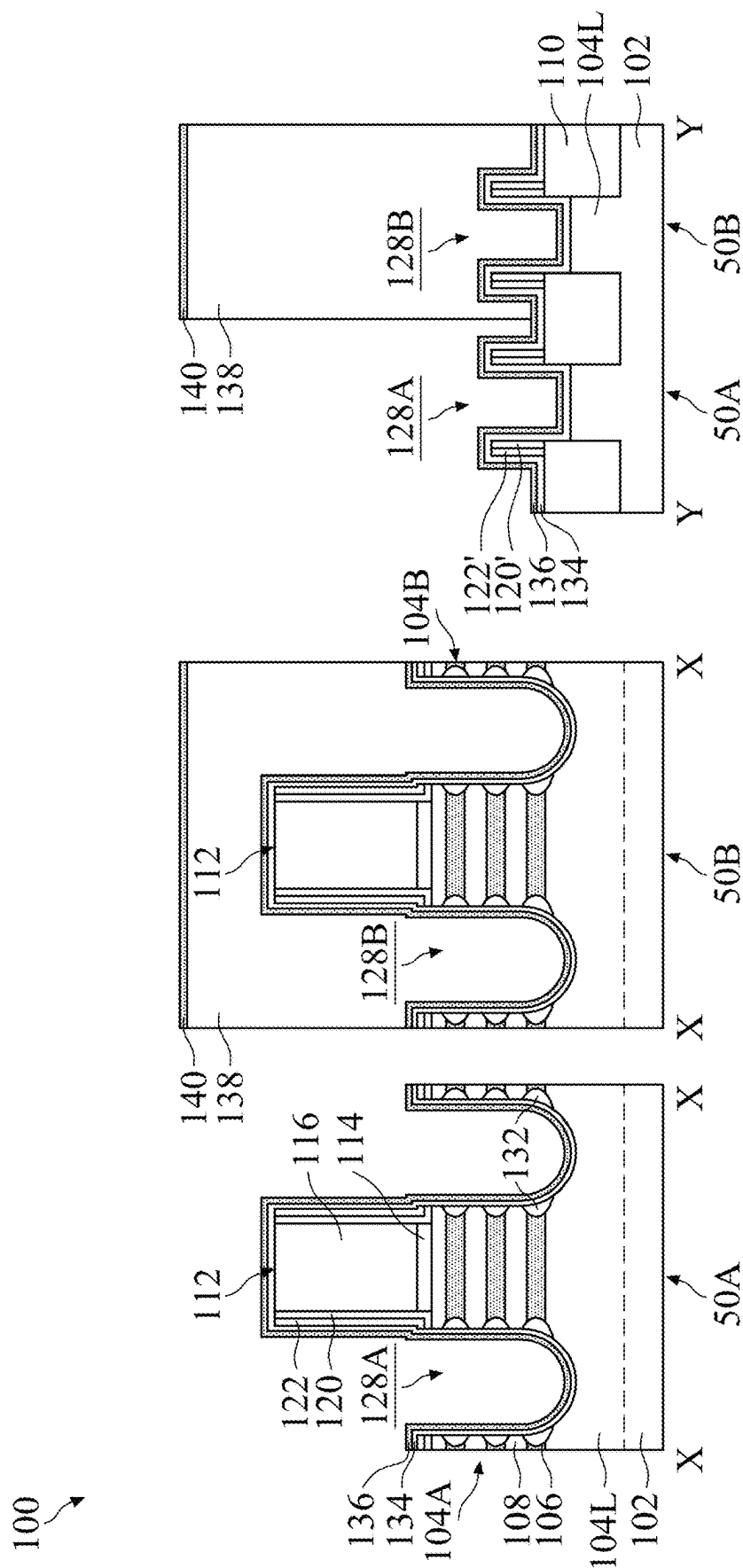


FIG. 2I-1

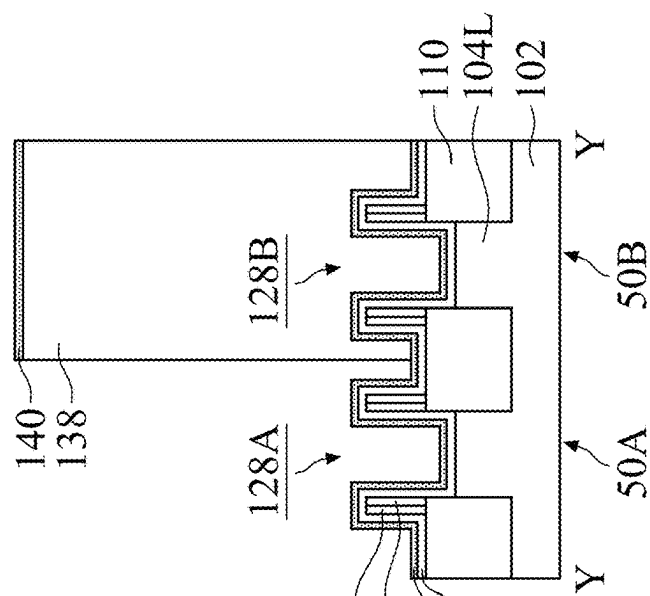


FIG. 2I-2

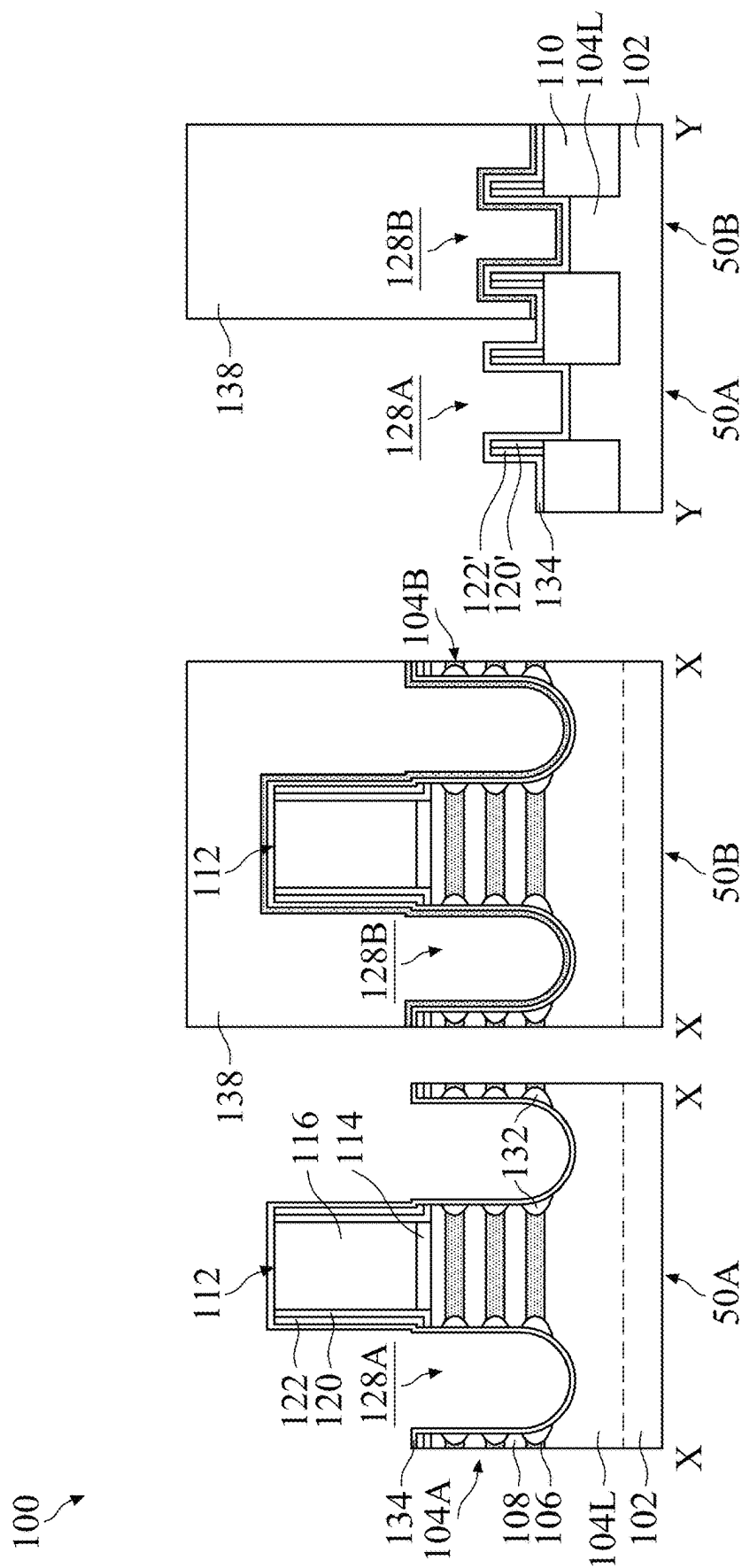


FIG. 2J-1

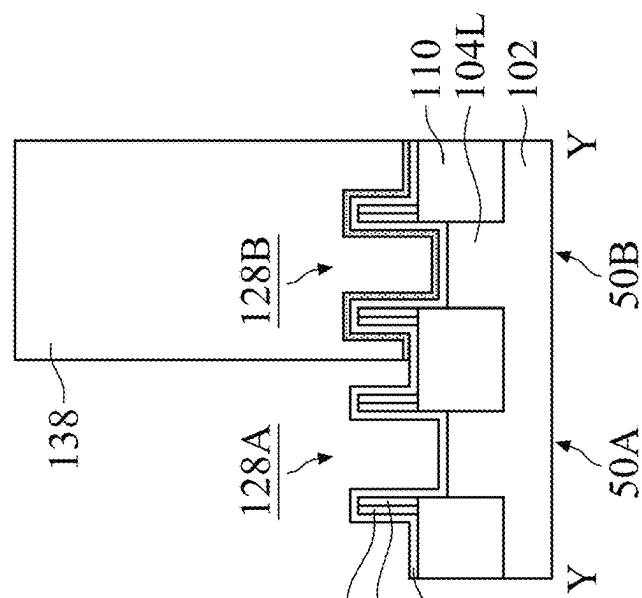


FIG. 2J-2

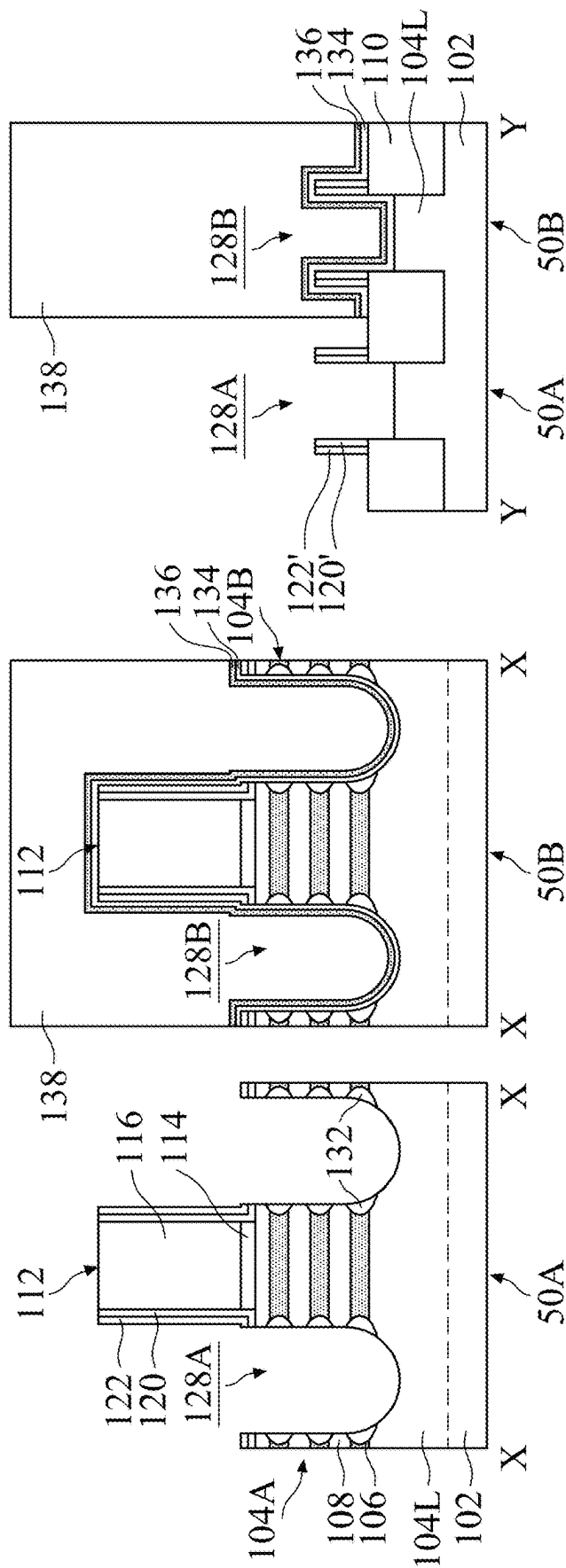


FIG. 2K-1

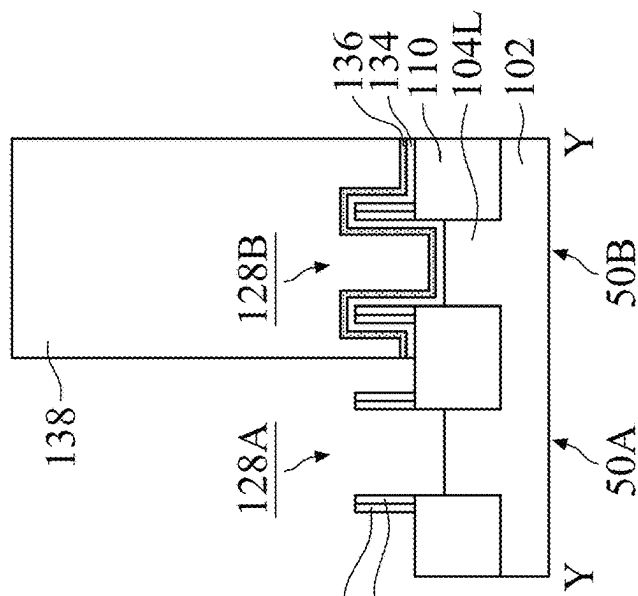


FIG. 2K-2

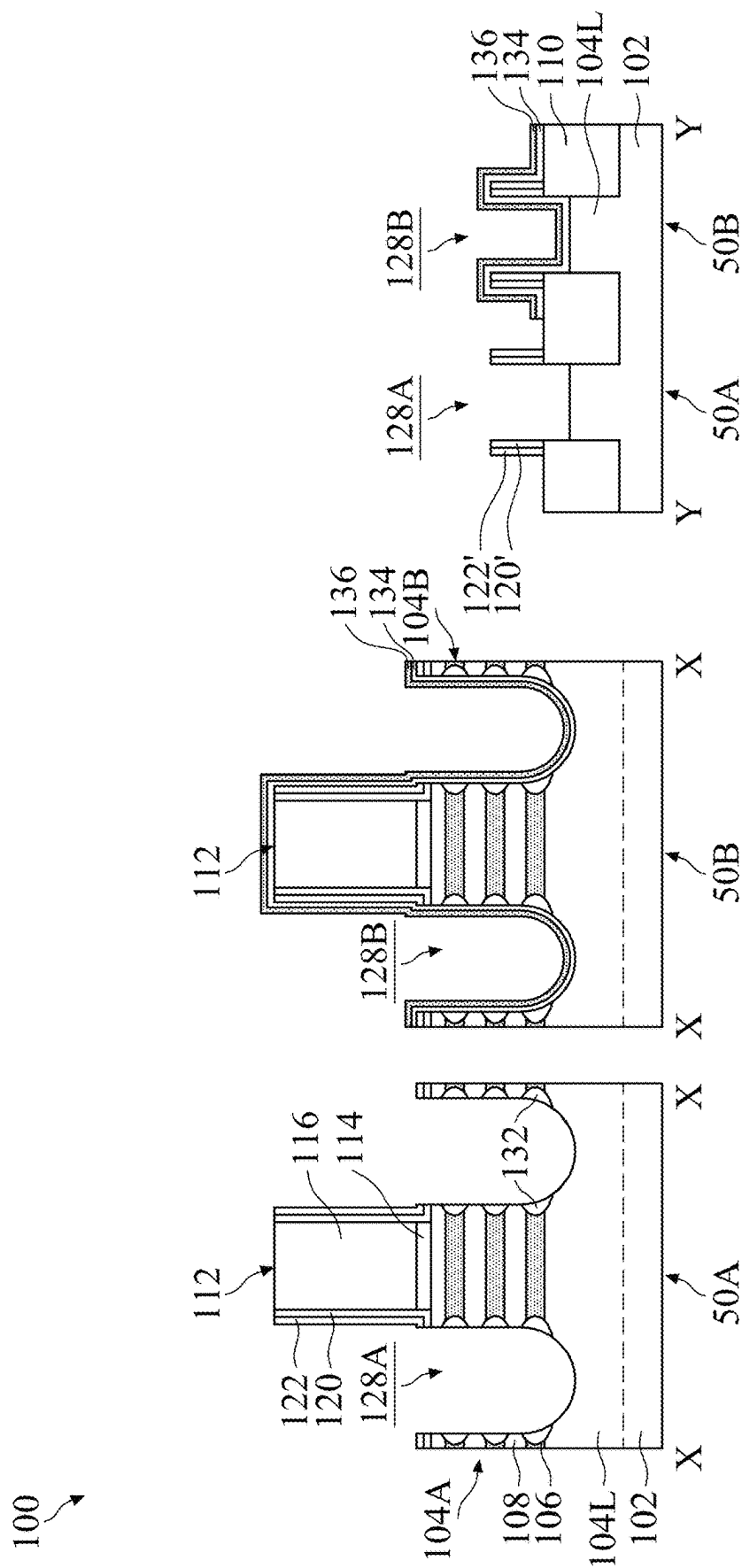


FIG. 2L-1

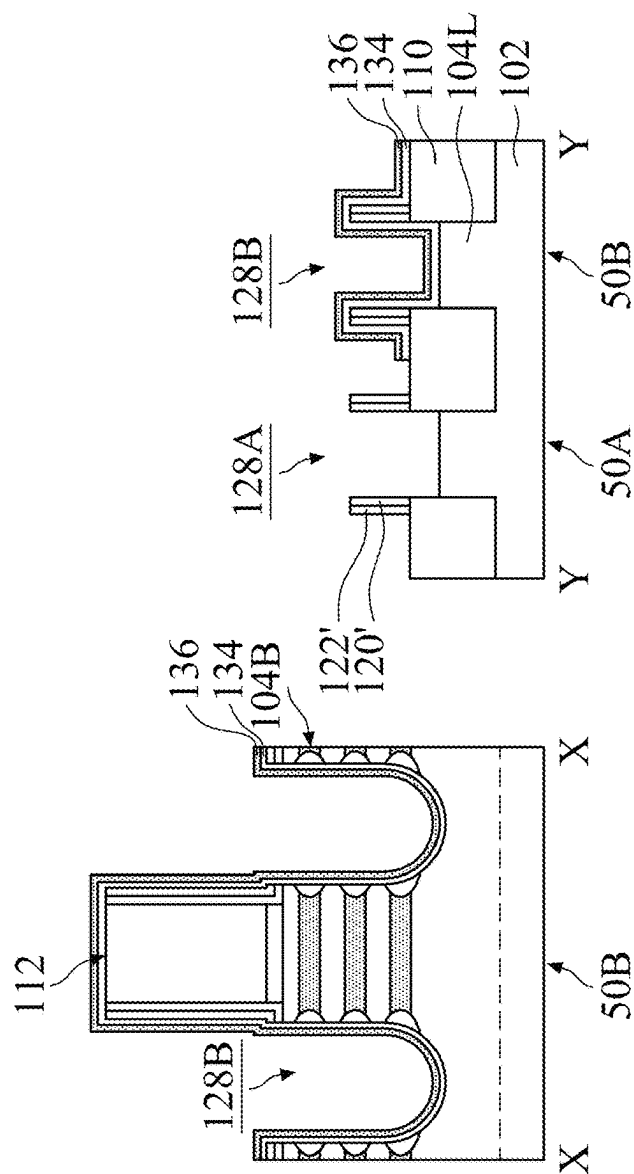
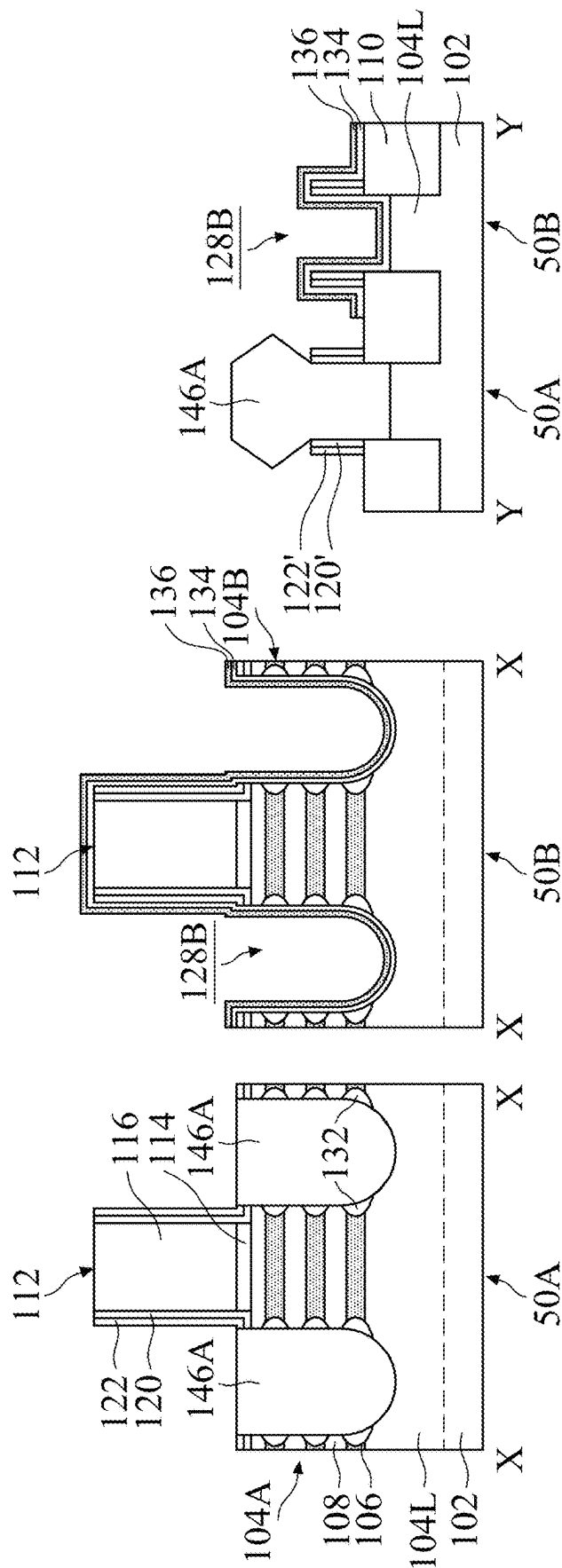


FIG. 2L-2



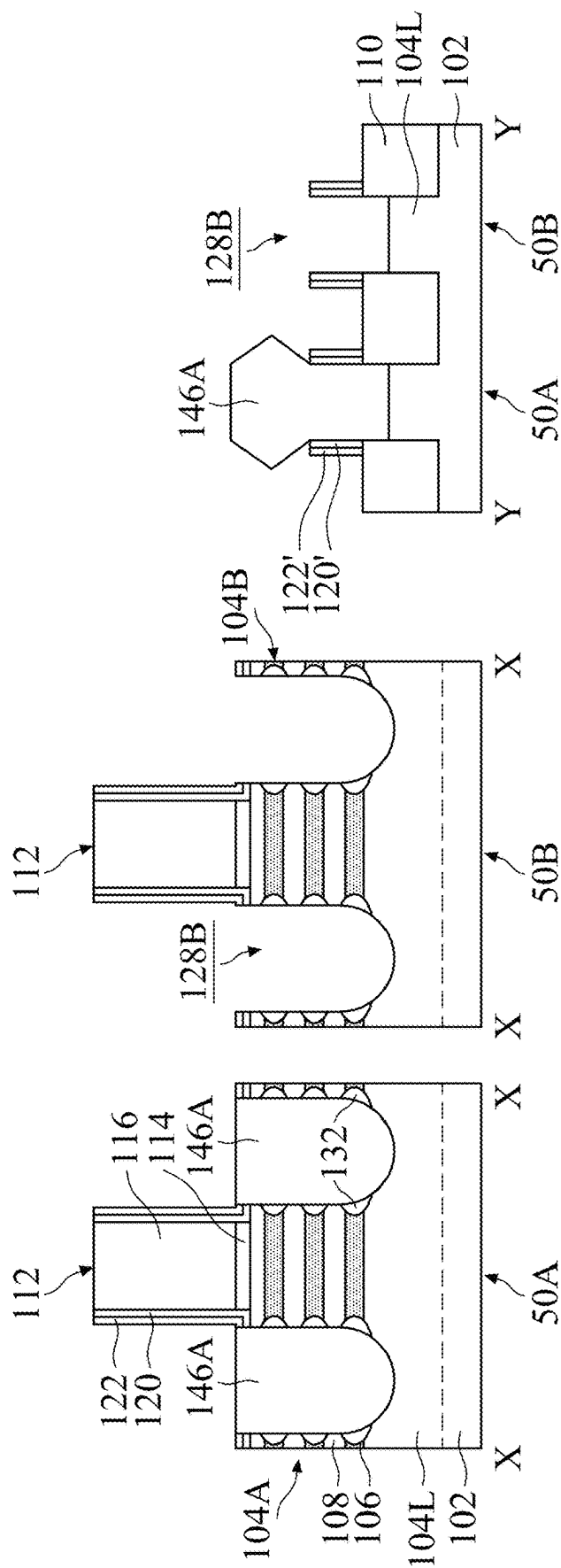


FIG. 2N-1

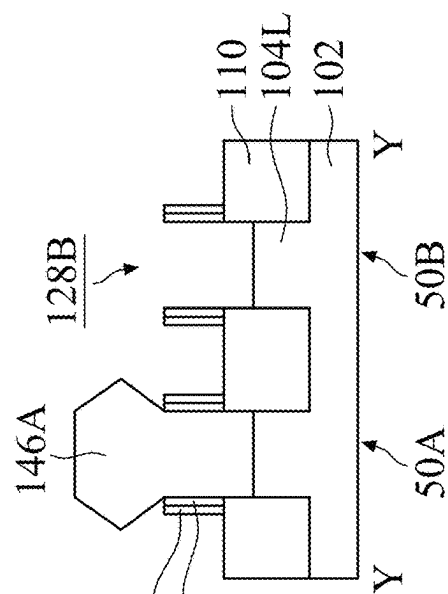


FIG. 2N-2

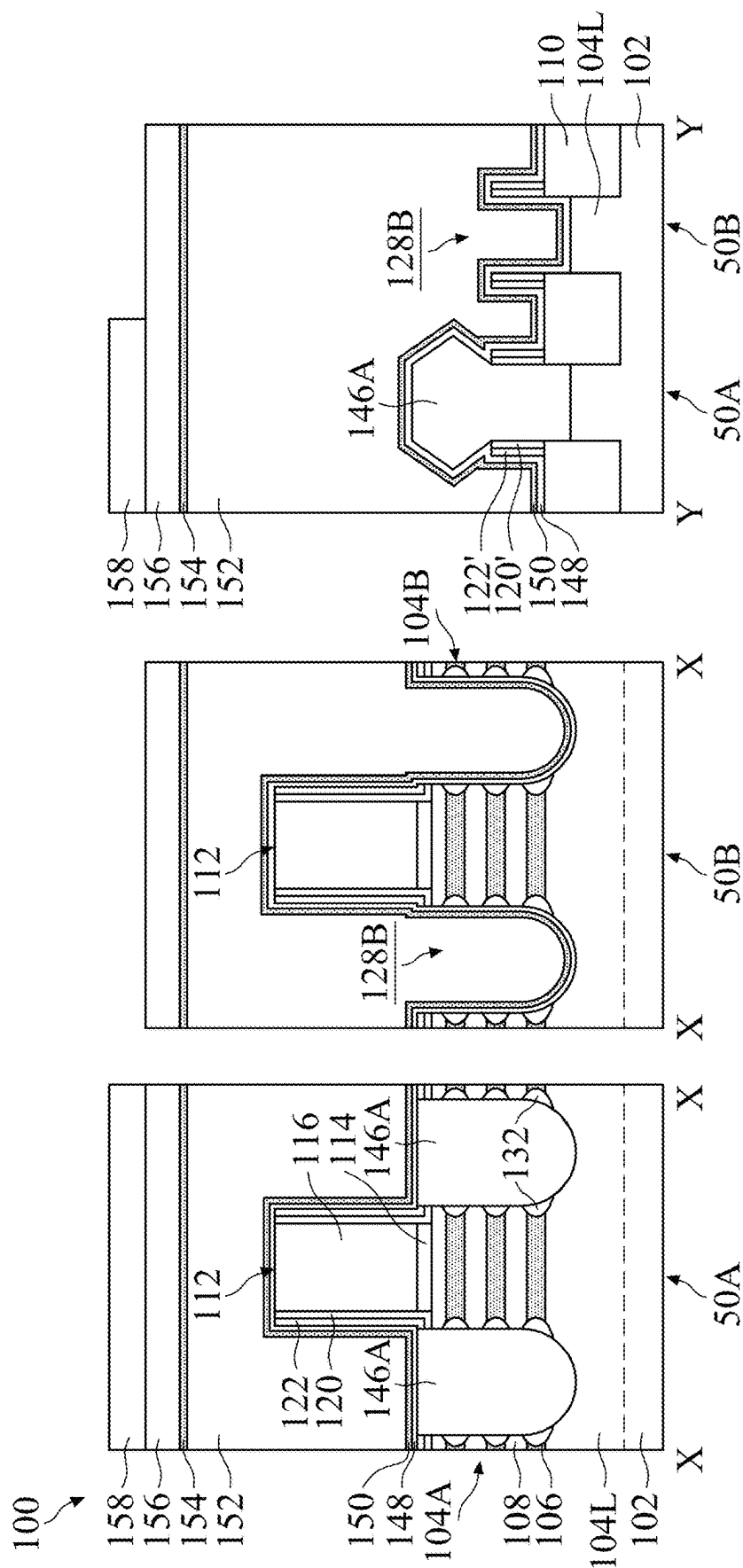


FIG. 20-2

FIG. 20-1

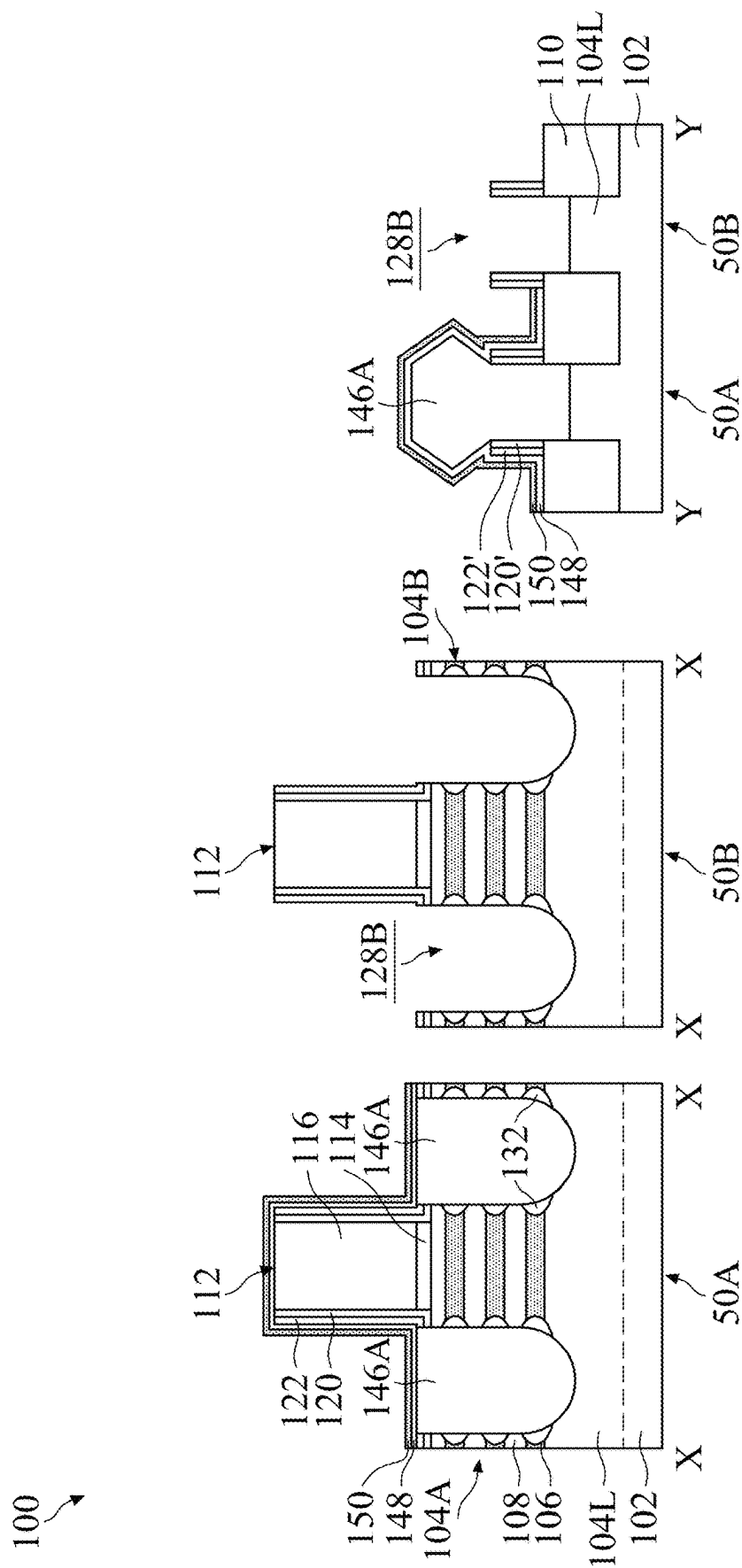


FIG. 2P-1

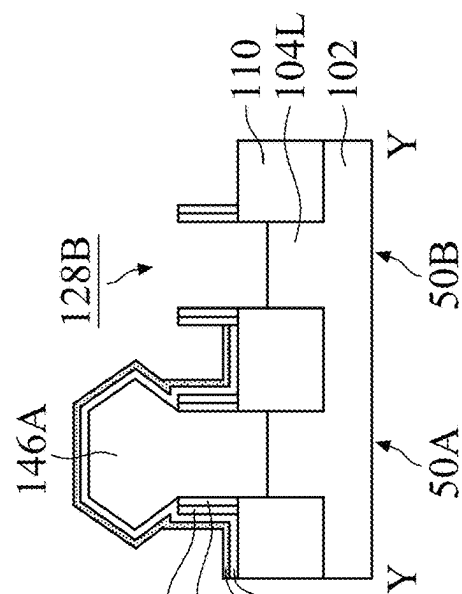


FIG. 2P-2

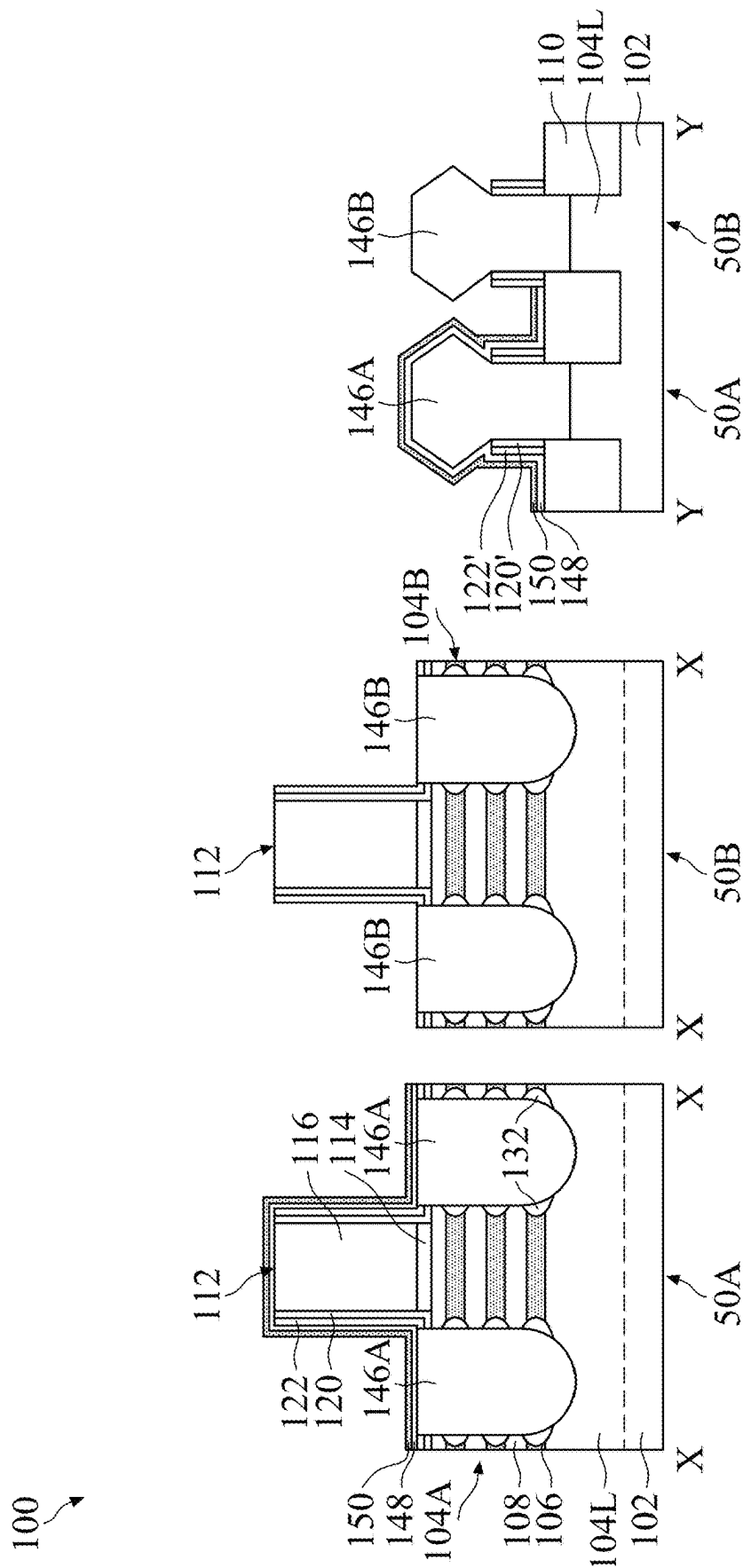
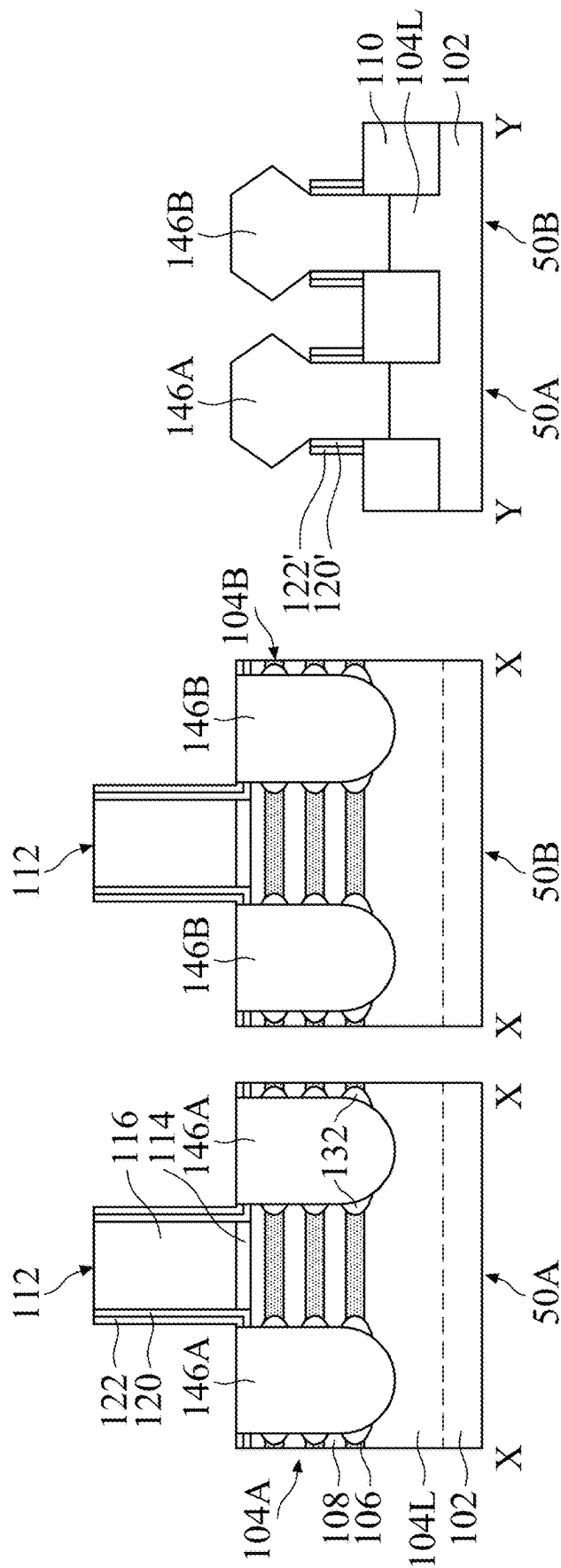
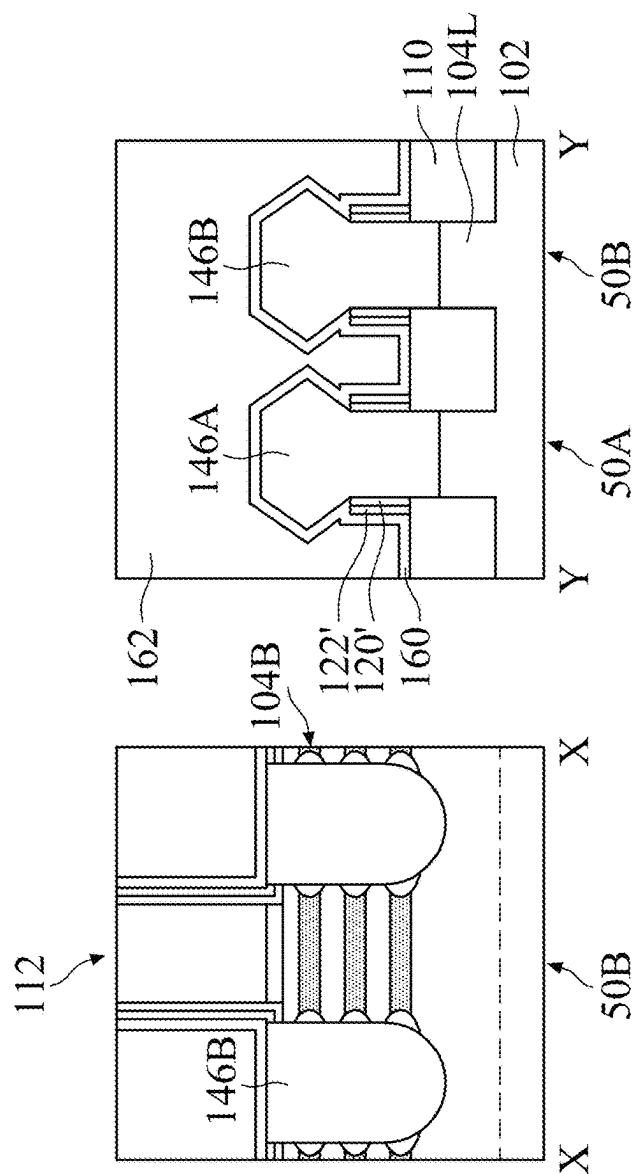
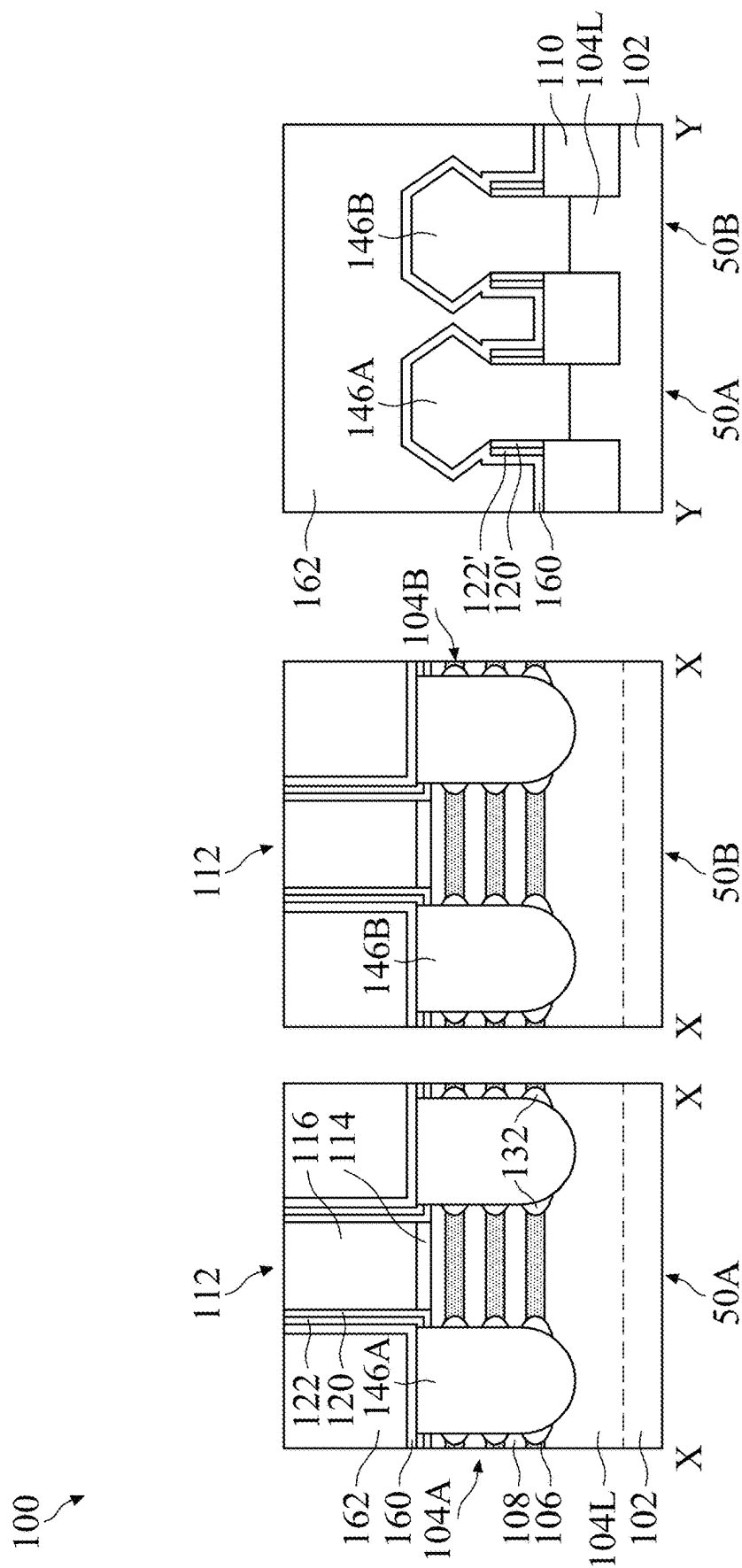


FIG. 2Q-2

FIG. 2Q-1





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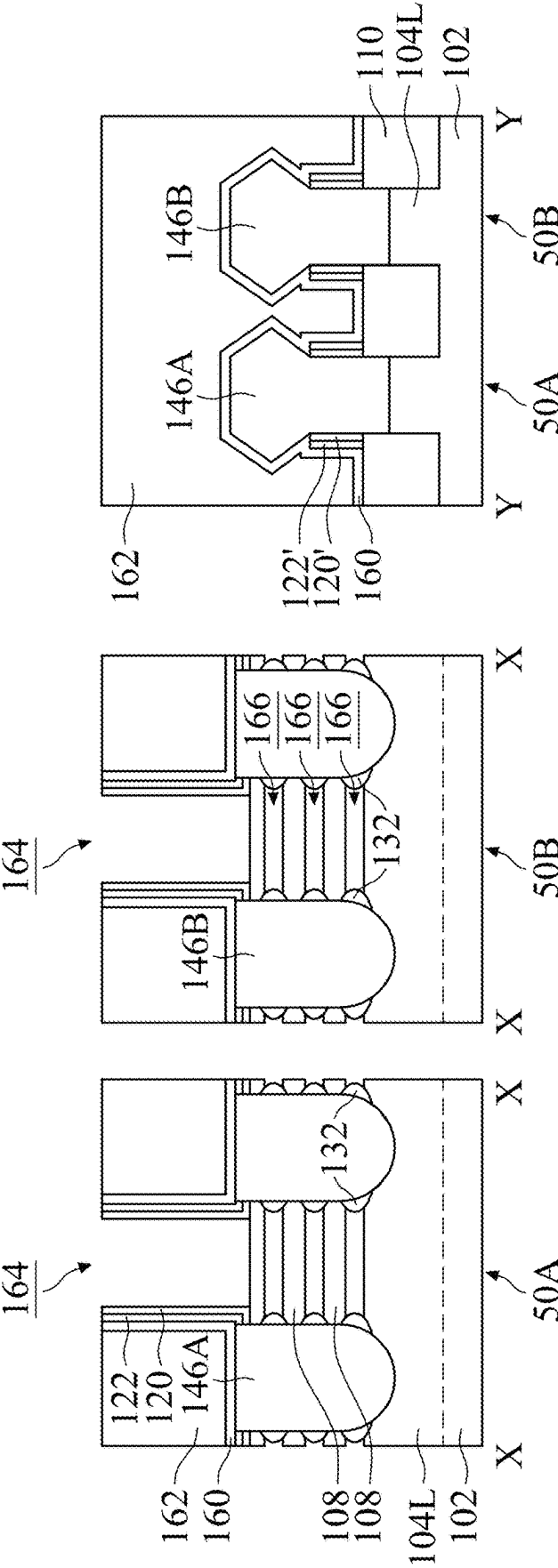


FIG. 2T-1

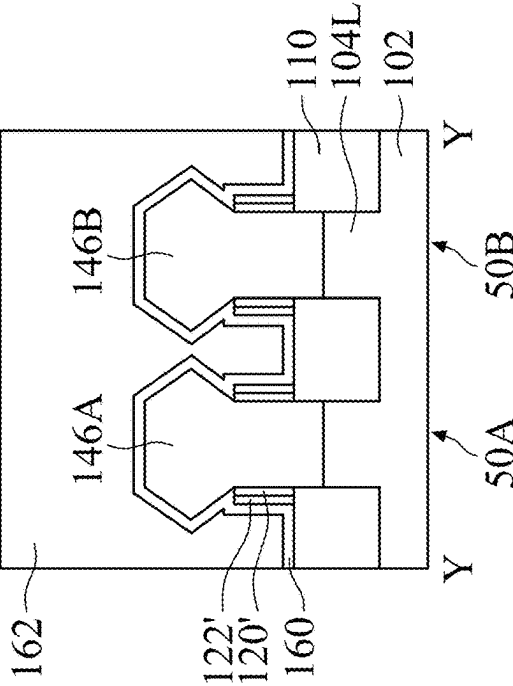


FIG. 2T-2

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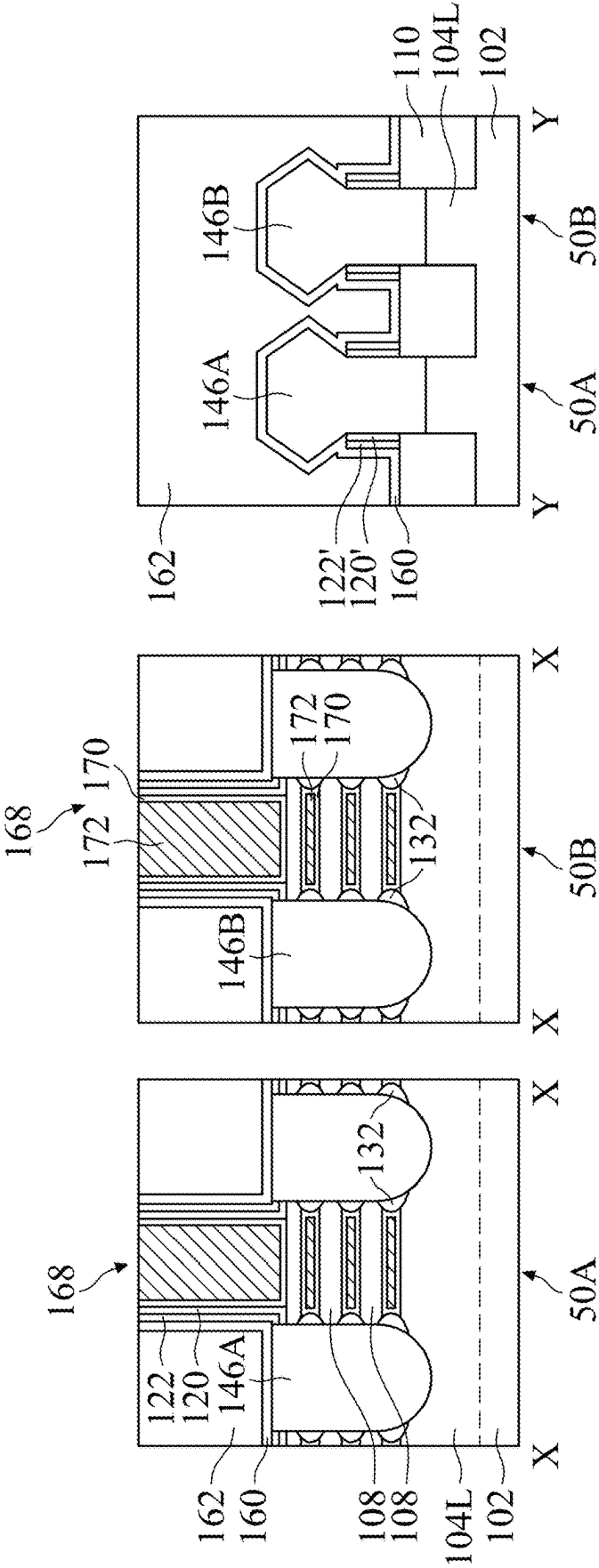


FIG. 2U-1

FIG. 2U-2

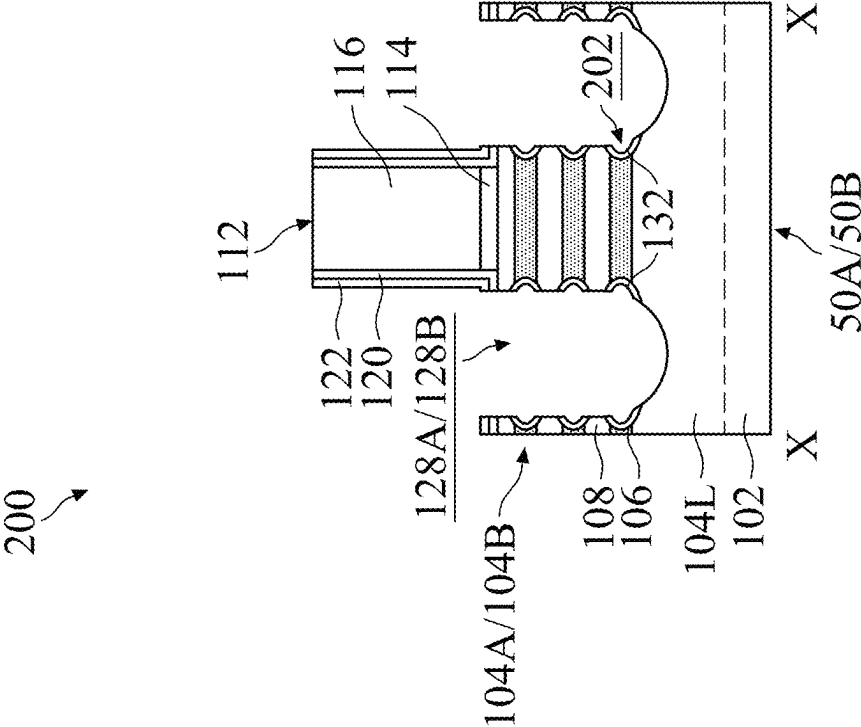


FIG. 3A

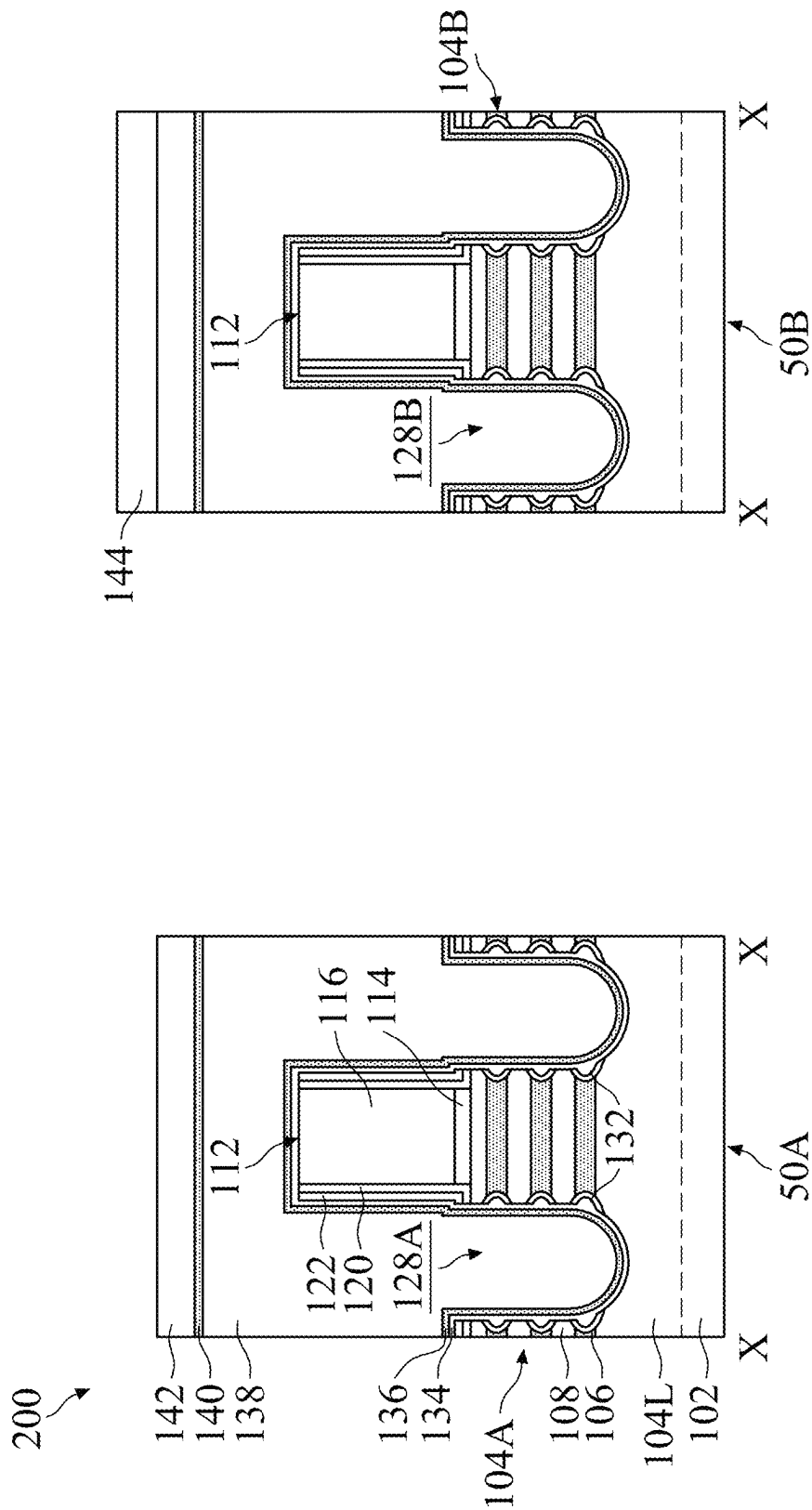


FIG. 3B

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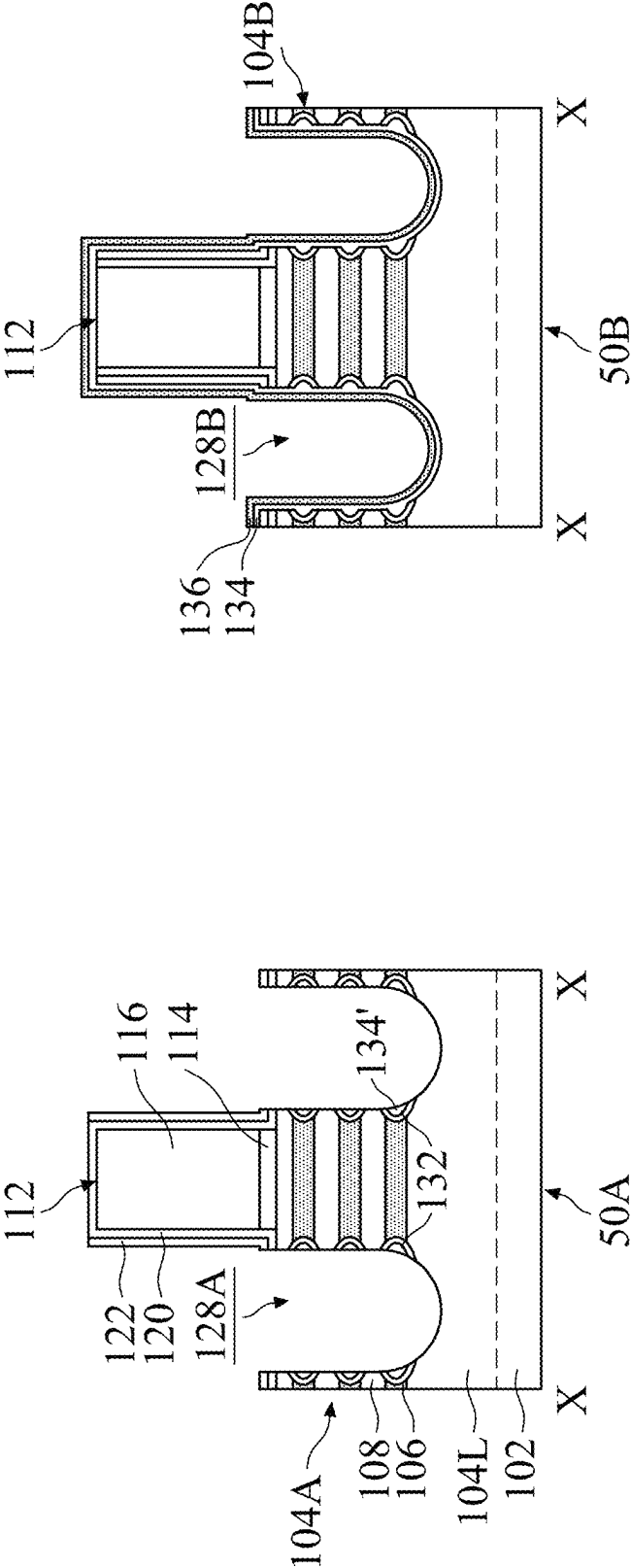


FIG. 3C

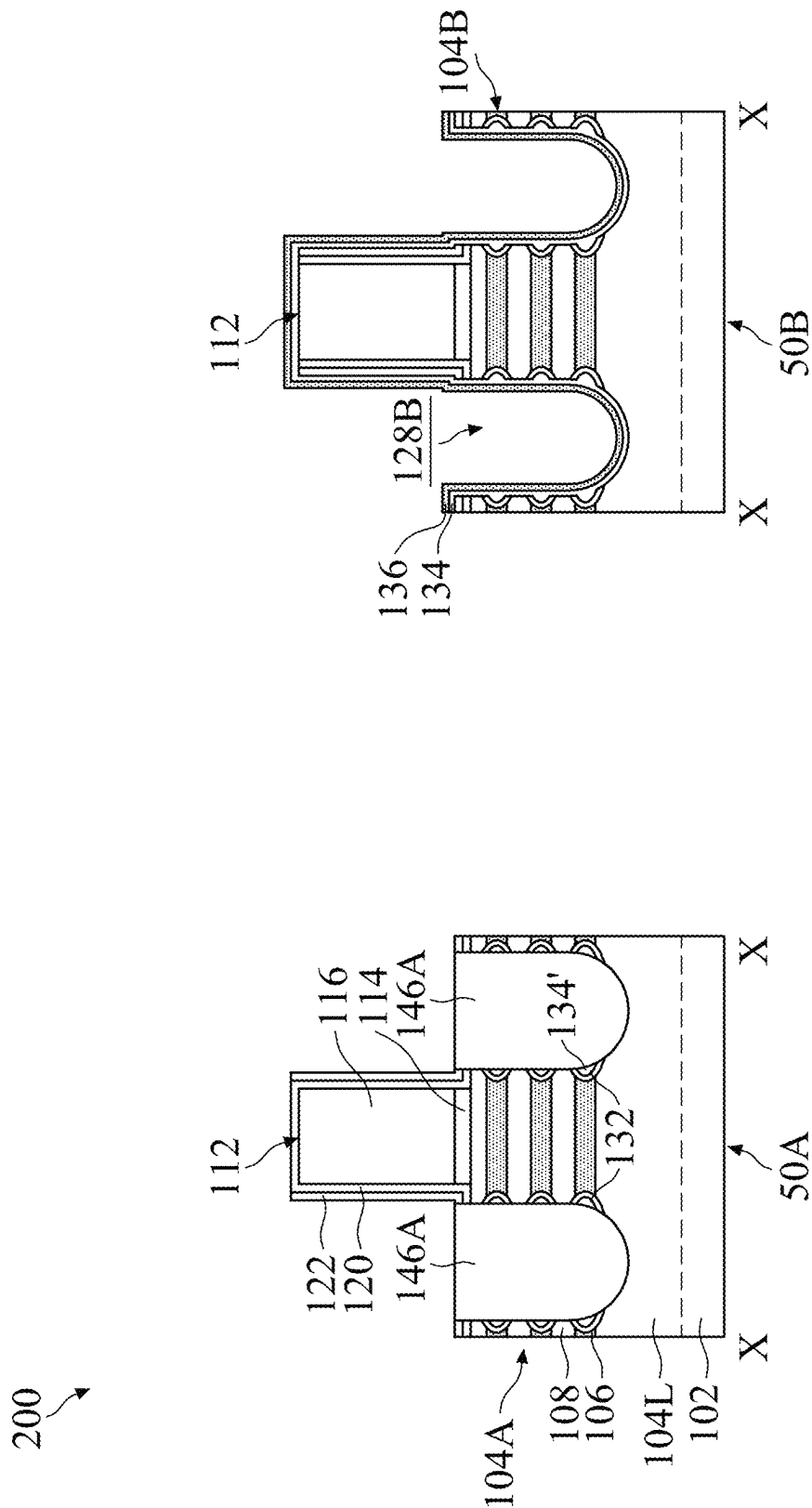


FIG. 3D

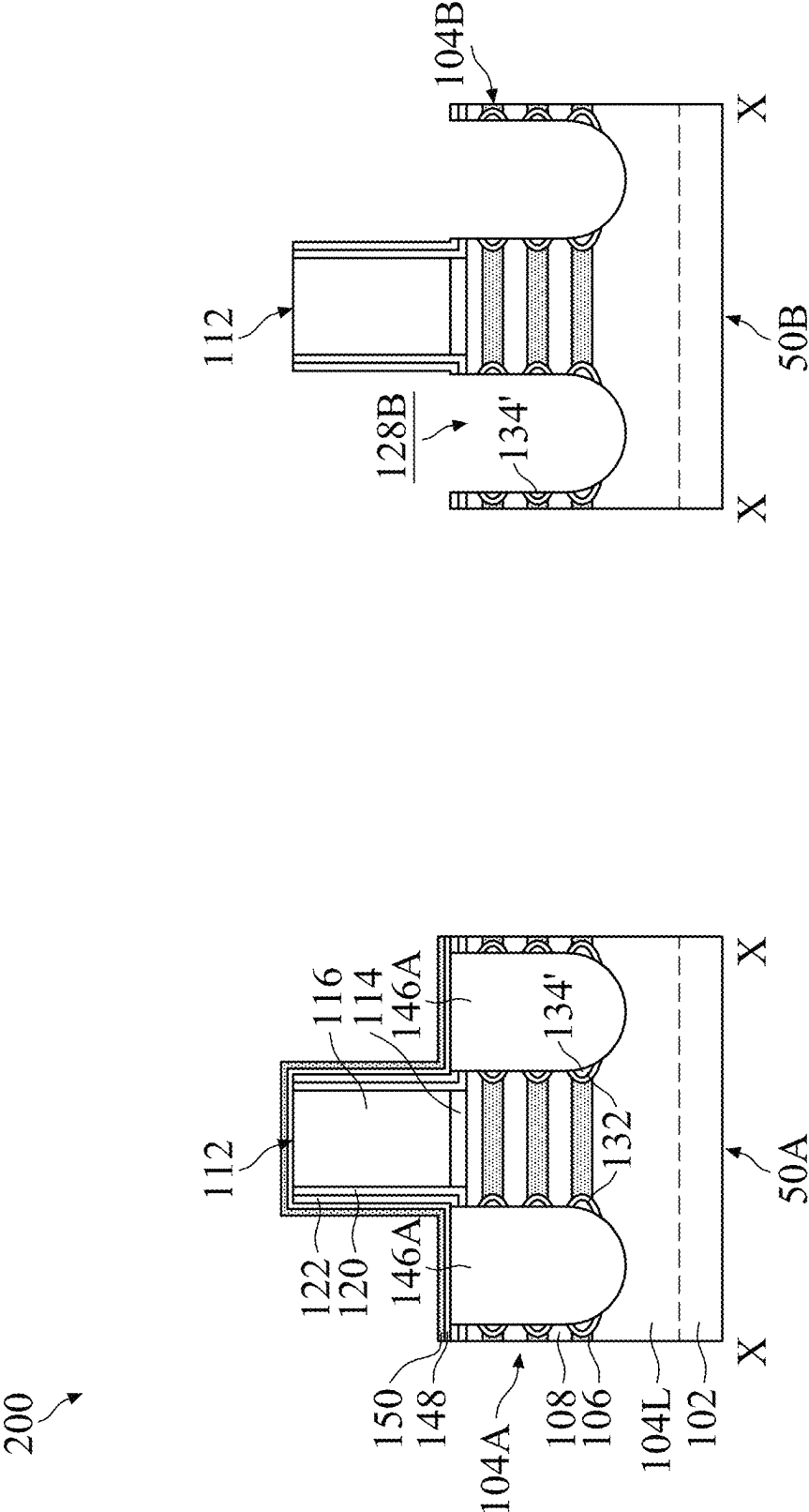


FIG. 3E

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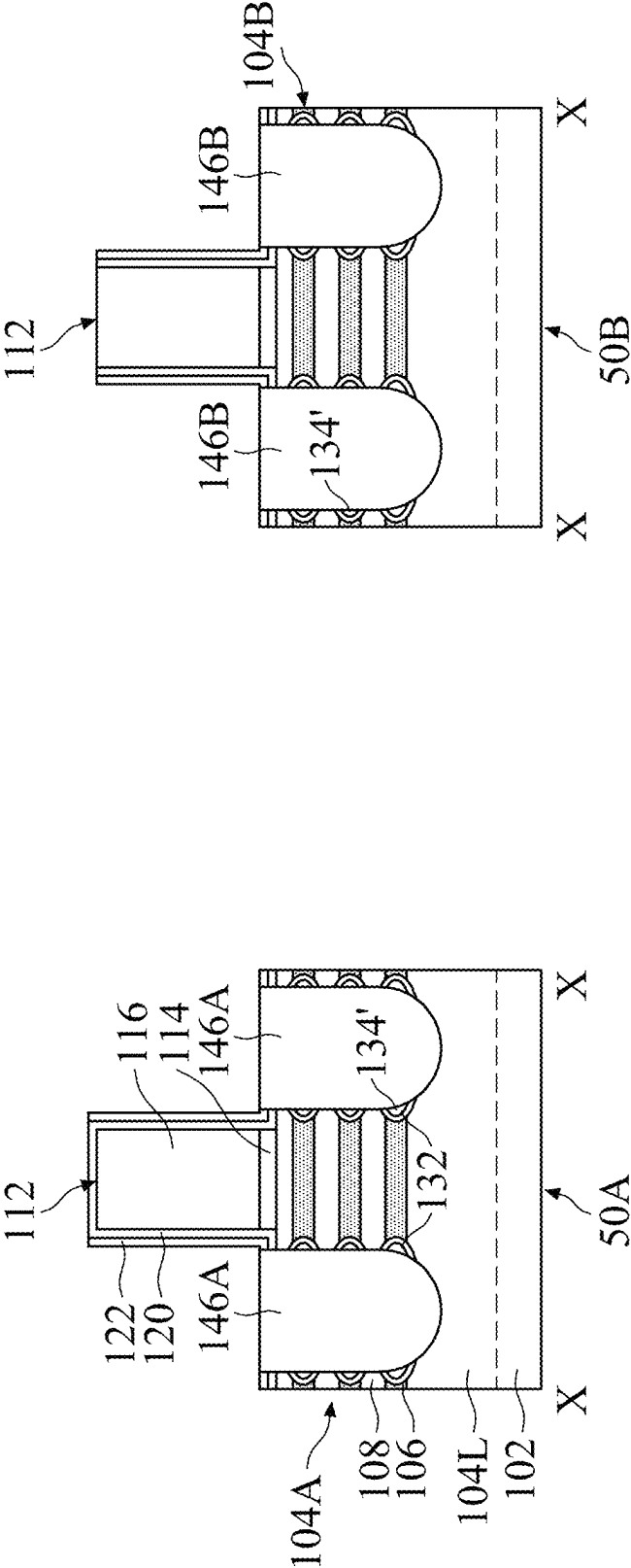


FIG. 3F

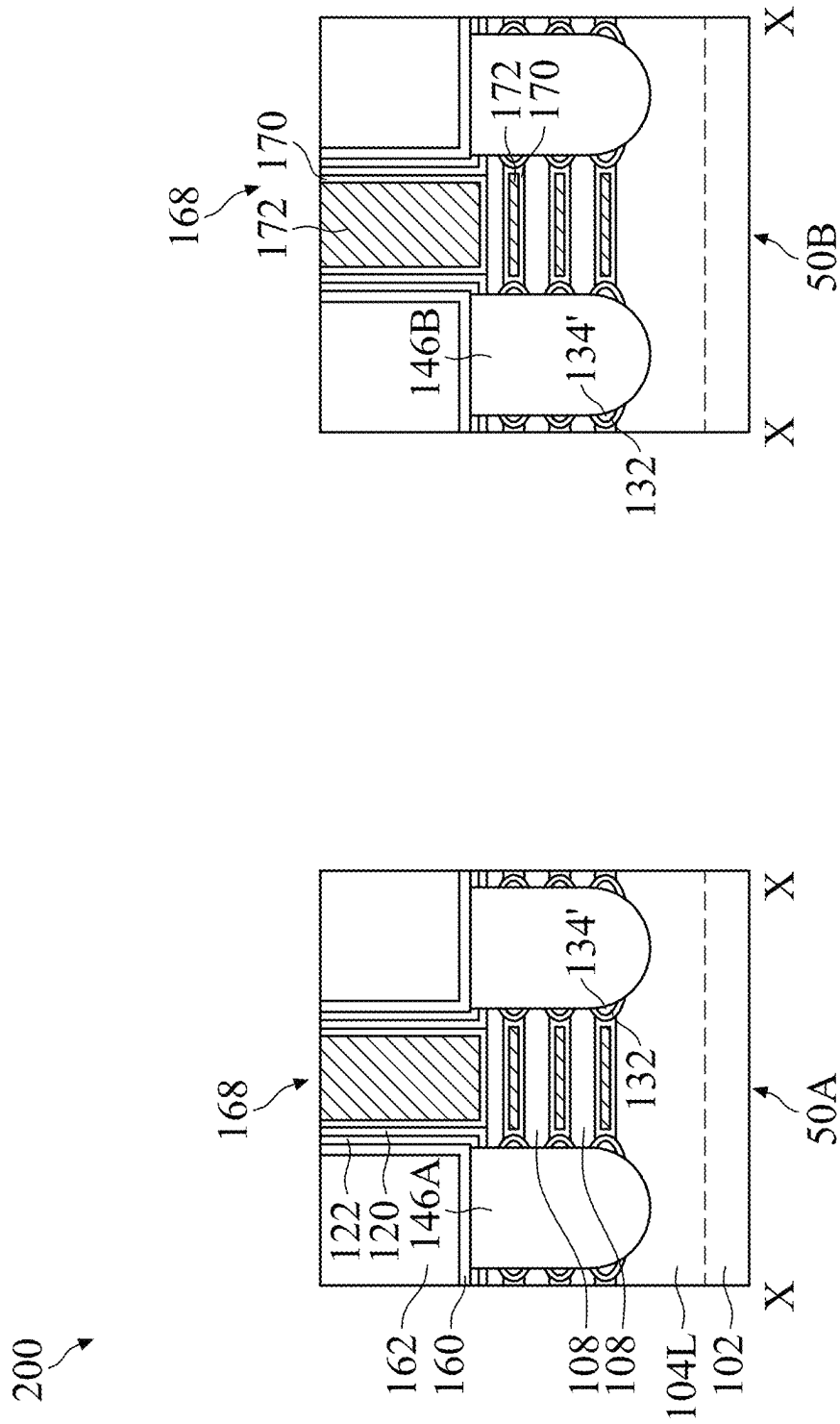


FIG. 3G

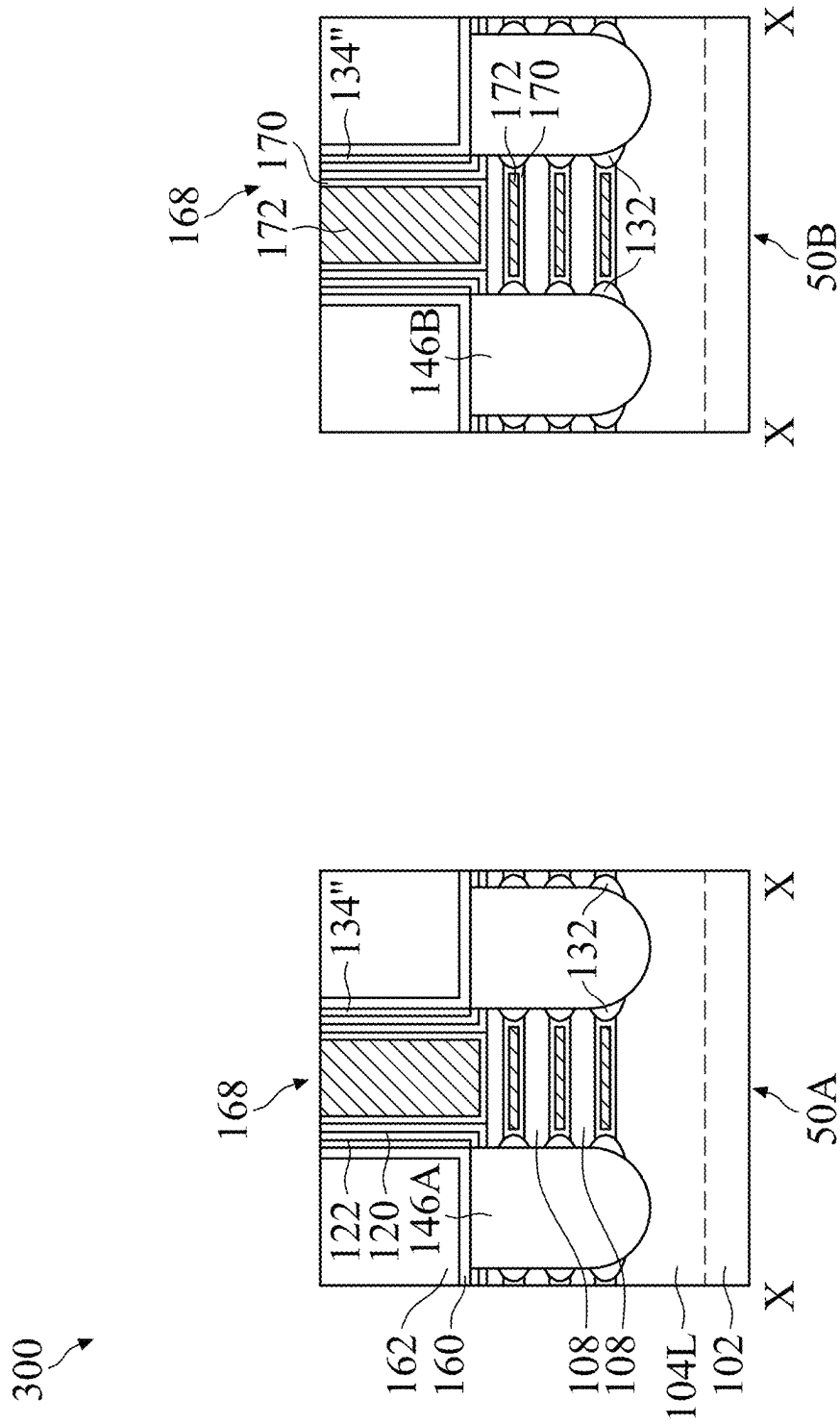


FIG. 4

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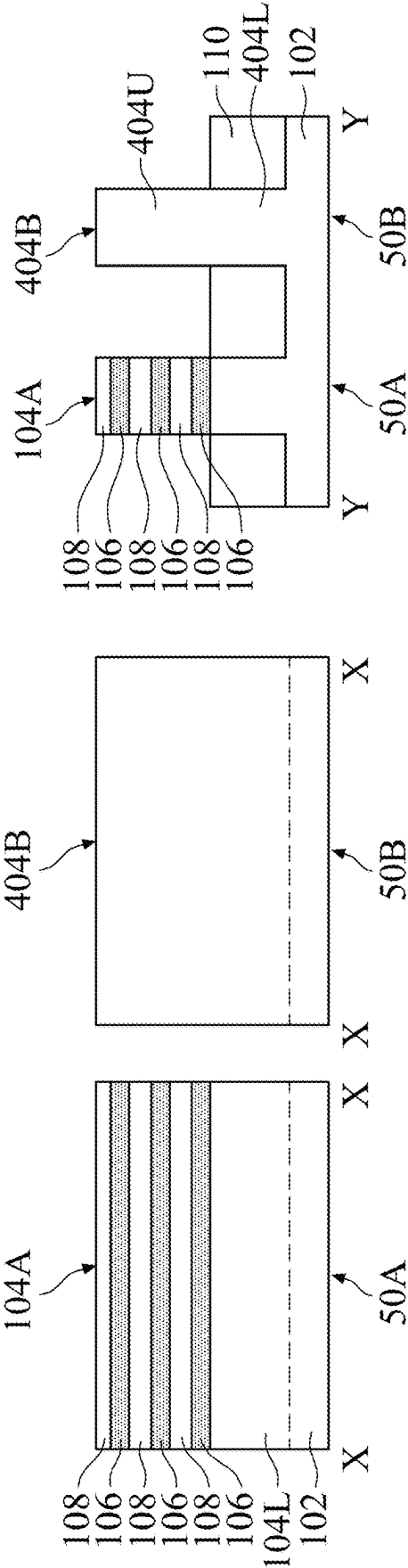


FIG. 5A-1

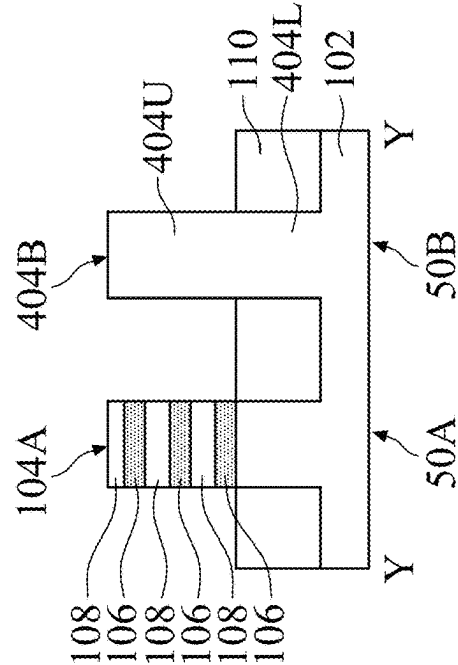


FIG. 5A-2

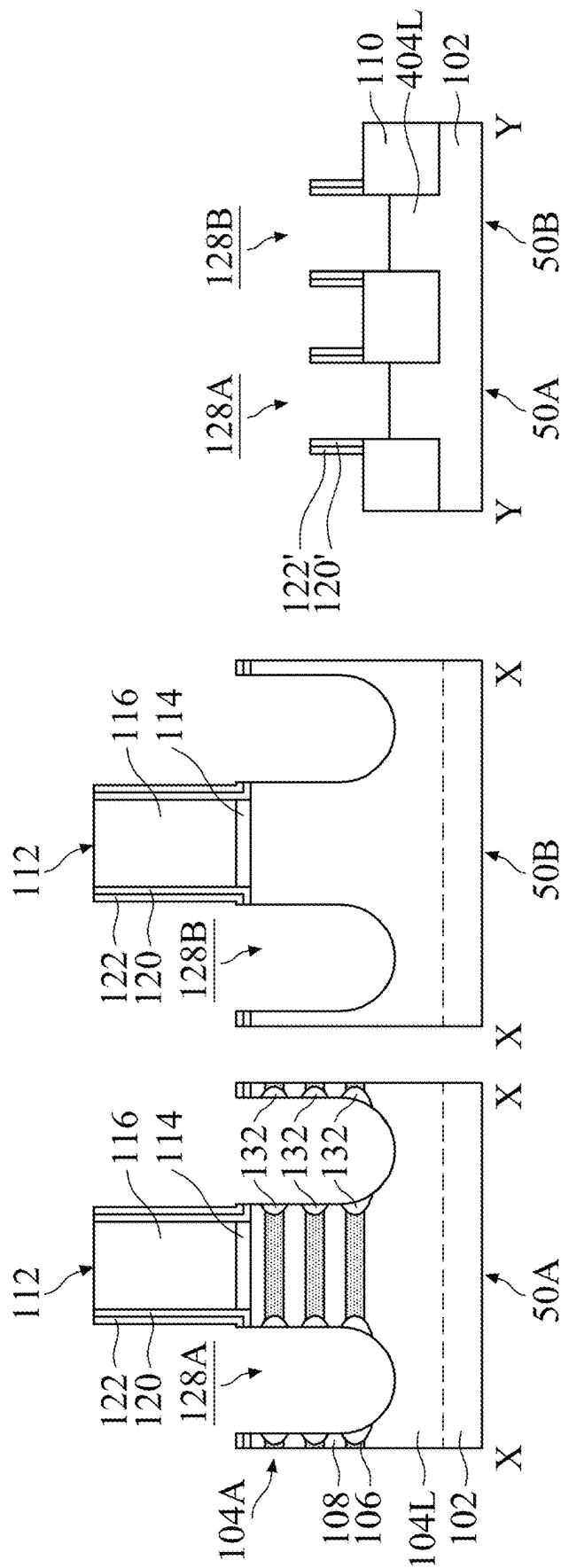


FIG. 5B-1

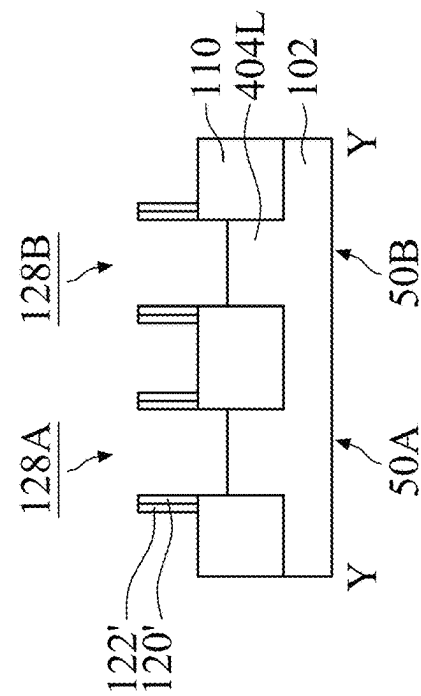
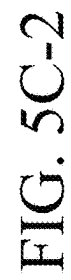
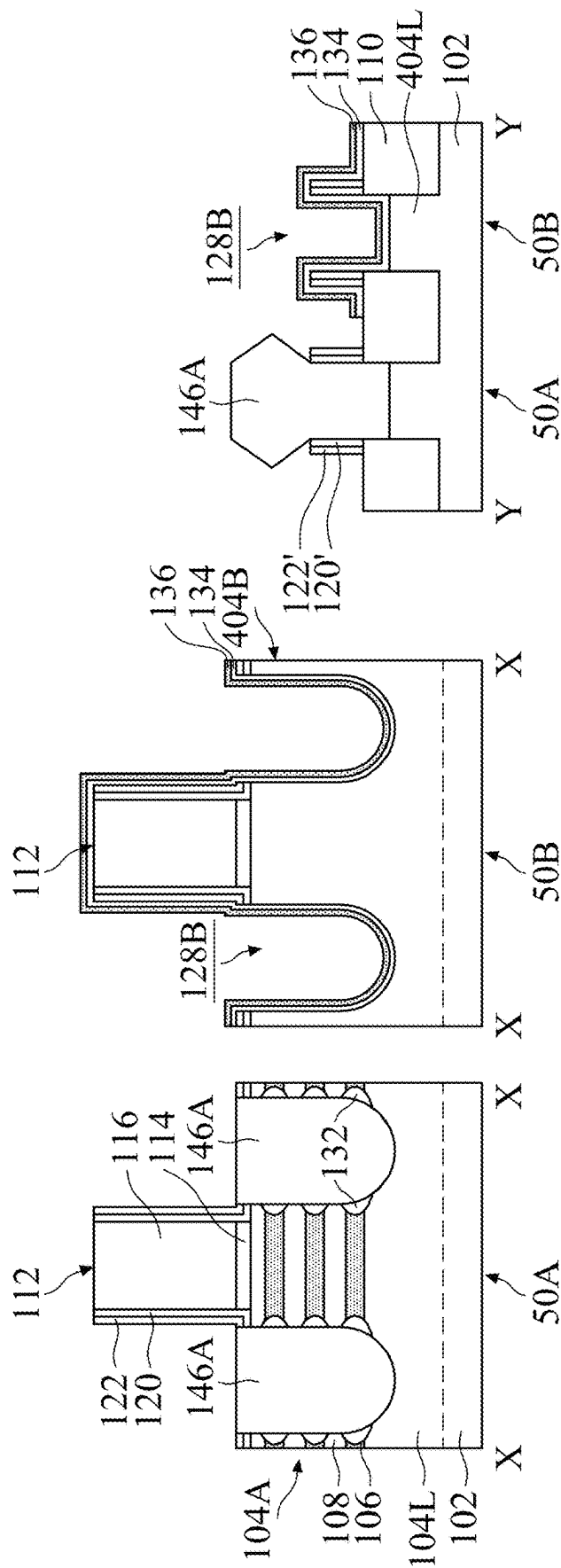


FIG. 5B-2



400

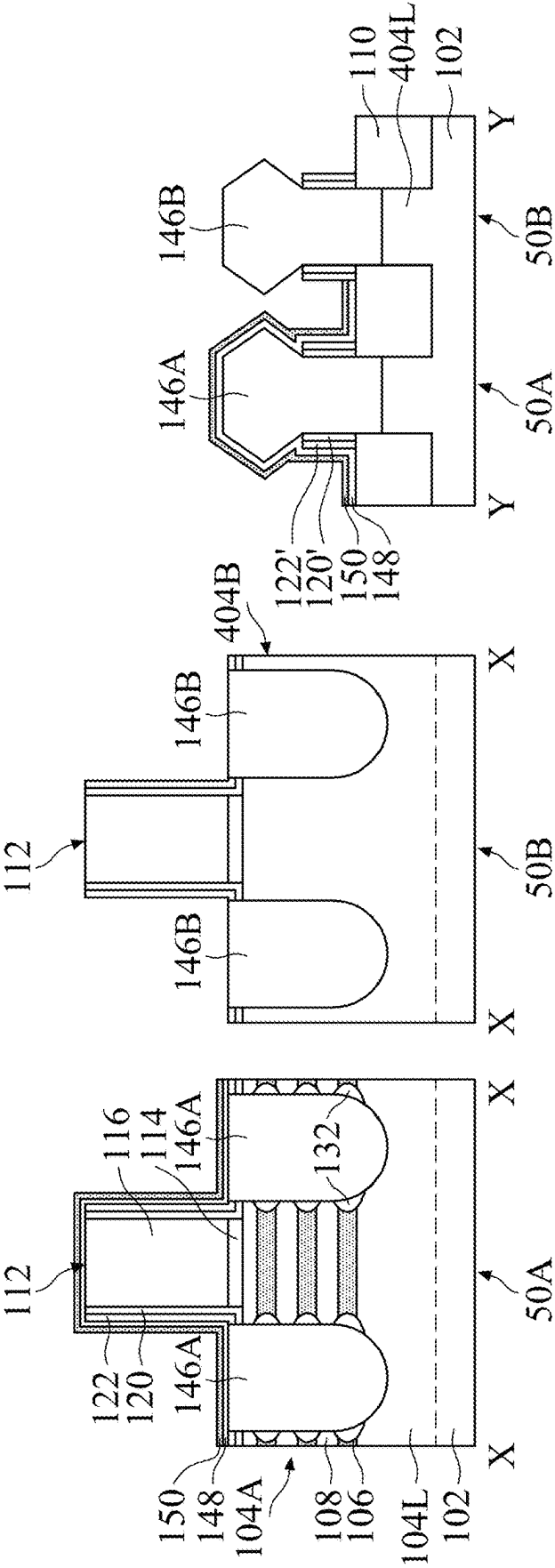


FIG. 5D-1

FIG. 5D-2

400

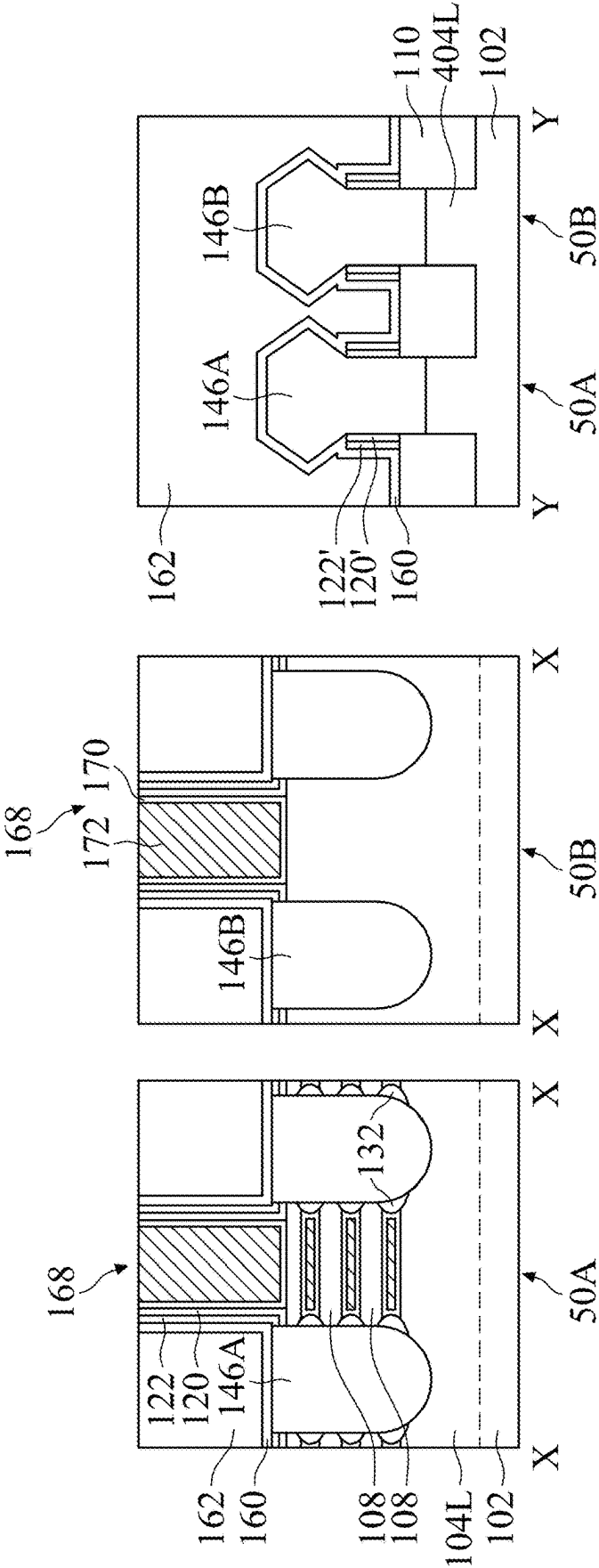


FIG. 5E-1

FIG. 5E-2

1

METHOD FOR FORMING SEMICONDUCTOR STRUCTURE

RELATED APPLICATIONS

The present application is a continuation application of U.S. application Ser. No. 17/237,249, filed on Apr. 22, 2021, entitled of "METHOD FOR FORMING SEMICONDUCTOR STRUCTURE WITH FINS USING A MULTILAYER MASK STRUCTURE FOR ETCHING TO FORM NANOSTRUCTURES," which is incorporated herein by reference in its entirety.

BACKGROUND

The electronics industry is experiencing an ever-increasing demand for smaller and faster electronic devices which are simultaneously able to support a greater number of increasingly complex and sophisticated functions. Accordingly, there is a continuing trend in the semiconductor industry to manufacture low-cost, high-performance, and low-power integrated circuits (ICs). So far, these goals have been achieved in large part by scaling down semiconductor IC dimensions (e.g., minimum feature size) and thereby improving production efficiency and lowering associated costs. However, such miniaturization has introduced greater complexity into the semiconductor manufacturing process. Thus, the realization of continued advances in semiconductor ICs and devices calls for similar advances in semiconductor manufacturing processes and technology.

Recently, multi-gate devices have been introduced in an effort to improve gate control by increasing gate-channel coupling, reduce OFF-state current, and reduce short-channel effects (SCEs). One such multi-gate device that has been introduced is the gate-all around transistor (GAA). The GAA device gets its name from the gate structure which can extend around the channel region providing access to the channel on two or four sides. GAA devices are compatible with conventional complementary metal-oxide-semiconductor (CMOS) processes and their structure allows them to be aggressively scaled-down while maintaining gate control and mitigating SCEs. In conventional processes, GAA devices provide a channel in a silicon nanowire. However, integration of fabrication of the GAA features around the nanowire can be challenging. For example, while the current methods have been satisfactory in many respects, continued improvements are still needed.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a perspective view of a semiconductor structure, in accordance with some embodiments of the disclosure.

FIGS. 2A-1 through 2U-2 are cross-sectional views illustrating the formation of a semiconductor structure at various intermediate stages, in accordance with some embodiments of the disclosure.

FIGS. 3A through 3G are cross-sectional views illustrating the formation of a semiconductor structure at various intermediate stages, in accordance with some embodiments of the disclosure.

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FIG. 4 is a cross-sectional view illustrating a semiconductor structure, in accordance with some embodiments of the disclosure.

FIGS. 5A-1 through 5E-2 are cross-sectional views illustrating the formation of a semiconductor structure at various intermediate stages, in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Some variations of the embodiments are described. Throughout the various views and illustrative embodiments, like reference numerals are used to designate like elements. It should be understood that additional operations can be provided before, during, and after the method, and some of the operations described can be replaced or eliminated for other embodiments of the method.

Further, when a number or a range of numbers is described with "about," "approximate," and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within +/-10% of the number described or other values as understood by person skilled in the art. For example, the term "about 5 nm" encompasses the dimension range from 4.5 nm to 5.5 nm.

The gate all around (GAA) transistor structures described below may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, smaller pitches than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

Embodiments of a method for forming a semiconductor structure are provided. The aspect of the present disclosure is directed to separately forming source/drain features in different device regions. The method may include forming a silicon-containing dielectric mask layer and a high-k dielectric mask layer over the silicon-containing dielectric mask layer, patterning the dielectric mask layers into a source/drain mask structure that covers a device region and exposes another device region, and then growing source/drain features in the exposed device region. The embodiments of the

present disclosure use the silicon-containing dielectric mask layer as the bottom layer of the source/drain mask structure, which may provide a good balance between the loss of the low-k spacers (such as the gate spacer layers and/or the inner spacers) and the amount of residue of the source/drain mask structure. Therefore, the process window of patterning the source/drain mask structure is enlarged.

FIG. 1 is a perspective view of a semiconductor structure, in accordance with some embodiments. A semiconductor structure **100** is received or provided, as shown in FIG. 1, in accordance with some embodiments. The semiconductor structure **100** includes a substrate **102** and a fin structure **104** over the substrate **102**, in accordance with some embodiments. Although one fin structure **104** is illustrated in FIG. 1, more than one fin structures **104** may be formed over a single device region and/or multiple device regions of the semiconductor structure **100**.

For a better understanding of the semiconductor structure, X-Y-Z coordinate reference is provided in the figures of the present disclosure. X-axis and Y-axis are generally orientated along the lateral (or horizontal) directions that are parallel to the main surface of the substrate **102**. Y-axis is transverse (e.g., substantially perpendicular) to the X-axis. Z-axis is generally oriented along the vertical direction that is perpendicular to the main surface of the substrate **102** (or the X-Y plane).

The fin structure **104** includes a lower fin element **104L** formed from a portion of the substrate **102** and an upper fin element formed from an epitaxial stack including alternating first semiconductor layers **106** and second semiconductor layer **108**, in accordance with some embodiments. The fin structure **104** extends in X direction, in accordance with some embodiments. That is, the fin structure **104** has a longitudinal axis parallel to X direction, in accordance with some embodiments. X direction may also be referred to as the channel-extending direction. The current of the resulting semiconductor device (i.e., GAAFET) flows in X direction through the channel.

The fin structure **104** includes a channel region CH and source/drain regions SD, where the channel region CH is defined between the source/drain regions SD, in accordance with some embodiments. In this disclosure, a source/drain refers to a source and/or a drain. It is noted that in the present disclosure, a source and a drain are interchangeably used and the structures thereof are substantially the same. FIG. 1 shows one channel region CH and two source/drain regions SD for illustrative purpose and is not intended to be limiting. The number of the channel region CH and the source/drain regions may be dependent on the semiconductor device design demand and/or performance consideration. Gate structure or gate stack (not shown) will be formed with a longitudinal axis parallel to Y direction and extending across and/or surrounds the channel region CH of the fin structure **104**. Y direction may also be referred to as a gate-extending direction.

FIG. 1 further illustrates a reference cross-section that is used in later figures. Cross-section X-X is in a plan parallel to the longitudinal axis (X direction) and through the fin structure **104**, in accordance with some embodiments. Cross-section Y-Y is in a plan parallel to the longitudinal axis (Y direction) and across the source/drain region SD of the fin structure **104**, in accordance with some embodiments.

FIGS. 2A-1 through 2U-2 are cross-sectional views illustrating the formation of a semiconductor structure at various intermediate stages, in which the figures ending with “-1” (such as FIG. 2A-1) are cross-sectional views corresponding to cross-section X-X shown in FIG. 1, and the figures ending

with “-2” (such as FIG. 2A-2) are cross-sectional views corresponding to cross-section Y-Y shown in FIG. 1, in accordance with some embodiments.

FIGS. 2A-1 and 2A-2 are cross-sectional views of a semiconductor structure **100** after the formation of fin structures **104A** and **104B** and an isolation structure **110**, in accordance with some embodiments.

A semiconductor structure **100** including a substrate **102** and fin structures **104A** and **104B** is received or provided, as shown in FIGS. 2A-1 and 2A-2, in accordance with some embodiments. The substrate **102** may be a portion of a semiconductor wafer, a semiconductor chip (or die), and the like. In some embodiments, the substrate **102** is a silicon substrate. In some embodiments, the substrate **102** includes an elementary semiconductor such as germanium; a compound semiconductor such as gallium nitride (GaN), silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); an alloy semiconductor such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or a combination thereof. Furthermore, the substrate **102** may optionally include an epitaxial layer (epi-layer), may be strained for performance enhancement, may include a silicon-on-insulator (SOI) structure, and/or have other suitable enhancement features.

The substrate **102** includes a first device region **50A** where a fin structure **104A** is formed and a second device region **50B** where a fin structure **104B** are formed, as shown in FIGS. 2A-1 and 2A-2, in accordance with some embodiments. In some embodiments, the first device region **50A** is a p-type device region and used for forming p-type semiconductor devices (e.g., p-channel nanostructure transistors such as p-channel gate-all-around transistors). In some embodiments, the second device region **50B** is an n-type device region and used for forming n-type semiconductor devices (e.g., n-channel nanostructure transistors such as n-channel gate-all-around transistors). Although the first device region **50A** and the second device region **50B** are illustrated as being immediately adjacent to each other, the first device region **50A** may be physically separated from the second device region **50B**. Any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be disposed between the first device region **50A** and the second device region **50B**.

The formation of the fin structures **104A** and **104B** includes forming an epitaxial stack over the substrate **102** using an epitaxial growth process, in accordance with some embodiments. The epitaxial stack includes alternating first semiconductor layers **106** and second semiconductor layers **108**, in accordance with some embodiments. The epitaxial growth process may be molecular beam epitaxy (MBE), metal organic chemical vapor deposition (MOCVD), or vapor phase epitaxy (VPE), or another suitable technique.

In some embodiments, the first semiconductor layers **106** are made of a first semiconductor material and the second semiconductor layers **108** are made of a second semiconductor material. The first semiconductor material for the first semiconductor layers **106** has a different lattice constant than the second semiconductor material for the second semiconductor layers **108**, in accordance with some embodiments. In some embodiments, the first semiconductor material and the second semiconductor material have different oxidation rates and/or etching selectivity. In some embodiments, the first semiconductor layers **106** are made of SiGe, where the percentage of germanium (Ge) in the SiGe is in a range from about 20 atomic % to about 50 atomic %, and the second semiconductor layers **108** are made of pure or

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substantially pure silicon. In some embodiments, the first semiconductor layers **106** are $\text{Si}_{1-x}\text{Ge}_x$, where x is more than about 0.3, or Ge ($x=1.0$) and the second semiconductor layers **108** are Si or $\text{Si}_{1-y}\text{Ge}_y$, where y is less than about 0.4, and $x>y$.

The first semiconductor layers **106** are configured as sacrificial layers and will be removed to form gaps to accommodate gate materials, and the second semiconductor layers **108** will form nanostructures (e.g., nanowires or nanosheets) that laterally extend between source/drain features and serve as the channel for the resulting semiconductor device (such as a nanostructure transistor), in accordance with some embodiments. As the term is used herein, “nanostructures” refers to semiconductor layers that have cylindrical shape, bar shaped and/or sheet shape. Gate stack (not shown) will be formed across and wrap around the nanostructures, in accordance with some embodiments.

In some embodiments, the thickness of each of the first semiconductor layers **106** is in a range from about 5 nm to about 20 nm. In some embodiments, the thickness of each of the second semiconductor layers **108** is in a range from about 5 nm to about 20 nm. The thickness of the second semiconductor layers **108** may be greater than, equal to, or less than the second protection layer **108**, which may depend on the amount of gate materials to be filled in spaces where the first semiconductor layers **106** are removed. Although three first semiconductor layers **106** and three second semiconductor layers **108** are shown in FIGS. 1A-1 and 1A-2, the numbers are not limited to three, and can be 1, 2, or more than 3, and is less than 20. By adjusting the number of the semiconductor layers, a driving current of the resulting GAAFET device can be adjusted.

The epitaxial stack including the first semiconductor layers **106** and the second semiconductor layers **108** are then patterned into the fin structure **104A** in the first device region **50A** and the fin structure **104B** in the second device region **50B**, in accordance with some embodiments. In some embodiments, the patterning process includes forming a patterned hard mask layer (not shown) over the epitaxial stack. An etching process is then performed to remove portions of the epitaxial stack and underlying substrate **102** uncovered by the patterned hard mask layer, thereby forming trenches and the fin structures **104A** and **104B** protruding from between the trenches, in accordance with some embodiments. The etching process may be an anisotropic etching process, e.g., dry plasma etching.

The portion of the substrate **102** protruding from between the trenches forms lower fin elements **104L** of the fin structures **104A** and **104B**, in accordance with some embodiments. A remainder of the epitaxial stack (including the first semiconductor layers **106** and the second semiconductor layers **108**) forms the upper fin elements of the fin structures **104A** and **104B** over the respective lower fin elements **104L**, in accordance with some embodiments.

In some embodiments, the fin structures **104A** and **104B** extend in X direction. That is, the fin structures **104A** and **104B** have longitudinal axes parallel to X direction, in accordance with some embodiments. FIG. 2A-2 shows one fin structure formed in each of the first device region **50A** and the second device region **50B** for illustrative purpose and is not intended to be limiting. The number of the fin structure **104A/104B** in the device region **50A/50B** may be dependent on the semiconductor device design demand and/or performance consideration.

An isolation structure **110** is formed to surround the lower fin elements **104L** of the fin structures **104A** and **104B**, as shown in FIG. 2A-2, in accordance with some embodiments.

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The isolation structure **110** is configured to electrically isolate active regions (e.g., the fin structures **104A** and **104B**) of the semiconductor structure **100** and is also referred to as shallow trench isolation (STI) feature, in accordance with some embodiments.

The formation of the isolation structure **110** includes forming an insulating material to overfill the trenches, in accordance with some embodiments. In some embodiments, the insulating material is made of silicon oxide, silicon nitride, silicon oxynitride (SiON), another suitable insulating material, multilayers thereof, and/or a combination thereof. In some embodiments, the insulating material is deposited using includes CVD (such as low-pressure CVD (LPCVD), plasma-enhanced CVD (PECVD), high density plasma CVD (HDP-CVD), high aspect ratio process (HARP), or flowable CVD (FCVD)), atomic layer deposition (ALD), another suitable technique, and/or a combination thereof. In some embodiments, the insulating material may be bi-layered, for example, a lining layer and a bulk layer over the lining layer. The lining layer may repair damage during the etching process that forms the fin structures. The bulk layer may have a good gap-fill ability to fill the trenches without forming voids or seams therein.

A planarization process is performed on the insulating material to remove a portion of the insulating material above the patterned hard mask layer (not shown) until the patterned hard mask layer (not shown) is exposed, in accordance with some embodiments. In some embodiments, the patterned hard mask layer is also removed in the planarization process, and the upper surfaces of the fin structures **104A** and **104B** are exposed. The planarization may be chemical mechanical polishing (CMP), etching back process, or a combination thereof. The insulating material is then recessed by an etching process (such as dry plasma etching and/or wet chemical etching) until the upper fin elements of the fin structures **104A** or **104B** are exposed, in accordance with some embodiments. The recessed insulating material forms the isolation structure **110**.

FIGS. 2B-1 and 2B-2 are cross-sectional views of a semiconductor structure **100** after the formation of a dummy gate structure **112**, in accordance with some embodiments.

Dummy gate structure **112** is formed over the semiconductor structure **100**, as shown in FIG. 2B-1, in accordance with some embodiments. The dummy gate structure **112** extends across and surrounds the channel regions of the fin structures **104A** and **104B** to define the channel regions and the source/drain regions, in accordance with some embodiments. The dummy gate structure **112** is configured as a sacrificial structure and will be replaced with a final gate stack, in accordance with some embodiments. In some embodiments, the dummy gate structure **112** extends in Y direction. That is, the dummy gate structure **112** has a longitudinal axis parallel to the Y direction, in accordance with some embodiments. FIG. 2B-1 shows one dummy gate structure **112** for illustrative purpose and is not intended to be limiting. The number of the dummy gate structures **112** may be dependent on the semiconductor device design demand and/or performance consideration.

The dummy gate structure **112** includes a dummy gate dielectric layer **114** and a dummy gate electrode layer **116** formed over the dummy gate dielectric layer **114**, as shown in FIG. 2B, in accordance with some embodiments. In some embodiments, the dummy gate dielectric layer **114** is made of one or more dielectric materials, such as silicon oxide (SiO₂), silicon nitride (SiN), silicon oxynitride (SiON), HfO₂, HfZrO, HfSiO, HfTiO, HfAlO, and/or a combination thereof. In some embodiments, the dielectric material is

formed using ALD, CVD, thermal oxidation, physical vapor deposition (PVD), another suitable technique, and/or a combination thereof. In some embodiments, the dummy gate electrode layer **116** is made of semiconductor material such as polysilicon, poly-silicon germanium. In some embodiments, the dummy gate electrode layer **116** is made of a conductive material such as metallic nitrides, metallic silicides, metals, and/or a combination thereof. In some embodiments, the material for the dummy gate electrode layer **116** is formed using CVD, another suitable technique, and/or a combination thereof.

In some embodiments, the formation of the dummy gate structure **112** includes globally and conformally depositing a dielectric material for the dummy gate dielectric layer **114** over the semiconductor structure **100**, depositing a material for the dummy gate electrode layer **116** over the dielectric material, planarizing the material for the dummy gate electrode layer **116**, and patterning the dielectric material and the material for the dummy gate electrode layer **116** into the dummy gate structure **112**. The patterning process includes forming a patterned hard mask layer (not shown) over the material for the dummy gate electrode layer **116** to cover the channel regions of the fin structures **104A** and **104B**, in accordance with some embodiments. The material for the dummy gate electrode layer **116** and dielectric material, uncovered by the patterned hard mask layer, is etched away until the source/drain regions of the fin structures **104A** and **104B** are exposed, in accordance with some embodiments.

FIGS. **2C-1** and **2C-2** are cross-sectional views of a semiconductor structure **100** after the formation of gate spacer layers **120** and **122**, in accordance with some embodiments.

The gate spacer layers **120** and **122** are sequentially formed over the semiconductor structure **100**, as shown in FIGS. **2C-1** and **2C-2**, in accordance with some embodiments. The gate spacer layers **120** and **122** are formed along and cover the upper surface and the sidewalls of the dummy gate structure **112** (and the patterned mask layer if exist), the upper surfaces and the sidewalls of the upper fin elements of the fin structures **104A** and **104B**, and the upper surface of the isolation structure **110**, in accordance with some embodiments.

In some embodiments, the gate spacer layers **120** and **122** are made of a silicon-containing dielectric material, such as silicon oxide (SiO_2), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), and/or oxygen-doped silicon carbonitride (Si(O)CN). In some embodiments, the gate spacer layers **120** and **122** are globally and conformally deposited using atomic layer deposition (ALD), CVD (such as LPCVD, PECVD, HDP-CVD, HARP, and FCVD), another suitable method, and/or a combination thereof.

In some embodiments, the gate spacer layers **120** and **122** are made of low-k dielectric materials. For example, the dielectric constant (k) values of the gate spacer layers **120** and **122** may be lower than the k-value of silicon oxide (SiO_2), such as lower than 4.2, equal to or lower than about 3.9, such as in a range from about 3.5 to about 3.9. In some embodiments, the gate spacer layer **120** and the gate spacer layer **122** are made of different materials and have different dielectric constant values. For example, the gate spacer layers **120** is a SiOCN layer and the gate spacer layers **122** is a Si(O)CN layer. The oxygen concentration in the SiOCN layer may be greater than the oxygen concentration in the Si(O)CN layer. In some embodiments, the gate spacer layer **120** has a thickness in a range from about 10 angstroms ($\text{\AA}$)

to about 60 $\text{\AA}$. In some embodiments, the gate spacer layer **122** has a thickness in a range from about 15 $\text{\AA}$ to about 90 $\text{\AA}$.

FIGS. **2D-1** and **2D-2** are cross-sectional views of a semiconductor structure **100** after the formation of source/drain recesses **128A** and **128B**, in accordance with some embodiments.

One or more etching process is performed to remove horizontal portions of the gate spacer layers **120** and **122** and then recess the source/drain regions of the fin structures **104A** and **104B**, thereby forming source/drain recesses **128A** in the first device region **50A** and source/drain recesses **128B** in the second device region **50B**, as shown in FIGS. **2D-1** and **2D-2**, in accordance with some embodiments. The one or more etching process may be an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching or wet chemical etching, and/or a combination thereof.

For example, in some embodiments, a first etching process is performed to remove horizontal portions of the gate spacer layers **120** and **122** along lateral surfaces (such as the upper surface of the dummy gate structure **112**, the upper surface of the upper fin elements of the fin structures **104A** and **104B**, and the upper surface of the isolation structure **110**). After the first etching process, the source/drain regions of the fin structures **104A** and **104B** are exposed, in accordance with some embodiments. Vertical portions of the gate spacer layers **120** and **122** are left remaining on the sidewalls of the dummy gate structure **112** to act as gate spacers, in accordance with some embodiments. The gate spacers are used to offset the subsequently formed source/drain features and separate the source/drain features from the gate structure, in accordance with some embodiments. Furthermore, vertical portions of the gate spacer layers **120** and **122** are also left remaining on the sidewalls of the upper fin elements of the fin structures **104A** and **104B** to act as fin spacer layers and denoted as **120'** and **122'**, as shown in FIG. **2D-2**, in accordance with some embodiments.

Afterward, a second etching process is performed using the gate spacers **120** and **122** and the dummy gate structure **112** as etch mask to recess the source/drain regions of the fin structures **104A** and **104B** such that the source/drain recess **128A/128B** are formed self-aligned on the opposite sides of the dummy gate structure **112**, in accordance with some embodiments. In some embodiments, the first and second etching processes are performed without an additional photolithography process. The source/drain recesses **128A** and **128B** pass through the upper fin element of the fin structure **104**, in accordance with some embodiments. The bottom surfaces of the source/drain recesses **128A** and **128B** may extend to a position substantially below the upper surface of the isolation structure **110**, in accordance with some embodiments.

FIGS. **2E-1** and **2E-2** are cross-sectional views of a semiconductor structure **100** after the formation of notches **130**, in accordance with some embodiments.

An etching process is performed on the semiconductor structure **100** to laterally recess, from the source/drain recesses **128A** and **128B**, the first semiconductor layers **106** of the fin structures **104A** and **104B** to form notches **130**, as shown in FIG. **2E-2**, in accordance with some embodiments. In some embodiments, in the etching process, the first semiconductor layers **106** have a greater etching rate (or etching amount) than the second semiconductor layers **108**, thereby forming notches **144** between adjacent second semiconductor layers **108** and between the lowermost second

semiconductor layer **108** and the lower fin element **104L**. In some embodiments, the etching process is an isotropic etching such as dry chemical etching, remote plasma etching, wet chemical etching, another suitable technique, and/or a combination thereof. In some embodiments, the notches **130** extend, from the source/drain regions toward the channel region, directly below the gate spacer layers **120** and **122**, in accordance with some embodiments. In some embodiments, the notches **130** has sidewalls that are curved, (i.e., concave surface of the first semiconductor layers toward the channel region). In some embodiments, the end portions of the second semiconductor layers **108** of the fin structures **104A** and **104B** are recessed by the etching process.

FIGS. **2F-1** and **2F-2** are cross-sectional views of a semiconductor structure **100** after the formation of inner spacer layers **132**, in accordance with some embodiments.

Inner spacer layers **132** are formed in the notches **130** to abut the recessed side surfaces of the first semiconductor layers **106**, as shown in FIG. **2F-1**, in accordance with some embodiments. The inner spacer layers **132** interpose subsequently formed source/drain features and gate stack to avoid the source/drain features and the gate stack from being in direct contact and are configured to reduce the parasitic capacitance between the metal gate stack and the source/drain features (i.e. C_{gs} and C_{gd}), in accordance with some embodiments.

In some embodiments, the inner spacer layers **132** are made of a silicon-containing dielectric material, such as silicon oxide (SiO_2), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), and/or oxygen-doped silicon carbonitride (Si(O)CN). In some embodiments, the inner spacer layers **132** are made of low-k dielectric materials. For example, the dielectric constant (k) values of the inner spacer layers **132** may be lower than a k -value of silicon oxide (SiO), such as lower than 4.2, equal to or lower than about 3.9, such as in a range from about 3.5 to about 3.9.

In some embodiments, the inner spacer layers **132** are formed by globally and conformally depositing a dielectric material for the inner spacer layers **132** over the semiconductor structure **100** to fill the notches **130**, and then etching back the dielectric material to remove the dielectric material outside the notches **130**. The dielectric material remaining in the notches **130** forms the inner spacer layers **132**, in accordance with some embodiments. In some embodiments, the deposition process includes ALD, CVD (such as PECVD, LPCVD or HARP), another suitable technique, and/or a combination thereof. In some embodiments, the etching-back process includes an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching or wet chemical etching, and/or a combination thereof. In some embodiments, seams (not shown) may be formed within the inner spacer layers **132** due to the conformal deposition.

FIGS. **2G-1** through **2L-2** illustrate the formation of a source/drain mask structure for the second device region **50B**. FIGS. **2G-1** and **2G-2** are cross-sectional views of a semiconductor structure **100** after the formation of dielectric mask layers **134** and **136**, in accordance with some embodiments.

A first dielectric mask layer **134** is formed over the semiconductor structure **100**, as shown in FIGS. **2G-1** and **2G-2**, in accordance with some embodiments. The first dielectric mask layer **134** is formed along and covers the bottom surfaces and the sidewalls of the source/drain

recesses **128A** and **128B**, the sidewalls and the upper surfaces of the gate spacer layer **122** and **120**, and the upper surface of the dummy gate electrode layer **116**, in accordance with some embodiments. The first dielectric mask layer **134** is also formed along and covers the upper surfaces and the sidewalls of the fin spacer layers **120'** and **122'** and the upper surface of the isolation structure **110**, in accordance with some embodiments.

In some embodiments, the first dielectric mask layer **134** is made of a silicon-containing dielectric material, such as silicon oxide (SiO_2), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), another suitable silicon-containing dielectric material, and/or a combination thereof. In some embodiments, the first dielectric mask layer **134** is globally and conformally deposited using ALD, CVD (such as LPCVD, PECVD, HDP-CVD, HARP, and FCVD), another suitable method, and/or a combination thereof.

In some embodiments, the first dielectric mask layer **134** is made of a different material than the gate spacer layers **120** and **122** and the inner spacers **132**. The first dielectric mask layer **134** has a different etching selectivity than the gate spacer layers **120** and **122** and the inner spacers **132**. In some embodiments, the first dielectric mask layer **134** is made of a low- k dielectric material, e.g., having k -value lower than about 7. In an embodiment, the first dielectric mask layer **134** is made of silicon oxide (SiO_2). The first dielectric mask layer **134** has a higher dielectric constant value than the gate spacer layers **120** and **122** and the inner spacers **132**, in accordance with some embodiments. In some embodiments, the first dielectric mask layer **134** has a thickness T_1 in a range from about 10 Å to about 50 Å.

A second dielectric mask layer **136** is formed over the first dielectric mask layer **134**, as shown in FIGS. **2G-1** and **2G-2**, in accordance with some embodiments. The first and second dielectric mask layers **134** and **136** partially fill the source/drain recesses **128A** and **128B**, in accordance with some embodiments.

In some embodiments, the second dielectric mask layer **136** is made of a silicon-free dielectric material. In some embodiments, the second dielectric mask layer **136** is made of oxide of amphoteric metal, such as zinc oxide (ZnO), aluminum oxide (Al_2O_3), tin oxide (SnO), lead oxide (PbO), beryllium oxide (BeO), chromium oxide (CrO , Cr_2O_3 , Cr_2O_3 or Cr_3O_4), and/or a combination thereof. In some embodiments, the second dielectric mask layer **136** is globally and conformally deposited using ALD, CVD (such as LPCVD, PECVD, HDP-CVD, HARP, and FCVD), physical vapor deposition (PVD), another suitable method, or a combination thereof. In some embodiments, the second dielectric mask layer **136** is also formed on the backside and the sidewall (or bevel) of the wafer. After the deposition process, a clean process may be performed to remove the second dielectric mask layer **136** formed on the backside and the sidewall (or bevel) of the wafer. The second dielectric mask layer **136** formed on the frontside of the wafer is not cleaned, so there is no material loss.

In some embodiments, the second dielectric mask layer **136** is made of a different material than the first dielectric mask layer **134**. The second dielectric mask layer **136** has a different etching selectivity than the first dielectric mask layer **134**, in accordance with some embodiments. In some embodiments, the second dielectric mask layer **136** is made of a high- k dielectric material, e.g., having k -value greater than about 7. In an embodiment, the second dielectric mask layer **136** is made of aluminum oxide (Al_2O_3). In some embodiments, the second dielectric mask layer **136** has a higher dielectric constant value than the first dielectric mask

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layer 134, the gate spacer layers 120 and 122 and the inner spacers 132. In some embodiments, the second dielectric mask layer 136 has a thickness T2 in a range from about 10 Å to about 50 Å. In some embodiments, the thickness T2 of the second dielectric mask layer 136 is greater than, substantially the same as, or less than the thickness T1 of the first dielectric mask layer 134.

FIGS. 2H-1 and 2H-2 are cross-sectional views of a semiconductor structure 100 after the formation of a tri-layer mask structure, in accordance with some embodiments.

A fill layer 138 is formed over the second dielectric mask layer 136 and overfills the remainders of the source/drain recesses 128A and 128B, as shown in FIGS. 2H-1 and 2H-2, in accordance with some embodiments. The fill layer 138 is formed above the upper surface of the dummy gate structure 112, in accordance with some embodiments. The fill layer 138 may be used as a planarization layer to provide a substantially planar surface for a following photolithography process.

In some embodiments, the fill layer 138 is a bottom anti-reflective coating (BARC) layer such as an inorganic material or an organic material (e.g., polymer, oligomer, or monomer). In some embodiments, the fill layer 138 is made of organic material including carbon and oxygen, which is made of cross-linked photo-sensitive material. In some embodiments, the fill layer 138 is formed by spin-on coating process, a CVD process (such as LPCVD, PECVD, HDP-CVD, HARP or FCVD), another suitable method, or a combination thereof. The fill layer 138 has a different etching selectivity than the second dielectric mask layer 136, in accordance with some embodiments. In some embodiments, the fill layer 138 has a thickness T3 in a range from about 700 Å to about 1100 Å, measured from the bottom of the source/drain recesses 128A/128B.

A tri-layer mask structure is formed over the fill layer 138, in accordance with some embodiments. The tri-layer mask structure includes a third dielectric mask layer 140 as a bottom layer, a BARC layer 142 as a middle layer, and a top photoresist mask 144, as shown in FIGS. 2H-1 and 2H-2, in accordance with some embodiments. The third dielectric mask layer 140 is formed over the fill layer 138, in accordance with some embodiments. In some embodiments, the third dielectric mask layer 140 is made of a silicon-free dielectric material. In some embodiments, the third dielectric mask layer 140 is made of oxide of amphoteric metal, such as zinc oxide (ZnO), aluminum oxide (Al₂O₃), tin oxide (SnO), lead oxide (PbO), beryllium oxide (BeO), chromium oxide (CrO, Cr₂O₃, Cr₂O₃ or Cr₃O₄), and/or a combination thereof. In some embodiments, the third dielectric mask layer 140 is deposited using ALD, CVD (such as LPCVD, PECVD, HDP-CVD, HARP, and FCVD), PVD, another suitable method, or a combination thereof.

In some embodiments, the third dielectric mask layer 140 is made of a high-k dielectric material, e.g., having k-value greater than about 7. In some embodiments, the third dielectric mask layer 140 is made of the same material as the second dielectric mask layer 136. In an embodiment, the third dielectric mask layer 140 is made of aluminum oxide (Al₂O₃). The third dielectric mask layer 140 has a different etching selectivity than the underlying fill layer 138 and overlying BARC layer 142, in accordance with some embodiments. In some embodiments, the third dielectric mask layer 140 has a thickness T4 in a range from about 30 Å to about 70 Å. In some embodiments, the thickness T4 of the third dielectric mask layer 140 is greater than the thickness T1 of the first dielectric mask layer 134 and the thickness T2 of the second dielectric mask layer 136.

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The BARC layer 142 is formed over the fill layer 138, in accordance with some embodiments. The BARC layer 142 may be similar to or the same as the BARC material described above. In some embodiments, the BARC layer 142 has a thickness T5 in a range from about 150 Å to about 250 Å. In some embodiments, the thickness T5 of the BARC layer 142 is greater than the thickness T3 of the third dielectric mask layer 140.

The top photoresist mask 144 is formed over the BARC layer 142, in accordance with some embodiments. In some embodiments, the top photoresist mask 144 covers the second device region 50B of the substrate 102 and uncovered the first device region 50A, in accordance with some embodiments. In some embodiments, the top photoresist mask 144 has a thickness T6 in a range from about 700 Å to about 900 Å. In some embodiments, the thickness T6 of the top photoresist mask 144 is greater than the thickness T5 of the BARC layer 142.

The top photoresist mask 144 is formed by a photolithography process, in accordance with some embodiments. The photolithography process can include forming a photoresist layer over the BARC layer 142 (for example, by spin coating), performing a pre-exposure baking process, performing an exposure process using a mask (or a reticle), performing a post-exposure baking process, and performing a developing process. During the exposure process, the photoresist layer is exposed to radiation energy (such as ultraviolet (UV) light, deep UV (DUV) light, or extreme UV (EUV) light), where the mask blocks, transmits, and/or reflects radiation to the photoresist layer depending on a mask pattern of the mask and/or mask type, such that an image is projected onto the photoresist layer that corresponds with the mask pattern. Since the photoresist layer is sensitive to radiation energy, exposed portions of the resist layer chemically change, and exposed (or non-exposed) portions of the resist layer are dissolved during the developing process depending on characteristics of the resist layer and characteristics of a developing solution used in the developing process. After development, the patterned photoresist layer forms the top photoresist mask 144 which includes a resist pattern that corresponds with the mask.

FIGS. 2I-1 and 2I-2 are cross-sectional views of a semiconductor structure 100 after one or more etching process, in accordance with some embodiments.

One or more etching process is performed using the top photoresist mask 144 to remove portions of the BARC layer 142, the third dielectric mask layer 140 and the fill layer 138 in the first device region 50A of the substrate 102, thereby exposing the second dielectric mask layer 136 in the first regions 50A, as shown in FIGS. 2I-1 and 2I-2, in accordance with some embodiments. After the one or more etching process, a portion of the third dielectric mask layer 140 in the second device region 50B is exposed, in accordance with some embodiments.

For example, a first etching process may be performed to remove a portion of the BARC layer 142 in the first device region 50A until the third dielectric mask layer 140, which may be used as an etching stop layer in the first etching process, is exposed. In some embodiments, the first etching process may be an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching or wet chemical etching, an ashing process, and/or a combination thereof. After the first etching process, the top photoresist mask 144 in the second device region 50B may be entirely consumed, or alternatively removed by an additional ashing process, thereby exposing a portion of the BARC layer 142 in the

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second device region **50B**, which may be used as an etching mask for a subsequent etching process.

A second etching process may be then performed to remove a portion of the third dielectric mask layer **140** in the first device region **50A** until the fill layer **138**, which may be used as an etching stop layer in the second etching process, is exposed. In some embodiments, the second etching process uses a wet etching process using an acidic solution with a pH value of about 3 to about 6.8, or an alkaline solution with a pH value of about 7.2 to about 11. In some embodiments, the second etching process uses Caros solution, including sulfuric acid (H_2SO_4) and peroxide (H_2O_2). After the second etching process, the BARC layer **142** in the second device region **50B** may be removed by an additional etching process or ashing process, thereby exposing a portion of the third dielectric mask layer **140** in the second device region **50B**, which may be used as an etching mask for a subsequent etching process.

Afterward, a third etching process may be performed to remove a portion of the fill layer **138** in the first device region **50A** until the second dielectric mask layer **136**, which may be used as an etching stop layer in the third etching process, is exposed. In some embodiments, the third etching process may be an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching or wet chemical etching, an ashing process, and/or a combination thereof. After the third etching process, a portion of the third dielectric mask layer **140** in the second device region **50B** may remain substantially unetched and is used as an etching mask in a subsequent etching process.

FIGS. 2J-1 and 2J-2 are cross-sectional views of a semiconductor structure **100** the removal of a portion of the second dielectric mask layer **136** in the first device region **50A**, in accordance with some embodiments.

An etching process is performed to remove the portion of the second dielectric mask layer **136** in the first device region **50A**, thereby exposing a portion of the first dielectric mask layer **134** in the first device region **50A**, as shown in FIGS. 2J-1 and 2J-2, in accordance with some embodiments. During the etching process, the third dielectric mask layer **140** in the second device region **50B** is also removed, thereby exposing the fill layer **138** in the second device region **50B**, in accordance with some embodiment. The source/drain mask structure covers the second device region **50B** and partially fills the source/drain recess **128B**, in accordance with some embodiments.

In some embodiments, the etching process uses a wet etching process using an acidic solution with a pH value of about 3 to about 6.8, e.g., a dilute hydrogen chloride solution (HCl), a dilute hydrogen fluoride (HF) solution, another suitable solution, and/or a combination thereof. In an embodiment, the dilute hydrogen fluoride solution with the concentration of the HF in a range from about 0.1 vol. % to about 0.3 vol. % is used. In some embodiments, the etching process uses a wet etching process using an alkaline solution with a pH value of about 7.2 to about 11, e.g., a dilute ammonium hydroxide (NH_4OH) solution.

FIGS. 2K-1 and 2K-2 are cross-sectional views of a semiconductor structure **100** the removal of a portion of the first dielectric mask layer **134** in the first device region **50A**, in accordance with some embodiments.

An etching process is performed to remove the portion of the first dielectric mask layer **134** in the first device region **50A**, thereby exposing the source/drain region of the fin structure **104A**, the inner spacers **132**, the gate spacer layers **122** and **120**, the dummy gate electrode layer **116**, as shown

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in FIGS. 2K-1 and 2K-2, in accordance with some embodiments. In some embodiments, the isolation structure **110** and the fin spacer layers **120'** and **122'** in the first device region **50A** are also exposed. Portions of the first dielectric mask layer **134** and the second dielectric mask layer **136** in the second device region **50B** are collectively used as a source/drain mask structure, in accordance with some embodiments.

In some embodiments, the etching process uses a wet etching process using an acidic solution with a pH value of about 3 to about 6.8, e.g., a dilute hydrogen chloride solution (HCl) or a dilute hydrogen fluoride (HF) solution. In an embodiment, the dilute hydrogen fluoride solution with the concentration of the HF in a range from about 0.1 vol. % to about 0.3 vol. % is used. In some embodiments, the etching process uses a wet etching process using an alkaline solution with a pH value of about 7.2 to about 11, e.g., a dilute ammonium hydroxide (NH_4OH) solution. In some embodiments, the etching process used for removing the second dielectric mask layer **136** and the etching process used for removing the first dielectric mask layer **134** is a single, continuous etching process such that the second dielectric mask layer **136** and the first dielectric mask layer **134** are consecutively removed. In some embodiments, the etching process used for removing the second dielectric mask layer **136** and the etching process used for removing the first dielectric mask layer **134** are separate etching processes.

In some embodiments, the etching process used for removing the first dielectric mask layer **134** includes a main etching step in which most portion of the first dielectric mask layer **134** is removed from the first device region **50A**. Once some layers (such as the fin structure **104A**, the inner spacers **132**, the gate spacer layers **122** and **120**, the dummy gate electrode layer **116**, the isolation structure **110**, and/or the fin spacer layers **120'**) begin to be exposed from the first dielectric mask layer **134**, the main etching step is completed. The time period of the main etching step may be determined by time mode or endpoint mode.

In some embodiments, the etching process used for removing the first dielectric mask layer **134** also includes an over-etching step after the main etching step, in which the remainder of the first dielectric mask layer **134** is partially or entirely removed from the first device region **50A**. The time period of the over-etching step may be 5% to about 30% of the time period of the main etching step. In order to avoid excessive loss of low-k materials of the gate spacer layers **122** and **120** and/or the inner spacers **132** during the etching process used for removing the first dielectric mask layer **134**, the time period of the over-etching step may be determined by the difference in the etching selectivity between the first dielectric mask layer **134** and the gate spacer layers **122** and **120** (or the inner spacers **132**). The loss of low-k materials of the gate spacer layers **122** and **120** and/or the inner spacers **132** may degrade the performance of the resulting semiconductor device (e.g., speed). In some embodiments, the etching process used for removing the first dielectric mask layer **134** does not include an over-etching step, the remainder of the first dielectric mask layer **134** is left on the layers described above.

For example, when the difference in the etching selectivity between the first dielectric mask layer **134** and the gate spacer layers **122** and **120** (or the inner spacers **132**) is relatively large, the over-etching step may be performed until the remainder of the first dielectric mask layer **134** is entirely removed. When the difference in the etching selectivity between the first dielectric mask layer **134** and the gate spacer layers **122** and **120** (or the inner spacers **132**) is

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relatively small, a small amount of residue of the first dielectric mask layer **134** may be left on the source/drain region of the fin structure **104A**, the inner spacers **132**, and/or the gate spacer layers **122** and **120** after the over-etching step.

When a small amount of residue of the first dielectric mask layer **134** is left on the source/drain region of the fin structure **104A**, the epitaxial growth of a subsequently formed source/drain feature may not be negatively affected, because the first dielectric mask layer **134** is made of a silicon containing dielectric material (such as SiO). Furthermore, the residue of the first dielectric mask layer **134** left on the gate spacer layers **122** and **120** and/or the inner spacers **132** may not significantly increase the overall capacitance between a subsequently formed source/drain feature and the final gate stack, because the first dielectric mask layer **134** has a relatively low dielectric constant value.

In some cases in which a first dielectric mask layer **134** is not formed, when the residue of the second dielectric mask layer **136**, which is made of oxide of amphoteric metal (such as Al₂O₃) and having relatively high dielectric constant value, is left on the source/drain region, the epitaxial growth of a subsequently formed source/drain feature may be negatively affected. Furthermore, even if the amount of residue of the second dielectric mask layer **136** left on the gate spacer layers **122** and **120** and/or inner spacers **132** is small, the overall capacitance between subsequently formed source/drain feature and final gate stack may significantly increase, thereby degrading the performance of the resulting semiconductor device (such as speed). However, when an over-etching step is performed to entirely remove the residue of the second dielectric mask layer **136**, the loss of low-k materials of the gate spacer layers **122** and **120** and/or the inner spacers **132** may be excessive, which may also degrade the performance of the resulting semiconductor device (such as speed).

The embodiments of the present disclosure use the low-k first dielectric mask layer **134** as the bottom layer of the source/drain mask structure, and therefore the high-k second dielectric mask layer **136** can be substantially entirely removed. Furthermore, the low-k first dielectric mask **134** may provide a good balance between the loss of the low-k spacers and the amount of residue of the source/drain mask structure, thereby enlarging the process window of patterning the source/drain mask structure.

FIGS. 2L-1 and 2L-2 are cross-sectional views of a semiconductor structure **100** the removal of a portion of the fill layer **138** in the second device region **50B**, in accordance with some embodiments.

The fill layer **138** is removed from the second device region **50B** of the substrate **102** until the second dielectric mask layer **136** is exposed, as shown in FIGS. 2L-1 and 2L-2, in accordance with some embodiments. In some embodiments, the removal process may be an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching or wet chemical etching, an ashing process, and/or a combination thereof.

FIGS. 2M-1 and 2M-2 are cross-sectional views of a semiconductor structure **100** after the formation of source/drain features **146A**, in accordance with some embodiments.

Source/drain features **146A** are formed in the source/drain recesses **128A** at first device region **50A** of the semiconductor structure **100** using an epitaxial growth process, as shown in FIGS. 2M-1 and 2M-2, in accordance with some embodiments. The epitaxial growth process may be MBE, MOCVD, or VPE, another suitable technique, or a combi-

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nation thereof. A preclean process may be performed before the epitaxial growth process. For example, a wet clean using dilute ammonium hydroxide solution and/or SiCoNi clean process may be utilized. The source/drain features **146A** are grown on the source/drain region of the fin structures **104A**, in accordance with some embodiments. The source/drain features **146A** abut the second semiconductor layers **108** of the fin structures **104A** and the inner spacer layers **132**, in accordance with some embodiments.

In some embodiments wherein the fin structure **104A** is used to formed as a p-channel nanostructure transistor, the source/drain features **146A** are made of semiconductor material such as SiGe, Si, GaAs, another suitable semiconductor material, or a combination thereof. In some embodiments, the source/drain features **146A** are doped with a p-type dopant during the epitaxial growth process. For example, the p-type dopant may be boron (B) or BF₂. For example, the source/drain features **146A** may be the epitaxially grown SiGe doped with boron (B) to form silicon germanium:boron (SiGe:B) source/drain feature.

The second dielectric mask layer **136** of the source/drain mask structure may sufficiently prevent the epitaxial semiconductor material from being grown on the dielectric surface of the second dielectric mask layer **136** during the epitaxial growth process of source/drain features **146A**, in accordance with some embodiments. As such, the source/drain mask structure can be easily removed in a following process.

The growth of the source/drain features **146A** is initially confined by the fin spacer layers **120'** and **122'** such that the source/drain features **146A** has a narrow bottom portion, in accordance with some embodiments. Once source/drain features **146A** grow to protrude from the fin spacer layers **120'** and **122'**, the source/drain features **146A** may grow to have facet surfaces that have specific crystalline orientations such that the source/drain features **146A** has a wider upper portion. Although the source/drain features **146A** are illustrated as having the facet surfaces, the source/drain features **146A** may have a curved surface in some other embodiments.

FIGS. 2N-1 and 2N-2 are cross-sectional views of a semiconductor structure **100** the removal of portions of the second dielectric mask layer **136** and the first dielectric mask layer **134** in the second device region **50B**, in accordance with some embodiments.

The second dielectric mask layer **136** and the first dielectric mask layer **134** are removed from the second device region **50B** of the substrate **102**, thereby exposing the source/drain region of the fin structure **104B**, the inner spacers **132**, the gate spacer layers **122** and **120**, and the dummy gate electrode layer **116**, as shown in FIGS. 2N-1 and 2N-2, in accordance with some embodiments. In some embodiments, the isolation structure **110** and the fin spacer layers **120'** and **122'** in the second device region **50B** are also exposed.

In some embodiments, the removal process of the second dielectric mask layer **136** and the first dielectric mask layer **134** may be an etching process using Caros solution, including sulfuric acid (H₂SO₄) and peroxide (H₂O₂). In some embodiments, the removal process of the second dielectric mask layer **136** may be an etching process which may be similar to or the same as that described above with respect to FIGS. 2J-1 and 2J-2. In some embodiments, the removal process of the first dielectric mask layer **134** may be an etching process which may be similar to or the same as that described above with respect to FIGS. 2K-1 and 2K-2. During the etching process, the loss of the epitaxial semi-

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conductor material of the source/drain features **146A** is low, due to a large difference in the etching selectivity between the dielectric mask layers **134** and **136** and the source/drain features **146A**.

Similarly, when the small amount of residue of the first dielectric mask layer **134** is left on the source/drain region of the fin structure **104B**, the epitaxial growth of a subsequently formed source/drain feature may not be negatively affected. Furthermore, the residue of the first dielectric mask layer **134** left on the gate spacer layers **122** and **120** and/or the inner spacers **132** may not significantly increase the overall capacitance between a subsequently formed source/drain feature and the final gate stack.

FIGS. **20-1** through **2P-2** illustrate the formation of a source/drain mask structure for the first device region **50A**. The source/drain mask structure for the first device region **50A** may be formed using a similar method to the method for forming the source/drain mask structure for the second device region **50B** described above in FIGS. **2G-1** through **2L-1**. FIGS. **20-1** and **20-2** are cross-sectional views of a semiconductor structure **100** after the formation of a tri-layer mask structure, in accordance with some embodiments.

A fourth dielectric mask layer **148**, a fifth dielectric mask layer **150**, a fill layer **152**, and a tri-layer mask structure are sequentially formed over the semiconductor structure **100**, as shown in FIGS. **20-1** and **20-2**, in accordance with some embodiments. The tri-layer mask structure includes a sixth dielectric mask layer **154** as a bottom layer, a BARC layer **156** as a middle layer, and a top photoresist mask **158**, in accordance with some embodiments. In some embodiments, the top photoresist mask **158** covers the first device region **50A** of the substrate **102** and uncovered the second device region **50B**, in accordance with some embodiments.

The materials and the formation methods of the fourth dielectric mask layer **148** and the fifth dielectric mask layer **150** may be similar to or the same as those of the first dielectric mask layer **134** and the second dielectric mask layer **136** described above with respect to FIGS. **2G-1** and **2G-2**. The materials and the formation methods of the fill layer **152**, the sixth dielectric mask layer **154**, the BARC layer **156**, and the top photoresist mask **158** may be similar to or the same as the fill layer **138**, the third dielectric mask layer **140**, the BARC layer **142**, and the top photoresist mask **144** described above with respect to FIGS. **2H-1** and **2H-2**.

FIGS. **2P-1** and **2P-2** are cross-sectional views of a semiconductor structure **100** after formation of a source/drain mask structure for the first device region **50A**, in accordance with some embodiments.

Multiple etching processes are performed on the semiconductor structure **100** using the top photoresist mask **158** to remove the BARC layer **156**, the sixth dielectric mask layer **154**, the fill layer **152**, the fifth dielectric mask layer **150**, and the fourth dielectric mask layer **148** from the second device region **50B**, thereby exposing the source/drain region of the fin structure **104B**, the inner spacers **132**, the gate spacer layers **122** and **120**, the dummy gate electrode layer **116**, as shown in FIGS. **2P-1** and **2P-2**, in accordance with some embodiments. In some embodiments, the isolation structure **110** and the fin spacer layers **120'** and **122'** in the second device region **50B** are also exposed.

The etching processes of removing the BARC layer **156**, the sixth dielectric mask layer **154** and the fill layer **152** may be similar to or the same as the etching processes described above with respect to FIGS. **2I-1** through **2I-2**. The etching process used for removing the fifth dielectric mask layer **150** may be similar to or the same as the etching processes described above with respect to FIGS. **2J-1** through **2J-2**.

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The etching process used for removing the fourth dielectric mask layer **148** may be similar to or the same as the etching processes described above with respect to FIGS. **2K-1** through **2K-2**.

Afterward, the fill layer **152** is removed from the first device region **50B**, in accordance with some embodiments. The etching process used for removing the fill layer **152** may be similar to or the same as the etching processes described above with respect to FIGS. **2L-1** through **2L-2**.

Similarly, after the etching process used for removing the fourth dielectric mask layer **148**, if the small amount of residue of the fourth dielectric mask layer **148** is left on the source/drain region of the fin structure **104B**, the epitaxial growth of a subsequently formed source/drain feature may not be negatively affected, because the fourth dielectric mask layer **148** is made of a silicon containing dielectric material (such as SiO). Furthermore, the residue of the fourth dielectric mask layer **148** left on the gate spacer layers **122** and **120** and/or the inner spacers **132** may not significantly increase the overall capacitance between a subsequently formed source/drain feature and the final gate stack, because the fourth dielectric mask layer **148** has a relatively low dielectric constant value.

The embodiments of the present disclosure use the low-k fourth dielectric mask layer **148** as the bottom layer of the source/drain mask structure, and therefore the high-k fifth dielectric mask layer **150** can be substantially entirely removed. Furthermore, the low-k fourth dielectric mask layer **148** may provide a good balance between the loss of the low-k spacers and the amount of residue of the source/drain mask structure, thereby enlarging the process window of patterning the source/drain mask structure.

FIGS. **2Q-1** and **2Q-2** are cross-sectional views of a semiconductor structure **100** after the formation of source/drain features **146B**, in accordance with some embodiments.

Source/drain features **146B** are formed in the source/drain recesses **128B** at second device region **50B** of the semiconductor structure **100** using an epitaxial growth process, as shown in FIGS. **2Q-1** and **2Q-2**, in accordance with some embodiments. The epitaxial growth process may be MBE, MOCVD, or VPE, another suitable technique, or a combination thereof. A preclean process may be performed before the epitaxial growth process. For example, a wet clean using dilute ammonium hydroxide solution and/or SiCoNi clean process may be utilized. The source/drain features **146B** are grown on the source/drain region of the fin structures **104B**, in accordance with some embodiments. The source/drain features **146B** abut the second semiconductor layers **108** of the fin structures **104B** and the inner spacer layers **132**, in accordance with some embodiments.

In some embodiments wherein the fin structure **104B** is used to formed as an n-channel nanostructure transistor, the source/drain features **146B** are made of semiconductor material such as SiP, SiAs, SiCP, SiC, Si, GaAs, another suitable semiconductor material, or a combination thereof. In some embodiments, the source/drain features **146B** are doped with an n-type dopant during the epitaxial growth process. For example, the n-type dopant may be phosphorous (P) or arsenic (As). For example, the source/drain features **146B** may be the epitaxially grown Si doped with phosphorous to form silicon:phosphor (Si:P) source/drain features and/or arsenic to form silicon:arsenic (Si:As) source/drain feature.

The fifth dielectric mask layer **150** of the source/drain mask structure may sufficiently prevent the epitaxial semiconductor material from being grown on the dielectric surface of the fifth dielectric mask layer **150** during the

epitaxial growth process of source/drain features **146B**, in accordance with some embodiments. As such, the source/drain mask structure can be easily removed in a following process. The source/drain features **146B** has a narrow bottom portion confined between the fin spacer layers **120'** and **122'** and a wider upper portion having facet surfaces. Although the source/drain features **146B** are illustrated as having the facet surfaces, the source/drain features **146A** may have a curved surface in some other embodiments.

FIGS. **2R-1** and **2R-2** are cross-sectional views of a semiconductor structure **100** the removal of portions of the fifth dielectric mask layer **150** and the fourth dielectric mask layer **148** in the first device region **50A**, in accordance with some embodiments.

The fifth dielectric mask layer **150** and the fourth dielectric mask layer **148** are removed from the first device region **50A** of the substrate **102**, thereby exposing the source/drain feature **146A**, the gate spacer layers **122** and **120**, and the dummy gate electrode layer **116**, as shown in FIGS. **2R-1** and **2R-2**, in accordance with some embodiments. In some embodiments, the isolation structure **110**, and the fin spacer layers **120'** and **122'** in the first device region **50A** are also exposed. The removal process may be similar to or the same as that described above with respect to FIGS. **2N-1** and **2N-2**. Similarly, when the small amount of residue of the fourth dielectric mask layer **148** is left on the gate spacer layers **122** and **120**, the overall capacitance between the source/drain feature **146A** and subsequently formed final gate stack may not significantly increase.

FIGS. **2S-1** and **2S-2** are cross-sectional views of a semiconductor structure **100** after the formation of a contact etching stop layer (CESL) **160** and an interlayer dielectric (ILD) layer **162**, in accordance with some embodiments.

A contact etching stop layer **160** is formed over the semiconductor structure **100**, as shown in FIGS. **2S-1** and **2S-2**, in accordance with some embodiments. The contact etching stop layer **160** extends along and covers the surfaces of the wider upper portions of the source/drain features **146A** and **146B**, the sidewalls of the gate spacer layers **122**, the sidewalls and the upper portions of the fin spacers **122'**, the upper portions of the fin spacers **120'**, and the upper surface of the isolation structure **110**, in accordance with some embodiments. Afterward, an interlayer dielectric layer **162** is formed over the contact etching stop layer **144**, as shown in FIGS. **2S-1** and **2S-2**, in accordance with some embodiments.

In some embodiments, the contact etching stop layer **160** is made of a dielectric material, such as silicon oxide (SiO_2), silicon nitride (SiN), silicon oxynitride (SiOC), silicon carbide (SiC), oxygen-doped silicon carbide (SiC:O), oxygen-doped silicon carbonitride (Si(O)CN), or a combination thereof. In some embodiments, a dielectric material for the contact etching stop layer **160** is globally and conformally deposited over the semiconductor structure **100** using CVD (such as LPCVD, PECVD, HDP-CVD, HARP, and FCVD), ALD, another suitable method, or a combination thereof.

In some embodiments, the interlayer dielectric layer **162** is made of a dielectric material, such as un-doped silicate glass (USG), or doped silicon oxide such as borophosphosilicate glass (BPSG), fluoride-doped silicate glass (FSG), phosphosilicate glass (PSG), borosilicate glass (BSG), and/or another suitable dielectric material. In some embodiments, the interlayer dielectric layer **162** is made of a different material than the contact etching stop layer **160**. In some embodiments, a dielectric material for the interlayer dielectric layer **162** is deposited using such as CVD (such as HDP-CVD, PECVD, HARP or FCVD), another suitable

technique, and/or a combination thereof. Afterward, the dielectric materials for the contact etching stop layer **160** and the interlayer dielectric layer **162** above the upper surfaces of the dummy gate electrode layer **116** are removed using such as CMP until the upper surfaces of the dummy gate structure **112** is exposed. In some embodiments, the upper surface of the interlayer dielectric layer **162** is substantially coplanar with the upper surfaces of the dummy gate electrode layers **116**.

FIGS. **2T-1** and **2T-2** are cross-sectional views of a semiconductor structure **100** after the formation of a gate trench **164** and gaps **166**, in accordance with some embodiments.

The dummy gate structure **112** is removed using an etching process to form a gate trench **164**, as shown in FIGS. **2T-1**, in accordance with some embodiments. The gate trench **164** exposes the channel regions CH of the fin structures **104A** and **104B**, in accordance with some embodiments. In some embodiments, the gate trench **164** also exposes the inner sidewalls of the gate spacer layer **120** facing the channel region, in accordance with some embodiments.

In some embodiments, the etching process includes one or more etching processes. For example, when the dummy gate electrode layer **116** is made of polysilicon, a wet etchant such as a tetramethylammonium hydroxide (TMAH) solution may be used to selectively remove the dummy gate electrode layers **116**. For example, the dummy gate dielectric layer **114** may be thereafter removed using a plasma dry etching, a dry chemical etching, and/or a wet etching.

The first semiconductor layers **106** of the fin structures **104A** and **104B** are removed using an etching process to form gaps **166**, as shown in FIG. **2T-1**, in accordance with some embodiments. The inner spacer layers **132** may be used as an etching stop layer in the etching process, which may protect the source/drain features **146A** and **146B** from being damaged. The gaps **166** are formed between the neighboring second semiconductor layers **108** and between the lowermost second semiconductor layer **108** and the lower fin elements **104L** of the fin structures **104A** and **104B**, in accordance with some embodiments. In some embodiments, the gaps **166** also expose the inner sidewalls of the inner spacer layers **132** facing the channel region.

After the etching process, the four main surfaces (only upper and bottom surfaces are shown in FIG. **2T-1**) of the second semiconductor layers **108** are exposed, in accordance with some embodiments. The exposed second semiconductor layers **108** form nanostructures that function as channel layers of the resulting semiconductor devices (e.g., a nanostructure transistor), in accordance with some embodiments.

In some embodiments, the etching process includes a selective wet etching process, such as APM (e.g., ammonia hydroxide-hydrogen peroxide-water mixture) etching process. In some embodiments, the wet etching process uses etchants such as ammonium hydroxide (NH_4OH), TMAH, ethylenediamine pyrocatechol (EDP), and/or potassium hydroxide (KOH) solutions. In some embodiments, the inner spacer layers **132** serve an etching stop layer in the etching process.

FIGS. **2U-1** and **2U-2** are cross-sectional views of a semiconductor structure **100** after the formation of a final gate stack **172**, in accordance with some embodiments.

A gate dielectric layer **170** is formed on and wraps around the exposed surfaces of the nanostructures **108**, as shown in FIG. **2U-1**, in accordance with some embodiments. The gate dielectric layer **170** is also formed along and covers the inner sidewalls of the gate spacer layers **120** facing the channel

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region, in accordance with some embodiments. The gate dielectric layer **170** is also formed along and covers the inner sidewalls of the inner spacer layers **132** facing the channel region, in accordance with some embodiments.

The gate dielectric layer **170** may include an interfacial layer and a high-k dielectric layer formed over the interfacial layer. The interfacial layer may be made of a chemically formed silicon oxide by one or more cleaning processes such as including ozone (O₃), ammonia hydroxide-hydrogen peroxide-water mixture, and/or hydrochloric acid-hydrogen peroxide-water mixture. The interfacial layer may be formed by oxidizing outer portions of the nanostructures **108** and formed on the exposed main surfaces of the nanostructures **108**, but not formed on the surface of dielectric layers (such as gate spacers and inner spacers). In some embodiments, the high-k dielectric layer is made of a dielectric material with high dielectric constant (k value), for example, greater than 3.9. In some embodiments, the high-k dielectric layer includes hafnium oxide (HfO₂), TiO₂, HfZrO, Ta₂O₃, HfSiO₄, ZrO₂, ZrSiO₂, LaO, AlO, ZrO, TiO, Ta₂O₅, Y₂O₃, SrTiO₃ (STO), BaTiO₃ (BTO), BaZrO, HfZrO, HfLaO, HfSiO, LaSiO, AlSiO, HfTaO, HfTiO, (Ba,Sr)TiO₃ (BST), Al₂O₃, Si₃N₄, oxynitrides (SiON), a combination thereof, or another suitable material. The high-k dielectric layer may be deposited using ALD, PVD, CVD, and/or another suitable technique.

A metal gate electrode layer **172** is formed over the gate dielectric layer **170** and fills remainders of the gate trench **164** and the gaps **166**, as shown in FIG. 2U-1, in accordance with some embodiments. The metal gate electrode layer **172** wraps around the nanostructures **108**, in accordance with some embodiments. In some embodiments, the metal gate electrode layer **172** is made of more than one conductive material, such as a metal, metal alloy, conductive metal oxide and/or metal nitride, another suitable conductive material, and/or a combination thereof. For example, the metal gate electrode layer **172** may be made of Ti, Ag, Al, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, TaN, Ru, Mo, Al, WN, Cu, W, Re, Ir, Co, Ni, another suitable conductive material, or multilayers thereof.

The metal gate electrode layer **172** may be a multi-layer structure with various combinations of a diffusion barrier layer, work function layers with a selected work function to enhance the device performance (e.g., threshold voltage) for n-channel FETs or p-channel FETs, a capping layer to prevent oxidation of work function layers, a glue layer to adhere work function layers to a next layer, and a metal fill layer to reduce the total resistance of gate stacks, and/or another suitable layer. The metal gate electrode layer **172** may be formed using ALD, PVD, CVD, e-beam evaporation, or another suitable process. The metal gate electrode layer **172** may be formed separately for n-channel nanostructure transistors (such as in the second device region **50B**) and n-channel nanostructure transistors (such as in the first device region **50A**), which may use different work function materials.

A planarization process such as CMP may be performed on the semiconductor structure **100** to remove the materials of the gate dielectric layer **170** and the metal gate electrode layer **172** formed above the upper surface of the interlayer dielectric layer **162**, in accordance with some embodiments. After the planarization process, the upper surface of the metal gate electrode layer **172** and the upper surface of the interlayer dielectric layer **162** are substantially coplanar, in accordance with some embodiments.

The gate dielectric layer **170** and the metal gate electrode layer **172** combine to form a final gate stack **168**, in

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accordance with some embodiments. In some embodiments, the final gate stack **168** extends in Y direction. That is, the final gate stack **168** has longitudinal axes parallel to Y direction, in accordance with some embodiments. The final gate stack **168** wraps around each of the nanostructures **108** and interposes between the source/drain features **146A** and between the source/drain feature **146B**, in accordance with some embodiments.

In the first device region **50A**, the final gate stack **168** combines with the source/drain features **146A** to form a nanostructure transistor, such as a p-channel nanostructure transistor, in accordance with some embodiments. In the second device region **50B**, the final gate stack **168** combines with the source/drain features **146B** to form a nanostructure transistor, such as an n-channel nanostructure transistor, in accordance with some embodiments. The final gate stack **168** may engage the channel region of the nanostructures **108**, such that current can flow between the source/drain feature **146A** and between the source/drain feature **146B** during operation.

It is understood that the semiconductor structure **100** may undergo further CMOS processes to form various features over the semiconductor structure **100**, such as a multilayer interconnect structure (e.g., contacts to gate and/or source/drain features, vias, lines, inter metal dielectric layers, passivation layers, etc.).

The embodiments of the present disclosure use a bilayered source/drain mask structure that includes a silicon-containing dielectric mask layer (such as the first dielectric mask layer **134** and the fourth dielectric mask layer **148**) and a high-k dielectric mask layer (such as the second dielectric mask layer **136** and the fifth dielectric mask layer **150**) over the silicon-containing dielectric mask layer. A good balance between the loss of the low-k spacers and the amount of residue of the source/drain mask structure may be provided, thereby enlarging the process window of patterning the source/drain mask structure. As a result, the yield and the performance of the resulting semiconductor device may be improved.

FIGS. 3A through 3G are cross-sectional views illustrating the formation of a semiconductor structure at various intermediate stages corresponding to cross-section X-X shown in FIG. 1, in accordance with some embodiments. The embodiments of FIGS. 3A through 3G are similar to that shown in FIGS. 2A-1 through 2U-2 except that the first dielectric mask layer **134** is left within the inner spacers **132**.

FIG. 3A illustrates a cross-sectional view of a semiconductor structure **200** which has been subjected the steps described above respect to FIGS. 2A-1 through 2F-2, in accordance with some embodiments.

Fin structures **104A** and **104B** are formed over the substrate **102**, and an isolation structure **110** (FIG. 2A-2) are formed to surround the lower fin elements **104L** of the fin structures **104A** and **104B**, in accordance with some embodiments. A dummy gate structure **112** is formed across and surrounds the channel region of the fin structures **104A** and **104B**, and gate spacer layers **120** and **122** are sequentially formed over the semiconductor structure **200**, in accordance with some embodiments. One or more etching process is performed to remove horizontal portions of the gate spacer layers **120** and **122** and recess the source/drain regions of the fin structures **104A** and **104B**, thereby forming source/drain recesses **128A** in the first device region **50A** and source/drain recesses **128B** in the second device region **50B**, in accordance with some embodiments. An etching process is performed on the semiconductor structure **200** to laterally recess, from the source/drain recesses **128A** and

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128B, the first semiconductor layers 106 of the fin structures 104A and 104B to form notches 130, and inner spacer layers 132 are formed in the notches 130 to about the first semiconductor layers 106 of the fin structures 104A and 104B, in accordance with some embodiments. Due to the conformal deposition of the inner spacer layers 132, seams 202 are formed within the inner spacer layers 132, as shown in FIG. 3A, in accordance with some embodiments.

FIG. 3B illustrates a cross-sectional view of a semiconductor structure 200 which has been subjected the steps described above respect to FIGS. 2G-1 through 2H-2, in accordance with some embodiments.

A first dielectric mask layer 134, the second dielectric mask layer 136, a fill layer 138, and a tri-layer mask structure including a third dielectric mask layer 140, a BARC layer 142, and a top photoresist mask 144 are formed over the semiconductor structure 200, in accordance with some embodiments. The first dielectric mask layer 134 is formed to fill the seams 202, as shown in FIG. 3B, in accordance with some embodiments.

FIG. 3C illustrates a cross-sectional view of a semiconductor structure 200 which has been subjected the steps described above respect to FIGS. 2I-1 through 2L-2, in accordance with some embodiments.

One or more etching process is performed using the top photoresist mask 144 to remove portions of the BARC layer 142, the third dielectric mask layer 140, the fill layer 138, the second dielectric mask layer 136, and the first dielectric mask layer 134 in the first device region 50A of the substrate 102, thereby exposing the source/drain region of the fin structure 104A, in accordance with some embodiments. After the etching process (FIGS. 2K-1 and 2K-2) of removing the first dielectric mask layer 134, the portion of the first dielectric mask layer 134 filling the seams 202 is left within the inner spacer 132 and are denoted as 134', in accordance with some embodiments. The remaining portions 134' of the first dielectric mask layer 134 may serve as a part of the inner spacers.

FIG. 3D illustrates a cross-sectional view of a semiconductor structure 200 which has been subjected the steps described above respect to FIGS. 2M-1 through 2M-2, in accordance with some embodiments.

Source/drain features 146A are formed in the source/drain recesses 128A at first device region 50A of the semiconductor structure 200 using an epitaxial growth process, in accordance with some embodiments.

FIG. 3E illustrates a cross-sectional view of a semiconductor structure 200 which has been subjected the steps described above respect to FIGS. 2N-1 through 2P-2, in accordance with some embodiments.

The mask structure for the second device region 50B is removed, in accordance with some embodiments. After the etching process (FIGS. 2N-1 and 2N-2) of removing the first dielectric mask layer 134, the portion of the first dielectric mask layer 134 filling the seams 202 is left within the inner spacer 132 and are denoted as 134', in accordance with some embodiments. The remaining portions 134' of the first dielectric mask layer 134 may serve as a part of the inner spacers.

A fourth dielectric mask layer 148, the fifth dielectric mask layer 150, a fill layer 152 (FIG. 2O-1), and a tri-layer mask structure (FIG. 2O-1) including a sixth dielectric mask layer 154, a BARC layer 156, and a top photoresist mask 158 are formed over the semiconductor structure 200, in accordance with some embodiments. One or more etching process is performed using the top photoresist mask 158 to remove portions of the BARC layer 156, the sixth dielectric

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mask layer 154, the fill layer 152, the fifth dielectric mask layer 150 and the fourth dielectric mask layer 148 in the second device region 50B of the substrate 102, thereby exposing the source/drain region of the fin structure 104B, in accordance with some embodiments.

FIG. 3F illustrates a cross-sectional view of a semiconductor structure 200 which has been subjected the steps described above respect to FIGS. 2Q-1 through 2R-2, in accordance with some embodiments.

Source/drain features 146B are formed in the source/drain recesses 128B at second device region 50B of the semiconductor structure 200 using an epitaxial growth process, in accordance with some embodiments. The mask structure for the first device region 50A is removed, in accordance with some embodiments.

FIG. 3G illustrates a cross-sectional view of a semiconductor structure 200 which has been subjected the steps described above respect to FIGS. 2S-1 through 2U-2, in accordance with some embodiments.

A contact etching stop layer 160 and an interlayer dielectric layer 162 are sequentially formed over the semiconductor structure 200, in accordance with some embodiments. The dummy gate structure 112 is removed using an etching process to form a gate trench 164 (FIG. 2T-1), in accordance with some embodiments. The first semiconductor layers 106 of the fin structures 104A and 104B are removed using an etching process to form gaps 166 (FIG. 2T-1), in accordance with some embodiments. A final gate structure 168 is formed to wraps around each of the nanostructures 108 (the second semiconductor layers 108 of the fin structure 104A), in accordance with some embodiments.

The remaining portions 134' of the first dielectric mask layer 134 are left on the inner spacers 132 may not negatively affect the epitaxial growth of the source/drain feature. Furthermore, the overall capacitance between the source/drain features and final gate stack may not significantly increase. In addition, because the remaining portions 134' of the first dielectric mask layer 134 fills the seams 202, the inner spacers 132 (and 134') may be more robust to resist the etching process used for forming the gaps 166 (FIG. 2U-1). As a result, the yield and the performance of the resulting semiconductor device may be improved.

FIG. 4 is a cross-sectional view illustrating a semiconductor structure 400, in accordance with some embodiments. The semiconductor structure 400 shown FIG. 4 is similar to the semiconductor structure 100 shown in FIG. 2U-1 except that the first dielectric mask layer 134 is left on the gate spacer layer 122.

After the etching process (FIGS. 2K-1 and 2K-2) of removing the first dielectric mask layer 134 from the first device region 50A, a portion of the first dielectric mask layer 134 is left on the gate spacer 122 in the first device region 50A, in accordance with some embodiments. And, after the etching process (FIGS. 2N-1 and 2N-2) of removing the first dielectric mask layer 134 from the second device region 50B, a portion of the first dielectric mask layer 134 is left on the gate spacer 122 in the second device region 50B, in accordance with some embodiments. The remaining portions of the first dielectric mask layer 134 may serve as a part of the gate spacer layers and are denoted as 134".

The remaining portions 134" of the first dielectric mask layer 134 left on the gate spacers 122 may not negatively affect the epitaxial growth of the source/drain feature. Furthermore, the overall capacitance between the source/drain features and final gate stack may not significantly increase.

FIGS. 5A-1 through 5A-2 are cross-sectional views illustrating the formation of a semiconductor structure at various

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intermediate stages, in which the figures ending with “-1” (such as FIG. 5A-1) are cross-sectional views corresponding to cross-section X-X shown in FIG. 1, and the figures ending with “-2” (such as FIG. 5A-2) are cross-sectional views corresponding to cross-section Y-Y shown in FIG. 1, in accordance with some embodiments. The embodiments of FIGS. 5A-1 through 5A-2 are similar to that shown in FIGS. 2A-1 through 2U-2 except that a GAAFET device is formed in the first device region 50A and a FinFET device is formed in the second device region 50B.

FIG. 5A illustrates a cross-sectional view of a semiconductor structure 400 after the formation of a fin structure 104A and a fin structure 404B and an isolation structure 110, in accordance with some embodiments.

The substrate 102 includes a first device region 50A where a fin structure 104A is formed and a second device region 50B where a fin structure 404B is formed, as shown in FIGS. 5A-1 and 5A-2, in accordance with some embodiments. In some embodiments, the first device region 50A is a p-type device region and used for forming p-type semiconductor devices (e.g., p-channel nanostructure transistors). In some embodiments, the second device region 50B is an n-type device region and used for forming n-type semiconductor devices such as n-channel FinFET.

The formation of the fin structure 104A and the fin structure 404B includes recessing an upper portion of the substrate 102 at the first device region 50A using an etching process to form a space in the first device region 50A, and then forming an epitaxial stack over the substrate 102 in the space using an epitaxial growth process, in accordance with some embodiments. The etching process may be an anisotropic etching process such as dry plasma etching, an isotropic etching process such as dry chemical etching, remote plasma etching or wet chemical etching, and/or a combination thereof. The material, composition and the formation method of the epitaxial stack may be similar to or the same as the epitaxial stack described above respect to FIGS. 2A-1 and 2A-2. A planarization process such as CMP or etching back process is performed to remove a portion of the epitaxial stack formed over the substrate 102 until the upper surface of the substrate 102 is exposed, in accordance with some embodiments. After the planarization process, the upper surface of the epitaxial stack and the upper surface of the substrate 102 are substantially coplanar, in accordance with some embodiments. Afterward, the epitaxial stack in the first device region 50A and the upper portion of the substrate 102 in the second device region 50B are patterned into the fin structures 104A in the first device region 50A and the fin structures 404B in the second device region 50B, in accordance with some embodiments. As such, the fin structures 404B and the upper portion of the substrate 102 are made of a single, continuous semiconductor material, in accordance with some embodiments.

An isolation structure 110 is formed to surround a lower fin element 104L of the fin structure 104A and a lower fin element 404L of the fin structure 404B, as shown in FIG. 5A-2, in accordance with some embodiments. The fin structure 404B includes an upper fin element protruding from between the isolation structure 110, in accordance with some embodiments. The material and the formation method of the isolation structure 110 may be similar to or the same as that of the isolation structure 110 as described above with respect to FIGS. 2A-1 and 2A-2.

FIGS. 5B-1 and 5B-2 illustrates a cross-sectional view of a semiconductor structure 400 which has been subjected the steps described above respect to FIGS. 2B-1 through 2F-2, in accordance with some embodiments.

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A dummy gate structure 112 is formed across and surrounds the channel region of the fin structure 104A and the channel region of the fin structure 404B, and gate spacer layers 120 and 122 are sequentially formed over the semiconductor structure 400, in accordance with some embodiments. One or more etching process is performed to removes horizontal portions of the gate spacer layers 120 and 122 and recess the source/drain regions of the fin structure 104A and the source/drain regions of the fin structure 404B, thereby forming source/drain recesses 128A in the first device region 50A and source/drain recesses 128B in the second device region 50B, in accordance with some embodiments. An etching process is performed on the semiconductor structure 400 to laterally recess, from the source/drain recesses 128A, the first semiconductor layers 106 of the fin structure 104A to form notches 130 (FIG. 2E-1), and inner spacer layers 132 are formed in the notches 130 to abut the first semiconductor layers 106 of the fin structure 104, in accordance with some embodiments.

FIGS. 5C-1 and 5C-2 illustrates a cross-sectional view of a semiconductor structure 400 which has been subjected the steps described above respect to FIGS. 2G-1 through 2M-2, in accordance with some embodiments.

A mask structure including a first dielectric mask layer 134 and a second dielectric mask layer 136 are formed to cover the second device region 50B, in accordance with some embodiments. Source/drain features 146A are formed in the source/drain recesses 128A at first device region 50A of the semiconductor structure 400 using an epitaxial growth process, in accordance with some embodiments.

FIGS. 5D-1 and 5D-2 illustrates a cross-sectional view of a semiconductor structure 400 which has been subjected the steps described above respect to FIGS. 2N-1 through 2Q-2, in accordance with some embodiments.

The mask structure for the second device region 50B is removed, in accordance with some embodiments. A mask structure including a fourth dielectric mask layer 148 and a fifth dielectric mask layer 150 are formed to cover the first device region 50A, in accordance with some embodiments. Source/drain features 146B are formed in the source/drain recesses 128B at second device region 50B of the semiconductor structure 400 using an epitaxial growth process, in accordance with some embodiments.

FIGS. 5E-1 and 5E-2 illustrates a cross-sectional view of a semiconductor structure 400 which has been subjected the steps described above respect to FIGS. 2R-1 through 2U-2, in accordance with some embodiments.

The mask structure for the first device region 50A is removed, in accordance with some embodiments. A contact etching stop layer 160 and an interlayer dielectric layer 162 are sequentially formed over the semiconductor structure 400, in accordance with some embodiments. The dummy gate structure 112 is removed using an etching process to form gate trench 164 (FIG. 2T-1), in accordance with some embodiments. The first semiconductor layers 106 of the fin structure 104A are removed using an etching process to form gaps 166 (FIG. 2T-1), in accordance with some embodiments. A final gate structure 168 is formed to wraps around each of nanostructures 108 (the second semiconductor layers 108 of the fin structure 104A) and the channel region of the fin structure 404B, thereby forming a p-channel GAA FET in the first device region 50A and an n-channel FinFET in the second device region 50B, in accordance with some embodiments.

The embodiments provide a hybrid structure including a GAA FET and a FinFET formed over the same semiconductor substrate (or wafer). Integrated circuits including the

hybrid structures may have better electrical performance and greater design flexibility. FIGS. 5A-2 through 5E-2 describes the embodiments of the p-channel GAA FET and the n-channel FinFET, but are not limited thereto. The aspect of the present disclosure for forming the source/drain mask structure may also contemplate the embodiments of a p-channel FinFET and an n-channel GAA FET, and/or p-channel FinFET and an n-channel FinFET.

As described above, the aspect of the present disclosure is directed to separately forming source/drain features in different device regions. Embodiments of the method include forming a silicon-containing dielectric mask layer (such as the first dielectric mask layer 134 or the fourth dielectric mask layer 148) and a high-k dielectric mask layer (such as the second dielectric mask layer 136 or the fifth dielectric mask layer 150) over the silicon-containing dielectric mask layer, patterning the dielectric mask layers into a source/drain mask structure that covers a device region and exposes another device region, and then growing source/drain features in the exposed device region. Therefore, the embodiments of the present disclosure use the silicon-containing dielectric mask layer as the bottom layer of the source/drain mask structure, which may provide a good balance between the loss of the low-k spacers (such as the gate spacer layers 120 and 122 and/or the inner spacers 132) and the amount of residue of the source/drain mask structure. Therefore, the process window of patterning the source/drain mask structure is enlarged.

Embodiments for forming a semiconductor structure may be provided. The method may include forming a first fin structure and a second fin structure, forming a first mask structure to cover a second fin structure. The first mask structure includes a first dielectric mask layer and a second dielectric mask layer over the first dielectric mask layer, and the first dielectric mask layer and the second dielectric mask layer are made of different materials. The method may also include forming a first source/drain feature in the first fin structure, and removing the first mask structure to expose the second fin structure. Therefore, a good balance between the loss of low-k spacers and the amount of residue of the source/drain mask structure is provided, thereby enlarging the process window of patterning the source/drain mask structure.

In some embodiments, a method for forming semiconductor structure is provided. The method includes forming a first active region and a second active region, etching the first active region and the second active region to form a first recess and a second recess, respectively, forming the first dielectric layer over the first active region and the second active region, forming a first fill layer over the first dielectric layer to overfill the first recess and the second recess, forming a first dielectric mask over the first fill layer, etching first portions of the first dielectric mask and the first fill layer over the first active region, removing a first portion of the first dielectric layer over the first active region, and forming a first source/drain feature on the first active region.

In some embodiments, a method for forming semiconductor structure is provided. The method includes forming a first dielectric layer over a first active region and a second active region, forming a second dielectric layer over the first dielectric layer, forming a patterned dielectric mask on the second dielectric layer over the second active region, removing a first portion of the second dielectric layer over the first active region, removing the patterned dielectric mask while removing the first portion of the second dielectric layer over the first active region, removing a first portion of the first

dielectric layer over the first active region, and growing a first epitaxial material on the first active region.

In some embodiments, a method for forming semiconductor structure is provided. The method includes forming a first active region and a second active region. Each of the first active region and the second active region includes first semiconductor layers and second semiconductor layers alternately stacked. The method further includes etching the first active region and the second active region to form a first recess and a second recess, respectively, laterally recessing the first active region and the second active region to form notches, forming inner spacers in the notches, forming a first dielectric layer in the first recess and the second recess, forming a second dielectric layer over the first dielectric layer, patterning the first dielectric layer and the second dielectric layer to form a first source/drain mask on the second active region, and forming a first source/drain feature on the first active region. The first dielectric layer is in direct contact with the inner spacers.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method for forming a semiconductor structure, comprising:
 - forming a first active region and a second active region;
 - etching the first active region and the second active region to form a first recess and a second recess, respectively;
 - forming the first dielectric layer over the first active region and the second active region;
 - forming a first fill layer over the first dielectric layer to overfill the first recess and the second recess;
 - forming a first dielectric mask over the first fill layer;
 - etching first portions of the first dielectric mask and the first fill layer over the first active region;
 - removing a first portion of the first dielectric layer over the first active region; and
 - forming a first source/drain feature on the first active region.
2. The method for forming the semiconductor structure as claimed in claim 1, further comprising:
 - removing a second portion of the first dielectric mask over the second active region while removing the first portion of the first dielectric layer over the first active region.
3. The method for forming the semiconductor structure as claimed in claim 1, further comprising:
 - removing a second portion of the first fill layer over the second active region to expose a second portion of the first dielectric layer over the second active region before forming the first source/drain feature.
4. The method for forming the semiconductor structure as claimed in claim 1, further comprising:
 - forming the second dielectric layer over the first active region and the second active region;
 - forming a second fill layer over the first dielectric layer to overfill the second recess;

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forming a second dielectric mask over the second fill layer;
 etching first portions of the second dielectric mask and the second fill layer over the second active region;
 removing a first portion of the second dielectric layer over the second active region; and
 forming a second source/drain feature on the second active region.

5. The method for forming the semiconductor structure as claimed in claim 1, further comprising:

forming a second dielectric layer over the first active region and the second active region, wherein the first dielectric layer is formed on the second dielectric layer; and
 removing a first portion of the second dielectric layer over the first active region before forming the first source/drain feature.

6. The method for forming the semiconductor structure as claimed in claim 5, wherein the first dielectric layer is made of an oxide of amphoteric metal, and the second dielectric layer is made of silicon oxide.

7. The method for forming the semiconductor structure as claimed in claim 5, wherein:

etching the first portion of the first dielectric layer comprises a wet etching process using an alkaline solution, and
 etching the first portion of the second dielectric layer comprises a wet etching process using an acidic solution.

8. The method for forming the semiconductor structure as claimed in claim 1, wherein each of the first active region and the second active region includes first semiconductor layers and second semiconductor layers alternately stacked.

9. The method for forming the semiconductor structure as claimed in claim 8, further comprising:

laterally recessing the first active region and the second active region to form notches; and
 forming inner spacers in the notches, wherein the first dielectric layer is formed on and in direct contact with the inner spacers.

10. A method for forming a semiconductor structure, comprising:

forming a first dielectric layer over a first active region and a second active region;
 forming a second dielectric layer over the first dielectric layer;
 forming a patterned dielectric mask on the second dielectric layer over the second active region;
 removing a first portion of the second dielectric layer over the first active region;
 removing the patterned dielectric mask while removing the first portion of the second dielectric layer over the first active region;
 removing a first portion of the first dielectric layer over the first active region; and
 growing a first epitaxial material on the first active region.

11. The method for forming the semiconductor structure as claimed in claim 10, wherein the first dielectric layer is made of silicon oxide, and the second dielectric layer is made of zinc oxide, aluminum oxide, tin oxide, lead oxide, beryllium oxide, or chromium oxide.

12. The method for forming the semiconductor structure as claimed in claim 10, further comprising:

removing second portions of the first dielectric layer and the second dielectric layer over the second active region; and

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growing a second epitaxial material on the second active region.

13. The method for forming the semiconductor structure as claimed in claim 12, wherein the first epitaxial material is doped with a p-type dopant, and the second epitaxial material is doped with an n-type dopant.

14. The method for forming the semiconductor structure as claimed in claim 10, further comprising:

alternately stacking first semiconductor layers and second semiconductor layers over a substrate; and
 patterning the first semiconductor layers and the second semiconductor layers into the first active region and the second active region.

15. The method for forming the semiconductor structure as claimed in claim 10, further comprising:

forming an isolation structure to surround the first active region and the second active region, wherein after removing a first portion of the first dielectric layer, second portions of the first dielectric layer and the second dielectric layer over the second active region terminate on a top surface of a portion of the isolation structure between the first active region and the second active region.

16. A method for forming a semiconductor structure, comprising:

forming a first active region and a second active region, wherein each of the first active region and the second active region includes first semiconductor layers and second semiconductor layers alternately stacked;
 etching the first active region and the second active region to form a first recess and a second recess, respectively;
 laterally recessing the first active region and the second active region to form notches;
 forming inner spacers in the notches;
 forming a first dielectric layer in the first recess and the second recess, wherein the first dielectric layer is in direct contact with the inner spacers;
 forming a second dielectric layer over the first dielectric layer;
 patterning the first dielectric layer and the second dielectric layer to form a first source/drain mask on the second active region; and
 forming a first source/drain feature on the first active region.

17. The method for forming the semiconductor structure as claimed in claim 16, wherein patterning the first dielectric layer and the second dielectric layer comprises:

forming a fill layer to overfill the first recess and the second recess;
 forming a patterned hard mask on the fill layer; and
 etching the fill layer, the second dielectric layer and the first dielectric layer.

18. The method for forming the semiconductor structure as claimed in claim 17, wherein the patterned hard mask is removed during the second dielectric layer is etched.

19. The method for forming the semiconductor structure as claimed in claim 17, wherein a portion of the first dielectric layer remains in the notches after etching the fill layer, the second dielectric layer and the first dielectric layer.

20. The method for forming the semiconductor structure as claimed in claim 16, further comprising:

removing the first source/drain mask;
 forming a second source/drain feature on the second active region;
 removing the first semiconductor layers; and

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forming a gate stack to surround the second semiconductor layers.

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